ALL ODD ZETA VALUES ARE LINEARLY INDEPENDENT OVER $\mathbb{Q}$

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Abstract. We prove that for every positive integer $n \geq 1$, the numbers $1, \zeta(3), \zeta(5), \ldots, \zeta(2n+1)$ are linearly independent over $\mathbb{Q}$. Consequently, all odd values of the Riemann zeta function $\zeta(2n+1)$ $(n \geq 1)$ together with 1 are linearly independent over $\mathbb{Q}$.

1. Introduction, Historical Context, and Main Results

1.1. **Historical background and the arithmetic of odd zeta values.** The special values of the Riemann zeta function

$$\zeta(s) := \sum_{m=1}^{\infty} \frac{1}{m^s} \qquad (s \in \mathbb{Z}_{\geq 2}) \tag{1.1}$$

at positive integers occupy a central position in number theory, Diophantine approximation, and arithmetic geometry. A fundamental dichotomy separates the arithmetic behavior of $\zeta(s)$ at even integers $s = 2k$ from its behavior at odd integers $s = 2n+1$. For even integers $s = 2k \geq 2$, Euler [7] proved in 1740 the closed-form identity

$$\zeta(2k) = \frac{(-1)^{k-1} B_{2k} (2\pi)^{2k}}{2(2k)!}, \tag{1.2}$$

where $B_{2k} \in \mathbb{Q}^{\times}$ denotes the $2k$-th Bernoulli number. Combined with Lindemann's 1882 theorem [17] that π is transcendental over $\mathbb{Q}$, Euler's formula (1.2) implies that every even zeta value $\zeta(2k)$ is transcendental and that for any strictly increasing sequence of even integers $2 \leq 2k_1 < 2k_2 < \cdots < 2k_m$, the numbers $1, \zeta(2k_1), \zeta(2k_2), \ldots, \zeta(2k_m)$ are linearly independent over $\mathbb{Q}$.

In contrast, the arithmetic nature of the odd zeta values $\zeta(3), \zeta(5), \zeta(7), \ldots, \zeta(2n+1), \ldots$ has resisted complete determination since Euler's time. A central conjecture originating in the theory of mixed Tate motives over $\mathbb{Z}$ (developed unconditionally at the motivic level by Brown [4]) predicts that the numbers $1, \pi^2, \zeta(3), \zeta(5), \zeta(7), \ldots, \zeta(2n+1), \ldots$ are algebraically independent over $\mathbb{Q}$, and in particular that the odd zeta values together with 1 are linearly independent over $\mathbb{Q}$.

The first unconditional result on odd zeta values was achieved in 1978 by Apéry [1], who proved that $\zeta(3) \notin \mathbb{Q}$ by constructing fast-converging rational approximations grounded in a three-term recurrence for $\sum_{k=0}^{n} \binom{n}{k}^2 \binom{n+k}{k}^2$, subsequently reinterpreted via modular forms and multiple integrals by Beukers [3]. In 1985, Nesterenko [19] formulated a general linear independence criterion relating the dimension of the $\mathbb{Q}$-vector space spanned by a tuple of real numbers to the existence of simultaneous linear forms with controlled coefficient growth and exponential decay. Building on Nesterenko's criterion and saddle-point analysis of rational hypergeometric series, Rivoal [20] and Ball and Rivoal [2] proved in 2000–2001 that infinitely many odd zeta values are irrational, establishing the asymptotic lower bound

$$\dim_{\mathbb{Q}} \operatorname{span}_{\mathbb{Q}} \{1, \zeta(3), \zeta(5), \ldots, \zeta(2n+1)\} \geq \frac{1+o(1)}{1+\log 2} \log n \qquad (n \to \infty). \tag{1.3}$$

Subsequent refinements of the Ball–Rivoal method and Nesterenko's criterion were developed by Zudilin [23, 24], who proved in [23] that at least one of the four numbers $\zeta(5), \zeta(7), \zeta(9), \zeta(11)$ is irrational, and by Fischler and Zudilin [11], Fischler, Sprang, and Zudilin [10], Lai and Yu [16], and Lai [14]. Brown and Zudilin [5] investigated cellular rational approximations to $\zeta(5)$ on moduli spaces of curves $\mathcal{M}_{0,n}$, while Calegari, Dimitrov, and Y. Tang [6] established the $\mathbb{Q}$-linear independence of $1, \zeta(2)$, and $L(2, \chi_{-3})$ using arithmetic holonomicity theorems.

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Recently, significant progress has been made through determinantal and auxiliary-polynomial methods that bypass Nesterenko's lower-bound loss when passing from a single family of linear forms to a square determinant. Fischler [9] combined Siegel's lemma with auxiliary polynomial multipliers in the Ball–Rivoal construction to improve the asymptotic dimension bound (1.3) to $0.21\sqrt{n}/\log n$ for large n. Tang [22] formalized a proof of the $\mathbb{Q}$-linear independence of $1, \zeta(2), \zeta(3)$, and Lai [15] proved the $\mathbb{Q}$-linear independence of $1, \zeta(3), \zeta(4)$ by constructing a $2N \times 2N$ determinant from a rational function $R_n(t)$ with double poles at $\{0, -1, \dots, -An\}$, triple zeros at $\{1, \dots, Bn\}$, and polynomial multipliers $t^i(t - An)^j$, combining reversed-Laplace saddle-point asymptotics with a p-adic valuation optimization over residue classes modulo p. Concurrently, Fauzan [8] studied the individual irrationality of $\zeta(5)$ via Hankel determinants of shifted negative-square rational kernels.

1.2. **Obstructions to multi-row elimination in the Ball–Rivoal method.** Establishing the unconditional $\mathbb{Q}$-linear independence of the full odd-zeta hierarchy $1, \zeta(3), \zeta(5), \dots, \zeta(2n+1)$ for every integer $n \geq 2$ encounters structural obstacles that do not arise for $\{1, \zeta(3)\}$ or $\{1, \zeta(3), \zeta(4)\}$. In Lai's proof for $\{1, \zeta(3), \zeta(4)\}$ [15], the target zeta values have consecutive weights 3 and 4 of opposite parity, so a kernel with double poles ($L = 2$) evaluated under a second-order differential operator produces only two zeta coefficients, one of which is eliminated across N rows of a $2N \times 2N$ matrix. By contrast, to reach $\zeta(2n+1)$ ($n \geq 2$) while eliminating all even zeta values $\zeta(2), \zeta(4), \dots, \zeta(2n)$ by reflection symmetry $z \mapsto -2AK - z$, the kernel must carry pole multiplicity $L_{\text{in}} = 2n + 1 \geq 5$ so that partial fraction decomposition and summation over the positive integers $\nu \geq 1$ produce a linear form

$$\mathcal{A}(P) = \sum_{j=1}^{n} S_{2j+1}(P)\zeta(2j+1) + \mathcal{R}_0(P) \tag{1.4}$$

in $1, \zeta(3), \zeta(5), \dots, \zeta(2n+1)$. To test a hypothetical nontrivial rational linear relation $\sum_{j=1}^{n} a_{2j+1}\zeta(2j+1) + c = 0$ ($a_{2n+1} \neq 0$) via a square $(N+R) \times (N+R)$ determinant, one must simultaneously eliminate the $n-1$ secondary odd zeta combinations

$$L^{(2j+1)}(P) := a_{2n+1}S_{2j+1}(P) - a_{2j+1}S_{2n+1}(P) \qquad (1 \leq j \leq n-1) \tag{1.5}$$

across R top rows while retaining N bottom series rows governed by $a_{2n+1}\mathcal{R}_0(P) - cS_{2n+1}(P)$. When one attempts this multi-row elimination using a uniform pole multiplicity $L = 2n + 1 \geq 5$ on $\{-2AK, \dots, 0\}$, three structural obstructions arise:

(1) *The Top-Block Rank and Tail-Prime Obstruction:* If all $2AK + 1$ poles $z \in \{-2AK, \dots, 0\}$ carry full multiplicity $L = 2n + 1 \geq 5$, each odd-zeta functional $S_{2j+1}(P)$ $(1 \leq j \leq n-1)$ is supported across the entire pole interval of half-width AK. The $n-1$ elimination operators $L^{(2j+1)}$ then have combined rank $R \asymp 2(n-1)AK$. Whenever the maximum column polynomial degree $M_{\max} = N + R - 1$ exceeds AK, primes in the tail interval $p \in (2AK, 2(N+R)K]$ divide the central factorial denominators $(2m)!$ of the polynomial basis without any compensating factorial factor from the kernel prefactor $((2AK)!)^L$, producing p-adic denominators that exceed the factorial prefactor.

(2) *The Leibniz Jet-Inclusion Obstruction:* At any pole $z = -k$ of multiplicity $L_k = 2n+1 \geq 5$, the highest-weight functional $A_{2n+1,k}(P)$ depends only on the value $P(r^2)$ at $r = |AK - k|$, whereas each lower odd-weight functional $A_{2j+1,k}(P)$ $(1 \leq j \leq n-1)$ depends via Leibniz's rule on the $2(n-j)$-jet $(P(r^2), P'(r^2), \dots, P^{(2n-2j)}(r^2))$ at $u = r^2$ (and up to order $n-j$ at $r = 0$). Because the 0-jet governing S_{2n+1} is a sub-jet of the higher-order jets governing $S_3, S_5, \dots, S_{2n-1}$, if the row multiplier polynomials $A_i(u)$ are allowed to have degree exceeding $2A_0K$, then polynomial multipliers $A(u)$ that vanish to high order at the inner pole nodes r^2 $(0 \leq r \leq A_0K)$ cause the row space of S_{2n+1} to fall inside the row space of the top elimination block L_{top}, eliminating the constant $c = -\sum_{j=1}^{n} a_{2j+1}\zeta(2j+1)$ from the Schur complement.

(3) *The Sub-Rank S_{2n+1} Free-Contour-Row Obstruction:* In an $(N+R) \times (N+R)$ Schur-complement matrix $M_K(\mathbf{a}, c)$, let R_{2n+1} denote the quotient rank $\operatorname{rank}(S_{2n+1,\text{bot}} \bmod \operatorname{row}(L_{\text{top}}))$. Whenever the number of bottom rows N exceeds R_{2n+1}, there exist $N_0 := N - R_{2n+1} > 0$ linearly independent combinations of the bottom rows on which S_{2n+1} (and

hence every odd zeta value $\zeta(3),\dots,\zeta(2n+1))$ is eliminated independently of the assumed relation. Those N_0 rows together with L_{top} form an $(R+N_0)\times(N+R)$ purely rational submatrix whose maximal minors are unconditional rational moment determinants that would be forced to vanish by Liouville's inequality if their net exponent were negative.

1.3. **Statement of the principal theorem and corollaries.** The purpose of this paper is to resolve all three obstructions simultaneously through a unified algebraic, combinatorial, and analytic construction: the windowed multi-pole kernel $\mathcal{F}_{2n+1,2}^{\mathrm{odd},[A_0,A]}(z;P)$ coupled with degree-cutoff jet separation on the central factorial polynomial basis $\mathcal{E}_m(u)$. This establishes our principal theorem.

Theorem 1.1 (Rational Linear Independence of Odd Zeta Values)**.** *For every positive integer $n\geq 1$, the $n+1$ numbers*

$$1,\ \zeta(3),\ \zeta(5),\ \zeta(7),\ \dots,\ \zeta(2n+1) \tag{1.6}$$

are linearly independent over $\mathbb{Q}$. Equivalently, for every integer $n\geq 1$ and any rational numbers $q_0,q_1,\dots,q_n\in\mathbb{Q}$,

$$q_0+q_1\zeta(3)+q_2\zeta(5)+\cdots+q_n\zeta(2n+1)=0 \implies q_0=q_1=\cdots=q_n=0, \tag{1.7}$$

so that

$$\dim_{\mathbb{Q}}\operatorname{span}_{\mathbb{Q}}\{1,\zeta(3),\zeta(5),\dots,\zeta(2n+1)\}=n+1. \tag{1.8}$$

Theorem 1.1 immediately implies the rational linear independence of the entire infinite sequence of odd zeta values, as well as the linear independence of every finite subcollection.

Corollary 1.2 (Linear Independence of the Infinite Sequence and Finite Subsets of Odd Zeta Values)**.** *The infinite set of real numbers*

$$\{1,\zeta(3),\zeta(5),\zeta(7),\zeta(9),\dots,\zeta(2n+1),\dots\} \tag{1.9}$$

is linearly independent over $\mathbb{Q}$. In particular:

(i) *Every odd zeta value $\zeta(2n+1)$ $(n\geq 1)$ is irrational, and for any distinct positive integers $n_1\neq n_2$, the ratio $\zeta(2n_1+1)/\zeta(2n_2+1)$ is irrational.*

(ii) *In the case $n=2$, the three numbers $1,\zeta(3),\zeta(5)$ are linearly independent over $\mathbb{Q}$, so $\dim_{\mathbb{Q}}\operatorname{span}_{\mathbb{Q}}\{1,\zeta(3),\zeta(5)\}=3$, $\zeta(5)\notin\mathbb{Q}$, and $\zeta(5)/\zeta(3)\notin\mathbb{Q}$.*

(iii) *For any integer $r\geq 1$ and any strictly increasing sequence of odd integers $3\leq 2n_1+1<2n_2+1<\cdots<2n_r+1$, the $(r+1)$-tuple $\{1,\zeta(2n_1+1),\zeta(2n_2+1),\dots,\zeta(2n_r+1)\}$ is linearly independent over $\mathbb{Q}$ and spans a $\mathbb{Q}$-vector space of dimension $r+1$.*

Proof. By definition, a linear relation over $\mathbb{Q}$ among elements of the infinite set (1.9) involves only finitely many non-zero rational coefficients, and hence lies inside $\operatorname{span}_{\mathbb{Q}}\{1,\zeta(3),\dots,\zeta(2n+1)\}$ for $n:=\max_i n_i$. By Theorem 1.1, every coefficient in such a relation vanishes, which establishes the $\mathbb{Q}$-linear independence of (1.9) and of every finite subset in (iii). Specializing (iii) to $r=1$ gives $\zeta(2n_1+1)\notin\mathbb{Q}$, specializing to $r=2$ gives $\zeta(2n_1+1)/\zeta(2n_2+1)\notin\mathbb{Q}$ in (i), and specializing Theorem 1.1 to $n=2$ yields (ii). □

Combining Theorem 1.1 and Corollary 1.2 with Euler's formula (1.2) and Lindemann's theorem [17] also yields unconditional dimension bounds for vector spaces spanned by mixed collections of even and odd zeta values.

Corollary 1.3 (Dimension and Codimension Bounds for Mixed Even–Odd Zeta Spaces)**.** *Let $r\geq 1$ and $m\geq 1$ be positive integers, let $3\leq 2n_1+1<2n_2+1<\cdots<2n_r+1$ be r distinct odd integers, and let $2\leq 2k_1<2k_2<\cdots<2k_m$ be m distinct even integers. Then:*

(i) *The mixed $\mathbb{Q}$-vector space spanned by 1, the r odd zeta values, and the m even zeta values satisfies*

$$\dim_{\mathbb{Q}}\operatorname{span}_{\mathbb{Q}}\{1,\zeta(2n_1+1),\dots,\zeta(2n_r+1),\zeta(2k_1),\dots,\zeta(2k_m)\}\geq\max(r+1,m+1). \tag{1.10}$$

More generally, for any subspace $W\subset V_{\mathrm{even}}:=\operatorname{span}_{\mathbb{Q}}\{1,\zeta(2k_1),\dots,\zeta(2k_m)\}$ satisfying $W\cap V_{\mathrm{odd}}=\mathbb{Q}\cdot 1$ (where $V_{\mathrm{odd}}:=\operatorname{span}_{\mathbb{Q}}\{1,\zeta(2n_1+1),\dots,\zeta(2n_r+1)\}$), we have $\dim_{\mathbb{Q}}(V_{\mathrm{odd}}+W)=r+\dim_{\mathbb{Q}}W$.

(ii) *Among any $r+1$ distinct even integers $2 \le 2k_1 < \cdots < 2k_{r+1}$, at least one even zeta value $\zeta(2k_j)$ $(1 \le j \le r+1)$ satisfies $\zeta(2k_j) \notin V_{\mathrm{odd}}$, so that the $(r+2)$-tuple $\{1, \zeta(2n_1+1), \dots, \zeta(2n_r+1), \zeta(2k_j)\}$ is linearly independent over $\mathbb{Q}$.*

(iii) *Among any $m+1$ distinct odd integers $3 \le 2n_1+1 < \cdots < 2n_{m+1}+1$, at least one odd zeta value $\zeta(2n_i+1)$ $(1 \le i \le m+1)$ satisfies $\zeta(2n_i+1) \notin V_{\mathrm{even}}$, so that the $(m+2)$-tuple $\{1, \zeta(2k_1), \dots, \zeta(2k_m), \zeta(2n_i+1)\}$ is linearly independent over $\mathbb{Q}$.*

Proof. By Euler's identity (1.2), for each $k \ge 1$ we have $\zeta(2k) = \rho_k \pi^{2k}$ with rational prefactor $\rho_k = (-1)^{k-1} B_{2k} 2^{2k-1}/(2k)! \in \mathbb{Q}^\times$. By Lindemann's theorem [17], π is transcendental over $\mathbb{Q}$, so no non-zero polynomial in π^2 with rational coefficients can vanish; hence for any strictly increasing sequence of even integers $2 \le 2k_1 < \cdots < 2k_t$, the numbers $1, \zeta(2k_1), \dots, \zeta(2k_t)$ are linearly independent over $\mathbb{Q}$. In particular, $\dim_{\mathbb{Q}} V_{\mathrm{even}} = m+1$. By Corollary 1.2(iii), the numbers $1, \zeta(2n_1+1), \dots, \zeta(2n_r+1)$ are linearly independent over $\mathbb{Q}$, so $\dim_{\mathbb{Q}} V_{\mathrm{odd}} = r+1$. Since the sum space $V_{\mathrm{odd}} + V_{\mathrm{even}}$ contains both V_{odd} and V_{even} as $\mathbb{Q}$-subspaces, we have $\dim_{\mathbb{Q}}(V_{\mathrm{odd}} + V_{\mathrm{even}}) \ge \max(\dim_{\mathbb{Q}} V_{\mathrm{odd}}, \dim_{\mathbb{Q}} V_{\mathrm{even}}) = \max(r+1, m+1)$, and if $W \subset V_{\mathrm{even}}$ satisfies $W \cap V_{\mathrm{odd}} = \mathbb{Q} \cdot 1$, the Grassmannian dimension formula gives $\dim_{\mathbb{Q}}(V_{\mathrm{odd}}+W) = \dim_{\mathbb{Q}} V_{\mathrm{odd}} + \dim_{\mathbb{Q}} W - \dim_{\mathbb{Q}}(V_{\mathrm{odd}} \cap W) = (r+1) + \dim_{\mathbb{Q}} W - 1 = r + \dim_{\mathbb{Q}} W$, which proves (i).

To prove (ii), suppose for contradiction that $\zeta(2k_j) \in V_{\mathrm{odd}}$ for every $j \in \{1, 2, \dots, r+1\}$. Since $\{1, \zeta(2n_1+1), \dots, \zeta(2n_r+1)\}$ is a basis of V_{odd}, each $\zeta(2k_j)$ can be expanded uniquely as

$$\zeta(2k_j) = c_{j,0} \cdot 1 + \sum_{i=1}^{r} c_{j,i} \zeta(2n_i+1) \qquad (c_{j,0}, c_{j,1}, \dots, c_{j,r} \in \mathbb{Q},\ 1 \le j \le r+1). \tag{1.11}$$

Equivalently, in the quotient $\mathbb{Q}$-vector space $V_{\mathrm{odd}}/(\mathbb{Q} \cdot 1)$ (which has dimension $\dim_{\mathbb{Q}} V_{\mathrm{odd}} - 1 = r$ with basis cosets $[\zeta(2n_1+1)], \dots, [\zeta(2n_r+1)]$), each coset $[\zeta(2k_j)] \in V_{\mathrm{odd}}/(\mathbb{Q} \cdot 1)$ $(1 \le j \le r+1)$ has coordinate vector $\mathbf{v}_j := (c_{j,1}, c_{j,2}, \dots, c_{j,r}) \in \mathbb{Q}^r$. Since $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{r+1}$ are $r+1$ vectors in the r-dimensional $\mathbb{Q}$-vector space $\mathbb{Q}^r \cong V_{\mathrm{odd}}/(\mathbb{Q} \cdot 1)$, they are linearly dependent over $\mathbb{Q}$: there exist rational numbers $\lambda_1, \lambda_2, \dots, \lambda_{r+1} \in \mathbb{Q}$, not all zero, such that

$$\sum_{j=1}^{r+1} \lambda_j \mathbf{v}_j = \mathbf{0} \quad \text{in } \mathbb{Q}^r, \qquad \text{that is,} \qquad \sum_{j=1}^{r+1} \lambda_j c_{j,i} = 0 \quad \text{for every } 1 \le i \le r. \tag{1.12}$$

Multiplying (1.11) by λ_j, summing over $1 \le j \le r+1$, and substituting (1.12) gives

$$\sum_{j=1}^{r+1} \lambda_j \zeta(2k_j) = \left(\sum_{j=1}^{r+1} \lambda_j c_{j,0} \right) \cdot 1 + \sum_{i=1}^{r} \left(\sum_{j=1}^{r+1} \lambda_j c_{j,i} \right) \zeta(2n_i+1) = \lambda_0 \cdot 1, \tag{1.13}$$

where $\lambda_0 := \sum_{j=1}^{r+1} \lambda_j c_{j,0} \in \mathbb{Q}$. Rearranging (1.13) yields the rational linear relation $-\lambda_0 \cdot 1 + \sum_{j=1}^{r+1} \lambda_j \zeta(2k_j) = 0$ with $(\lambda_1, \dots, \lambda_{r+1}) \ne (0, \dots, 0)$, which contradicts the $\mathbb{Q}$-linear independence of $1, \zeta(2k_1), \dots, \zeta(2k_{r+1})$. Therefore, there exists at least one index $j^* \in \{1, \dots, r+1\}$ such that $\zeta(2k_{j^*}) \notin V_{\mathrm{odd}}$. Adjoining $\zeta(2k_{j^*})$ to the basis $\{1, \zeta(2n_1+1), \dots, \zeta(2n_r+1)\}$ of V_{odd} proves (ii).

To prove (iii), suppose for contradiction that $\zeta(2n_i+1) \in V_{\mathrm{even}}$ for every $i \in \{1, 2, \dots, m+1\}$, so that $\zeta(2n_i+1) = d_{i,0} \cdot 1 + \sum_{j=1}^{m} d_{i,j} \zeta(2k_j)$ with $d_{i,j} \in \mathbb{Q}$. The $m+1$ vectors $\mathbf{w}_i := (d_{i,1}, \dots, d_{i,m}) \in \mathbb{Q}^m$ $(1 \le i \le m+1)$ lie in $\mathbb{Q}^m$ and hence satisfy a nontrivial relation $\sum_{i=1}^{m+1} \mu_i \mathbf{w}_i = \mathbf{0}$ with $\mu_1, \dots, \mu_{m+1} \in \mathbb{Q}$ not all zero. Summing $\mu_i \zeta(2n_i+1)$ over $1 \le i \le m+1$ eliminates all even zeta values $\zeta(2k_1), \dots, \zeta(2k_m)$ and yields $-\mu_0 \cdot 1 + \sum_{i=1}^{m+1} \mu_i \zeta(2n_i+1) = 0$ $(\mu_0 := \sum_{i=1}^{m+1} \mu_i d_{i,0} \in \mathbb{Q})$, which contradicts the $\mathbb{Q}$-linear independence of $1, \zeta(2n_1+1), \dots, \zeta(2n_{m+1}+1)$ from Corollary 1.2(iii). Thus at least one index $i^* \in \{1, \dots, m+1\}$ satisfies $\zeta(2n_{i^*}+1) \notin V_{\mathrm{even}}$, and adjoining $\zeta(2n_{i^*}+1)$ to the basis $\{1, \zeta(2k_1), \dots, \zeta(2k_m)\}$ of V_{even} proves (iii). □

1.4. **Overview of the proof and organization of the paper.** We now summarize the logical structure of the proof and the organization of the paper:

(1) *Central Factorial Polynomials and Total Positivity (Section 2):* We work throughout with the reflection-invariant variable $u := (z + AK)^2$, which is unchanged under $z \mapsto -2AK - z$. In Section 2, we study the normalized central factorial polynomials $\mathcal{E}_m(u) := \frac{2}{(2m)!} \prod_{t=0}^{m-1} (u - t^2)$. We prove that $\mathcal{E}_m(r^2) = \binom{r+m}{2m} + \binom{r+m-1}{2m} \in \mathbb{Z}$ vanishes for all $0 \le r < m$

and equals 1 at $r = m$, and establish $\mathrm{lcm}(1,\dots,2m)^{2q}\frac{1}{q!}\mathcal{E}_m^{(q)}(r^2) \in \mathbb{Z}$ (Proposition 2.2), derive the first-derivative identity $\mathcal{E}'_m(r^2) = \frac{(1+\delta_{r,0})(-1)^{m-1-r}}{2m(2m-1)\binom{2m-2}{m-1+r}}$, the second-derivative formula with rational harmonic telescoping $T(m,r)$, and the arbitrary-order derivative formula and degree-$2m$ linear-w factorization (Propositions 2.3 and 2.4, Theorem 2.5), and establish strict total positivity of $(\mathcal{E}_{r_j}(x_i^2))_{1\le i,j\le M}$ on $x_1 > r_M$ via an explicit non-intersecting lattice-path construction and the Lindström–Gessel–Viennot theorem (Theorem 2.6).

(2) *Windowed Multi-Pole Kernel and Support Compression (Section 3):* In Section 3, we introduce the windowed rational kernel $\mathcal{F}_{2n+1,2}^{\mathrm{odd},[A_0,A]}(z;P)$ on $\{-2AK,\dots,0\}$, which carries odd multiplicity $L_{\mathrm{in}} = 2n+1$ on the inner window $|k-AK| \le A_0K$ ($A_0 \ll A$) and multiplicity $L_{\mathrm{out}} = 2$ on the outer shoulders $A_0K < |k-AK| \le AK$. Odd reflection symmetry $\mathcal{F}(-2AK-z;P) = -\mathcal{F}(z;P)$ causes all even-weight pole sums $S_{2j}(P) = 0$ and the residue sum $S_1(P) = 0$ to vanish, while the outer multiplicity $L_{\mathrm{out}} = 2 < 3$ forces all odd zeta functionals $S_3(P), S_5(P), \dots, S_{2n+1}(P)$ to vanish identically at every outer pole $A_0K < |k-AK| \le AK$ (Theorem 3.3). Consequently, all $n-1$ top elimination operators are supported exclusively on the inner window of half-width A_0K, compressing their combined rank to $R = (n-1)(2A_0K+1)$ while allowing the outer barrier $A = 2.5(C+R_c)$ to absorb all central factorial denominators and eliminate all tail primes $p > 2AK$. We also establish the properties of the reflection-symmetric rising-factorial zero barrier $\Omega_B(u)^{M_0}$ (Theorem 3.5).

(3) *Degree-Cutoff Jet Separation and Confluent Vandermonde Determinant (Section 4):* In Section 4, we analyze the local Leibniz jet expansion of $S_{2j+1}(P)$ at the inner nodes $0 \le r \le A_0K$ (Proposition 4.1) and construct the explicit Hermite interpolation basis on $\mathbb{Q}[u]_{\le 2A_0K}$ (Lemma 4.2). On the R tail column indices $J_{\mathrm{tail}} = \{N,\dots,N+R-1\}$ ($N = A_0K+1$), every column basis polynomial $\mathcal{E}_m(u)$ is divisible by $W_0(u) = \prod_{r=0}^{A_0K}(u-r^2)$, so $S_{2n+1}(\mathcal{E}_i\mathcal{E}_m) = 0$ identically on all tail columns $m \in J_{\mathrm{tail}}$. Restricting the row multipliers of each of the $n-1$ top blocks to $\mathcal{E}_0(u),\dots,\mathcal{E}_{2A_0K}(u)$ of dimension $2A_0K+1$ and the bottom block to $\mathcal{E}_0(u),\dots,\mathcal{E}_{A_0K}(u)$ of dimension $N = A_0K+1$, we prove by backward Hermite induction on $m = n-j \in \{1,\dots,n-1\}$ that the stacked $(N+R)\times(N+R)$ matrix $\binom{L_{\mathrm{top}}}{S_{2n+1,\mathrm{bot}}}$ is block-triangular on $J_{\mathrm{head}} \sqcup J_{\mathrm{tail}}$ and invertible over $\mathbb{Q}$ for every $n \ge 2$, ensuring $N_0 = 0$ identically (Theorem 4.3), and we derive the exact closed-form confluent Vandermonde determinant formula for $\det(S_{\mathrm{odd,top}}[:,J_{\mathrm{tail}}])$ (Theorem 4.4).

(4) *Block-Triangular Schur Reduction, p-Adic Valuation Bounds, and Non-Vanishing (Sections 5–7):* In Section 5, we establish the Schur-complement moment identity $\Delta_{N+R} = \det\binom{L_{\mathrm{top}}}{a_{2n+1}\mathcal{A}_{\mathrm{bot}}}$ (Proposition 5.2), the finite-K block-triangular column reduction and homogeneous scalar cancellation in $\Gamma = (L_{\mathrm{top}}[:,J_{\mathrm{tail}}])^{-1}L_{\mathrm{top}}[:,J_{\mathrm{head}}]$ together with osculatory Hermite decoupling (Theorem 5.3), and the Cauchy–Binet series expansion and archimedean potential bound (Theorem 5.4). In Section 6, we prove the entry-wise p-adic valuation bound (Lemma 6.1), the inner-window p-adic integrality lemma showing that for all primes $p > 2A_0K$ the inner-window factor $U_{\mathrm{in},k}(w)$ has p-integral Taylor coefficients of all orders (Proposition 6.2), midpoint harmonic centering (Theorem 6.3), outer boundary divisibility $(r-A_0K)\binom{r+A_0K}{2A_0K} \mid \mathrm{lcm}(1,\dots,(A+A_0)K)$ and finite cancellation of the central factorial prefactor (Proposition 6.4), and the strict negativity of the net exponent across all $n \ge 1$ (Theorem 6.5). In Section 7, we prove the non-vanishing of Δ_{N+R} for all sufficiently large $K \ge K_0(n,\mathbf{a})$ via the Euler sine-square eigenvalue law $\widetilde{\Lambda}_{\mathrm{Euler}}(\mathbf{a}) \ne 0$ and block-triangular Schur reduction (Theorem 7.1), and complete the inductive proof of Theorem 1.1. Section 8 presents the computational verification suite `reproducibility2_final.py` and numerical tables from $n = 1$ to $n = 10000$.

2. Central Factorial Polynomials and Total Positivity

Throughout this paper, let $K \in \mathbb{Z}_{\geq 1}$ denote the asymptotic scaling integer, and let $A > 0$ be a rational scaling parameter such that $AK \in \mathbb{Z}_{\geq 1}$. We work with the reflection-invariant variable

$$u := (z + AK)^2, \tag{2.1}$$

which is unchanged under the reflection involution $z \mapsto -2AK - z$ across the midpoint $z = -AK$ of the pole interval $\{-2AK, -2AK + 1, \ldots, 0\}$.

Definition 2.1 (Normalized Central Factorial Polynomial Basis)**.** For each integer $m \geq 0$, define the normalized central factorial polynomial $\mathcal{E}_m(u) \in \mathbb{Q}[u]$ of degree m in u by

$$\mathcal{E}_0(u) := 1, \qquad \mathcal{E}_m(u) := \frac{2}{(2m)!} \prod_{t=0}^{m-1} (u - t^2) \quad (m \geq 1). \tag{2.2}$$

The normalization factor $\frac{2}{(2m)!}$ in (2.2) ensures that $\mathcal{E}_m(u)$ takes integer values at every integer square $u = r^2$ ($r \in \mathbb{Z}_{\geq 0}$) and equals 1 at $r = m$.

Proposition 2.2 (Integer-Square Integrality and Binomial Representation)**.** *For every integer $r \geq 0$ and every integer $m \geq 1$, the evaluation of $\mathcal{E}_m(u)$ at $u = r^2$ satisfies the exact binomial identity*

$$\mathcal{E}_m(r^2) = \binom{r+m}{2m} + \binom{r+m-1}{2m} \in \mathbb{Z}. \tag{2.3}$$

In particular:

(i) *$\mathcal{E}_m(r^2) = 0$ for all integers $0 \leq r < m$, and $\mathcal{E}_m(m^2) = 1$ for all $m \geq 0$;*

(ii) *For every integer $q \geq 0$ and every $r \in \mathbb{Z}$, the q-th Taylor coefficient of $\mathcal{E}_m(u)$ at $u = r^2$ satisfies*

$$\operatorname{lcm}(1, 2, \ldots, 2m)^{2q} \cdot \frac{1}{q!} \mathcal{E}_m^{(q)}(r^2) \in \mathbb{Z}. \tag{2.4}$$

Proof. Fix $m \geq 1$ and $r \geq 0$. When $0 \leq r < m$, the factor with index $t = r$ in the product $\prod_{t=0}^{m-1}(r^2 - t^2)$ is $r^2 - r^2 = 0$, so $\mathcal{E}_m(r^2) = 0$. Simultaneously, for $0 \leq r < m$ we have $0 \leq r + m - 1 < r + m < 2m$, so both binomial coefficients $\binom{r+m}{2m}$ and $\binom{r+m-1}{2m}$ vanish, confirming (2.3) for $0 \leq r < m$.

Now suppose $r \geq m \geq 1$. Separating the $t = 0$ factor $r^2 - 0^2 = r^2$ and factoring each remaining term $r^2 - t^2 = (r - t)(r + t)$ ($1 \leq t \leq m - 1$), where $\prod_{t=1}^{m-1}(r - t) = \frac{(r-1)!}{(r-m)!}$ and $\prod_{t=1}^{m-1}(r + t) = \frac{(r+m-1)!}{r!}$, gives

$$\begin{aligned} \prod_{t=0}^{m-1} (r^2 - t^2) &= r^2 \left(\prod_{t=1}^{m-1} (r - t) \right) \left(\prod_{t=1}^{m-1} (r + t) \right) = r^2 \cdot \frac{(r-1)!}{(r-m)!} \cdot \frac{(r+m-1)!}{r!} \\ &= r \cdot \frac{(r+m-1)!}{(r-m)!} = \frac{r}{r+m} \cdot \frac{(r+m)!}{(r-m)!}. \end{aligned}$$

Multiplying by $\frac{2}{(2m)!}$ and using the identity $2r = (r + m) + (r - m)$ yields

$$\mathcal{E}_m(r^2) = \frac{2r}{r+m} \cdot \frac{(r+m)!}{(2m)!(r-m)!} = \frac{(r+m)+(r-m)}{r+m} \binom{r+m}{2m} = \binom{r+m}{2m} + \binom{r+m-1}{2m} \in \mathbb{Z},$$

which establishes (2.3). Setting $r = m \geq 1$ in (2.3) gives $\mathcal{E}_m(m^2) = \binom{2m}{2m} + \binom{2m-1}{2m} = 1 + 0 = 1$, and $\mathcal{E}_0(0) = 1$ by Definition 2.1, proving (i).

To prove (ii), for $q = 0$ we have $\frac{1}{0!}\mathcal{E}_m^{(0)}(r^2) = \mathcal{E}_m(r^2) \in \mathbb{Z}$ by (2.3), while for $q > m$ or $m = 0$ ($q \geq 1$), $\mathcal{E}_m(u)$ has degree $m < q$ in u, so $\frac{1}{q!}\mathcal{E}_m^{(q)}(r^2) = 0 \in \mathbb{Z}$. Now fix integers $m \geq 1$, $1 \leq q \leq m$, and $r \in \mathbb{Z}$. Since $r^2 = |r|^2$, we may replace r by $|r|$ and assume $r \geq 0$. Set $L_{2m} := \operatorname{lcm}(1, 2, \ldots, 2m)$, so that for every prime p, $e_p := v_p(L_{2m}) = \lfloor \frac{\log(2m)}{\log p} \rfloor \in \mathbb{Z}_{\geq 0}$. To prove $L_{2m}^{2q} \frac{1}{q!}\mathcal{E}_m^{(q)}(r^2) \in \mathbb{Z}$, it suffices to show that $v_p(\frac{1}{q!}\mathcal{E}_m^{(q)}(r^2)) \geq -2qe_p$ for every prime p.

If $p > 2m$, then $e_p = 0$ and $v_p((2m)!) = 0$, so $\mathcal{E}_m(u) = \frac{2}{(2m)!}\prod_{t=0}^{m-1}(u-t^2) \in \mathbb{Z}_{(p)}[u]$ has p-integral coefficients, whence $\frac{1}{q!}\mathcal{E}_m^{(q)}(r^2) \in \mathbb{Z}_{(p)}$ and $v_p(\frac{1}{q!}\mathcal{E}_m^{(q)}(r^2)) \geq 0 = -2qe_p$.

Now suppose $p \leq 2m$, so $e_p \geq 1$. Differentiating $\mathcal{E}_m(u) = \frac{2}{(2m)!}\prod_{t=0}^{m-1}(u-t^2)$ q times at $u = r^2$ by the product rule gives

$$\frac{1}{q!}\mathcal{E}_m^{(q)}(r^2) = \sum_{\substack{T\subset\{0,1,\ldots,m-1\}\\|T|=q}} \Pi_T(r), \qquad \Pi_T(r) := \frac{2}{(2m)!}\prod_{\substack{0\leq t\leq m-1\\ t\notin T}}(r-t)(r+t). \tag{2.5}$$

By the non-archimedean ultrametric inequality, it suffices to prove that $v_p(\Pi_T(r)) \geq -2qe_p$ for every q-element subset $T \subset \{0,1,\ldots,m-1\}$. Fix such a subset T. If $\Pi_T(r) = 0$, then $v_p(\Pi_T(r)) = +\infty \geq -2qe_p$. Assume now that $\Pi_T(r) \neq 0$, so $r - t \neq 0$ and $r + t \neq 0$ for all $t \in \{0,\ldots,m-1\} \setminus T$. Define the $2m$-element multiset of integers

$$\mathcal{M}_{m,r} := \bigsqcup_{t=0}^{m-1}\{r-t, r+t\} = \{r-m+1, r-m+2, \ldots, r+m-1\} \sqcup \{r\},$$

where r appears with multiplicity 2 (from $t = 0$) and each $r \pm t$ ($1 \leq t \leq m-1$) appears with multiplicity 1. The product $\prod_{t\notin T}(r-t)(r+t)$ is the product of the $2m-2q$ non-zero integers remaining when the $2q$ integers $\mathcal{D}_T := \bigsqcup_{t\in T}\{r-t, r+t\}$ are deleted from $\mathcal{M}_{m,r}$. Compare $\mathcal{M}_{m,r}$ with the block of $2m$ consecutive integers

$$\mathcal{I}_+ := \{r-m+1, r-m+2, \ldots, r+m-1, r+m\} = (\mathcal{M}_{m,r} \setminus \{r\}) \sqcup \{r+m\}.$$

For each integer $s \geq 1$, let $N_s(\mathcal{M}_{m,r} \setminus \mathcal{D}_T)$ denote the number of elements in the $(2m-2q)$-element multiset $\mathcal{M}_{m,r} \setminus \mathcal{D}_T$ that are divisible by p^s. Since every non-zero integer $a \in \mathcal{M}_{m,r} \setminus \mathcal{D}_T$ contributes $\sum_{s\geq 1}\mathbf{1}_{p^s|a} = v_p(a)$ to v_p, while Legendre's formula gives $v_p((2m)!) = \sum_{s=1}^{e_p}\lfloor 2m/p^s\rfloor$, we have

$$v_p(\Pi_T(r)) \geq v_p(2) + \sum_{s=1}^{e_p} N_s(\mathcal{M}_{m,r} \setminus \mathcal{D}_T) - \sum_{s=1}^{e_p}\left\lfloor \frac{2m}{p^s}\right\rfloor. \tag{2.6}$$

For each level $1 \leq s \leq e_p$, we compare $N_s(\mathcal{M}_{m,r})$ with $\lfloor 2m/p^s\rfloor$:

(1) Since $\mathcal{I}_+$ consists of $2m$ consecutive integers, it contains at least $\lfloor 2m/p^s\rfloor$ multiples of p^s; moreover, if $p^s \nmid 2m$ and the upper endpoint $r+m$ is divisible by p^s, write $2m = q_s p^s + r_s$ with $q_s = \lfloor 2m/p^s\rfloor$ and $1 \leq r_s < p^s$, so the multiples of p^s in $\mathcal{I}_+ = \{r+m-2m+1, \ldots, r+m\}$ are $(r+m) - jp^s$ for $0 \leq j \leq q_s$, giving $N_s(\mathcal{I}_+) = q_s + 1 = \lfloor 2m/p^s\rfloor + 1$.

(2) Consequently, replacing the endpoint $r+m$ of $\mathcal{I}_+$ by r to form $\mathcal{M}_{m,r}$ can yield $N_s(\mathcal{M}_{m,r}) < \lfloor 2m/p^s\rfloor$ only if $N_s(\mathcal{I}_+) = \lfloor 2m/p^s\rfloor$, $p^s \mid (r+m)$, and $p^s \nmid r$. By Step (1), $p^s \mid (r+m)$ and $N_s(\mathcal{I}_+) = \lfloor 2m/p^s\rfloor$ force $p^s \mid 2m$. Since $p^s \mid (r+m)$ and $p^s \mid 2m$, we have $p^s \mid (2(r+m) - 2m) = 2r$.

(3) If $p > 2$, then $p^s \mid 2r$ implies $p^s \mid r$, contradicting $p^s \nmid r$; hence $N_s(\mathcal{M}_{m,r}) \geq \lfloor 2m/p^s\rfloor$ for every $1 \leq s \leq e_p$ when $p > 2$.

(4) If $p = 2$ and $r \geq 1$, then $2^s \mid 2r$ and $2^s \nmid r$ can hold only for the single exponent $s = v_2(r)+1$, where $N_s(\mathcal{M}_{m,r}) = \lfloor 2m/2^s\rfloor - 1$; summing over $1 \leq s \leq e_2$ and adding $v_2(2) = 1$ gives $v_2(2) + \sum_{s=1}^{e_2} N_s(\mathcal{M}_{m,r}) \geq \sum_{s=1}^{e_2}\lfloor 2m/2^s\rfloor = v_2((2m)!)$. When $r = 0$, we must have $0 \in T$ since $\Pi_T(0) \neq 0$, so $T' := T \setminus \{0\} \subset \{1,\ldots,m-1\}$ has cardinality $|T'| = q-1$, and $\mathcal{M}_{m,0} \setminus \{0,0\} = \{\pm 1, \ldots, \pm(m-1)\}$ gives

$$v_p(\Pi_T(0)) = v_p\left(\frac{2((m-1)!)^2}{(2m)!}\right) - \sum_{t\in T'} 2v_p(t) = -v_p\left(\frac{m^2}{2}\binom{2m}{m}\right) - \sum_{t\in T'} 2v_p(t).$$

For every $1 \leq s \leq v_p(m)$, $p^s \mid m$ implies $\lfloor 2m/p^s\rfloor - 2\lfloor m/p^s\rfloor = 0$, while for $v_p(m) < s \leq e_p$ we have $\lfloor 2m/p^s\rfloor - 2\lfloor m/p^s\rfloor \leq 1$. Since $-v_p(2) \leq 0$ for every prime p and $p^{v_p(m)} \leq m < 2m$

implies $v_p(m) \le e_p$, Legendre's formula yields

$$v_p\left(\frac{m^2}{2}\binom{2m}{m}\right) = 2v_p(m) - v_p(2) + \sum_{s=v_p(m)+1}^{e_p}\left(\left\lfloor\frac{2m}{p^s}\right\rfloor - 2\left\lfloor\frac{m}{p^s}\right\rfloor\right)$$
$$\le 2v_p(m) + (e_p - v_p(m)) = e_p + v_p(m) \le 2e_p$$

(and for $p=2$, $2^{v_2(m)+1} \mid 2m \implies v_2(m)+1 \le e_2$ and $-v_2(2) = -1$ give the sharper bound $v_2(\frac{m^2}{2}\binom{2m}{m}) \le e_2 + v_2(m) - 1 \le 2e_2 - 2$). Since $|T'| = q-1$ with $v_p(t) \le e_p$ for each $1 \le t \le m-1$, we obtain $v_p(\Pi_T(0)) \ge -2e_p - 2(q-1)e_p = -2qe_p$.

Finally, for $r \ge 1$, deleting the $2q$ elements of $\mathcal{D}_T$ from $\mathcal{M}_{m,r}$ reduces N_s by at most $2q$ at each of the e_p levels $1 \le s \le e_p$. Substituting $N_s(\mathcal{M}_{m,r} \setminus \mathcal{D}_T) \ge N_s(\mathcal{M}_{m,r}) - 2q$ into (2.6) yields $v_p(\Pi_T(r)) \ge -2qe_p$, and the ultrametric inequality on (2.5) establishes (2.4). □

Inside the root interval $0 \le r < m$, the value $\mathcal{E}_m(r^2)$ vanishes identically, whereas the derivatives $\mathcal{E}_m^{(q)}(r^2)$ $(q \ge 1)$ govern the local Leibniz jet of the odd zeta functionals on the tail columns. We now establish closed-form expressions for the first, second, and general q-th derivatives of $\mathcal{E}_m(u)$ at $u = r^2$ for $0 \le r < m$.

Proposition 2.3 (First Derivatives at Integer Squares $0 \le r < m$)**.** *For every integer $m \ge 1$ and every integer r with $0 \le r < m$, the first derivative of $\mathcal{E}_m(u)$ at $u = r^2$ is given by the unified formula*

$$\mathcal{E}_m'(r^2) = \frac{(1+\delta_{r,0})(-1)^{m-1-r}(m-1+r)!(m-1-r)!}{(2m)!} = \frac{(1+\delta_{r,0})(-1)^{m-1-r}}{2m(2m-1)\binom{2m-2}{m-1+r}}$$
$$= \begin{cases} \dfrac{(-1)^{m-1-r}}{2m(m-r)\binom{2m-1}{m-r}}, & 1 \le r < m, \\ \dfrac{(-1)^{m-1}}{m^2\binom{2m-1}{m-1}}, & r = 0, \end{cases} \tag{2.7}$$

where $\delta_{r,0} = 1$ if $r = 0$ and $\delta_{r,0} = 0$ if $1 \le r < m$. In particular, $\mathcal{E}_m'(r^2) \ne 0$ for all $0 \le r < m$, and its reciprocal is a non-zero integer:

$$\frac{1}{\mathcal{E}_m'(r^2)} = \frac{(-1)^{m-1-r}}{1+\delta_{r,0}}\,2m(2m-1)\binom{2m-2}{m-1+r} \in \mathbb{Z}\setminus\{0\} \qquad (0 \le r < m). \tag{2.8}$$

Proof. Fix $m \ge 1$ and $0 \le r < m$, and write $\mathcal{E}_m(u) = (u - r^2)Q_{m,r}(u)$, where

$$Q_{m,r}(u) := \frac{2}{(2m)!}\prod_{\substack{0\le t<m\\ t\ne r}}(u - t^2).$$

Differentiating $\mathcal{E}_m(u) = (u-r^2)Q_{m,r}(u)$ once with respect to u and evaluating at $u = r^2$ leaves only the term where $u - r^2$ is differentiated:

$$\mathcal{E}_m'(r^2) = Q_{m,r}(r^2) = \frac{2}{(2m)!}\prod_{\substack{0\le t<m\\ t\ne r}}(r^2 - t^2) = \frac{2}{(2m)!}\left(\prod_{\substack{0\le t<m\\ t\ne r}}(r-t)\right)\left(\prod_{\substack{0\le t<m\\ t\ne r}}(r+t)\right). \tag{2.9}$$

We evaluate (2.9) in the two cases $r = 0$ and $1 \le r < m$:

(1) When $r = 0$, the product in (2.9) runs over $1 \le t \le m-1$, where $(0-t)(0+t) = -t^2$. Thus

$$\mathcal{E}_m'(0) = \frac{2}{(2m)!}\prod_{t=1}^{m-1}(-t^2) = \frac{2(-1)^{m-1}((m-1)!)^2}{(2m)!},$$

which matches the first expression in (2.7) since $1 + \delta_{0,0} = 2$. Using the factorial identity $(2m)! = 2m(2m-1)(2m-2)! = 2m(2m-1)\binom{2m-2}{m-1}((m-1)!)^2$ and noting that $(2m -$

$1)\binom{2m-2}{m-1} = m\binom{2m-1}{m-1}$, we obtain both

$$\mathcal{E}'_m(0) = \frac{(-1)^{m-1}}{m(2m-1)\binom{2m-2}{m-1}} = \frac{(-1)^{m-1}}{m^2\binom{2m-1}{m-1}}.$$

(2) When $1 \le r < m$, we split the first product $\prod_{t\neq r}(r-t)$ into the ranges $0 \le t \le r-1$ (where $r-t$ runs from r down to 1) and $r+1 \le t \le m-1$ (where $r-t = -(t-r)$ runs from -1 down to $-(m-1-r)$):

$$\prod_{\substack{0\le t<m\\ t\neq r}} (r-t) = \left(\prod_{t=0}^{r-1}(r-t)\right)\left(\prod_{t=r+1}^{m-1}(r-t)\right) = r!\,(-1)^{m-1-r}(m-1-r)!.$$

For the second product $\prod_{t\neq r}(r+t)$, the full product over $0 \le t \le m-1$ is $\prod_{t=0}^{m-1}(r+t) = \frac{(m-1+r)!}{(r-1)!}$, and dividing out the omitted $t=r$ factor $r+r=2r$ yields

$$\prod_{\substack{0\le t<m\\ t\neq r}} (r+t) = \frac{(m-1+r)!}{2r\,(r-1)!} = \frac{(m-1+r)!}{2\,r!}.$$

Substituting both products into (2.9), the factors $2\,r!$ cancel between numerator and denominator, leaving

$$\mathcal{E}'_m(r^2) = \frac{(-1)^{m-1-r}(m-1+r)!(m-1-r)!}{(2m)!}.$$

Since $(m-1+r)+(m-1-r) = 2m-2$ and $(2m)! = 2m(2m-1)(2m-2)!$, we have

$$\frac{(2m)!}{(m-1+r)!(m-1-r)!} = 2m(2m-1)\binom{2m-2}{m-1+r} = 2m(m-r)\binom{2m-1}{m-r},$$

where the last equality uses $(2m-1)\binom{2m-2}{m-1-r} = (m-r)\binom{2m-1}{m-r}$.

This establishes all equalities in (2.7) and (2.8). □

Proposition 2.4 (Second Derivatives and Harmonic Telescoping)**.** *For every integer $m \ge 1$ and every integer r with $0 \le r < m$, the second derivative of $\mathcal{E}_m(u)$ at $u = r^2$ satisfies*

$$\frac{1}{2}\mathcal{E}''_m(r^2) = \mathcal{E}'_m(r^2)T(m,r), \qquad T(m,r) := \sum_{\substack{0\le t<m\\ t\neq r}} \frac{1}{r^2-t^2}. \tag{2.10}$$

At $r=0$, we have $T(m,0) = -H^{(2)}_{m-1} := -\sum_{t=1}^{m-1}\frac{1}{t^2}$. For every $1 \le r < m$, the sum $T(m,r)$ telescopes into a rational function of m with $2r$ simple poles at $m \in \{-r+1,\dots,r\}$:

$$T(m,r) = \sum_{\substack{0\le t<m\\ t\neq r}} \frac{1}{r^2-t^2} = \frac{1}{4r^2} + \frac{1}{2r}\sum_{u=m-r}^{m+r-1}\frac{1}{u} \qquad (1 \le r < m). \tag{2.11}$$

In particular, for every fixed $r \ge 1$, $T(m,r) = \frac{1}{4r^2} + \frac{1}{m} + O(m^{-2})$ as $m \to \infty$.

Proof. Differentiating $\mathcal{E}_m(u) = (u-r^2)Q_{m,r}(u)$ twice with respect to u gives $\mathcal{E}''_m(u) = 2Q'_{m,r}(u) + (u-r^2)Q''_{m,r}(u)$. Evaluating at $u=r^2$ and applying logarithmic differentiation to $Q_{m,r}(u) = \frac{2}{(2m)!}\prod_{t\neq r}(u-t^2)$ yields

$$\frac{1}{2}\mathcal{E}''_m(r^2) = Q'_{m,r}(r^2) = Q_{m,r}(r^2)\sum_{\substack{0\le t<m\\ t\neq r}}\frac{1}{r^2-t^2} = \mathcal{E}'_m(r^2)T(m,r),$$

proving (2.10). For $r=0$, the sum over $1 \le t \le m-1$ of $(0-t^2)^{-1}$ is $-H^{(2)}_{m-1}$.

For $1 \le r < m$, we separate the $t = 0$ term $\frac{1}{r^2}$ from the terms $1 \le t \le m-1$ ($t \ne r$) and decompose each summand into partial fractions $\frac{1}{r^2-t^2} = \frac{1}{2r}(\frac{1}{r-t} + \frac{1}{r+t})$:

$$T(m,r) = \frac{1}{r^2} + \frac{1}{2r}\sum_{t=1}^{r-1}\frac{1}{r-t} + \frac{1}{2r}\sum_{t=r+1}^{m-1}\frac{1}{r-t} + \frac{1}{2r}\sum_{\substack{1\le t\le m-1\\ t\ne r}}\frac{1}{r+t}. \tag{2.12}$$

Changing the summation index in the three sums of (2.12):

(1) Setting $u = r - t$ in the first sum gives $\sum_{t=1}^{r-1}\frac{1}{r-t} = \sum_{u=1}^{r-1}\frac{1}{u} = H_{r-1}^{(1)}$.
(2) Setting $u = t - r$ in the second sum gives $\sum_{t=r+1}^{m-1}\frac{1}{r-t} = -\sum_{u=1}^{m-1-r}\frac{1}{u} = -H_{m-1-r}^{(1)}$.
(3) Setting $u = r + t$ in the third sum (which covers $r+1 \le u \le m+r-1$ with the term $t = r$, i.e. $u = 2r$, omitted) gives

$$\sum_{\substack{1\le t\le m-1\\ t\ne r}}\frac{1}{r+t} = \sum_{u=r+1}^{m+r-1}\frac{1}{u} - \frac{1}{2r} = H_{m+r-1}^{(1)} - H_r^{(1)} - \frac{1}{2r}.$$

Substituting these three expressions into (2.12) and noting that $H_{r-1}^{(1)} - H_r^{(1)} = -\frac{1}{r}$, we obtain

$$T(m,r) = \frac{1}{r^2} + \frac{1}{2r}\left(-\frac{1}{r} - \frac{1}{2r} + H_{m+r-1}^{(1)} - H_{m-1-r}^{(1)}\right) = \frac{1}{4r^2} + \frac{1}{2r}\sum_{u=m-r}^{m+r-1}\frac{1}{u},$$

which establishes (2.11). Finally, for fixed $r \ge 1$ as $m \to \infty$, expanding each summand $\frac{1}{m+j} = \frac{1}{m} - \frac{j}{m^2} + O(m^{-3})$ for $-r \le j \le r-1$ gives

$$\frac{1}{2r}\sum_{u=m-r}^{m+r-1}\frac{1}{u} = \frac{1}{2r}\sum_{j=-r}^{r-1}\left(\frac{1}{m} - \frac{j}{m^2} + O(m^{-3})\right) = \frac{1}{m} + \frac{1}{2m^2} + O(m^{-3}) = \frac{1}{m} + O(m^{-2}),$$

completing the proof of Proposition 2.4. □

Theorem 2.5 (Arbitrary-Order Derivative Formula and p-Adic Denominator Bound)**.** *Let $m \ge 1$ and $0 \le r < m$ be integers. For every integer $q \ge 1$, the normalized q-th derivative of $\mathcal{E}_m(u)$ at $u = r^2$ is given by*

$$\frac{1}{q!}\mathcal{E}_m^{(q)}(r^2) = \mathcal{E}_m'(r^2)\, e_{q-1}\left(\left\{\frac{1}{r^2-t^2} : 0 \le t < m,\ t \ne r\right\}\right), \tag{2.13}$$

where e_{q-1} denotes the $(q-1)$-th elementary symmetric polynomial in the $m-1$ variables $x_t := (r^2-t^2)^{-1}$ $(0 \le t < m, t \ne r)$, with $e_0 := 1$ and $e_d := 0$ for $d \ge m$. Furthermore, defining the power sums

$$T_d(m,r) := \sum_{\substack{0\le t<m\\ t\ne r}}\frac{1}{(r^2-t^2)^d} \qquad (d \ge 1), \tag{2.14}$$

the sequence $e_d(m,r) := e_d(\{(r^2-t^2)^{-1}\}_{t\ne r})$ satisfies Newton's recurrence

$$e_d(m,r) = \frac{1}{d}\sum_{j=1}^{d}(-1)^{j-1}T_j(m,r)e_{d-j}(m,r) \qquad (d \ge 1), \tag{2.15}$$

and for every prime p and every $q \ge 1$:

$$v_p\left(\frac{1}{q!}\mathcal{E}_m^{(q)}(r^2)\right) \ge v_p(\mathcal{E}_m'(r^2)) - 2(q-1)\left\lfloor\frac{\log(2m)}{\log p}\right\rfloor. \tag{2.16}$$

Moreover, for every integer $r \in \mathbb{Z}$ and every integer $d \ge 0$, the linear-shift Taylor coefficient $[w^d]\mathcal{E}_m((w+r)^2)$ satisfies the uniform valuation bound (2.23) below.

Proof. Write $\mathcal{E}_m(u)$ in the factored form around $u = r^2$:

$$\mathcal{E}_m(u) = (u-r^2)\cdot\frac{2}{(2m)!}\prod_{\substack{0\le t<m\\ t\ne r}}((r^2-t^2)+(u-r^2)) = (u-r^2)\mathcal{E}_m'(r^2)\prod_{\substack{0\le t<m\\ t\ne r}}\left(1+\frac{u-r^2}{r^2-t^2}\right). \tag{2.17}$$

Expanding the product of $m-1$ linear factors in powers of $u-r^2$ gives

$$\prod_{\substack{0\le t<m\\ t\neq r}}\left(1+\frac{u-r^2}{r^2-t^2}\right)=\sum_{d=0}^{m-1}e_d\left(\left\{\frac{1}{r^2-t^2}:0\le t<m,\ t\neq r\right\}\right)(u-r^2)^d.$$

Multiplying by $(u-r^2)\mathcal{E}_m'(r^2)$ and extracting the coefficient of $(u-r^2)^q$ ($q=d+1\ge 1$) establishes (2.13). Newton's recurrence (2.15) is the standard algebraic identity relating the elementary symmetric polynomials e_d to the power sums $T_j=\sum_{t\neq r}x_t^j$.

To establish (2.16), note first that for $q=1$ we have $e_0=1$ and $v_p(1)=0=-2(1-1)\lfloor\log(2m)/\log p\rfloor$, while for $q>m$ we have $e_{q-1}=0$ and $v_p(0)=+\infty$. Now fix $2\le q\le m$ and let $I_{m,r}:=\{0,1,\dots,m-1\}\setminus\{r\}$. By definition of the elementary symmetric polynomial,

$$e_{q-1}(m,r)=\sum_{\substack{S\subset I_{m,r}\\ |S|=q-1}}\pi_S,\qquad \pi_S:=\prod_{t\in S}\frac{1}{(r-t)(r+t)}. \tag{2.18}$$

For each index $t\in I_{m,r}$, since $0\le r\le m-1$, $0\le t\le m-1$, and $t\neq r$, the two integer factors $r-t$ and $r+t$ are non-zero integers satisfying $1\le|r-t|\le m-1<2m$ and $1\le r+t\le 2m-3<2m$. Hence $v_p(r-t)\le\lfloor\frac{\log(2m)}{\log p}\rfloor$ and $v_p(r+t)\le\lfloor\frac{\log(2m)}{\log p}\rfloor$, so $v_p((r-t)(r+t))\le 2\lfloor\frac{\log(2m)}{\log p}\rfloor$ for all $t\in I_{m,r}$. By additivity of v_p on products and the ultrametric inequality across $S\subset I_{m,r}$ ($|S|=q-1$), we obtain $v_p(e_{q-1}(m,r))\ge -2(q-1)\lfloor\frac{\log(2m)}{\log p}\rfloor$. Adding $v_p(\mathcal{E}_m'(r^2))$ according to (2.13) establishes (2.16).

Furthermore, when $\mathcal{E}_m(u)$ is expanded directly in the linear shift variable $w=z+k$ around $u=(w+r)^2$ ($r=AK-k$, $|r|\le AK$), factoring $(w+r)^2-t^2=(w+r-t)(w+r+t)$ expresses $\mathcal{E}_m((w+r)^2)$ as a product of $2m$ linear factors in w:

$$\mathcal{E}_m((w+r)^2)=\frac{2}{(2m)!}(w+r)\prod_{t=-(m-1)}^{m-1}(w+r-t). \tag{2.19}$$

For $0\le|r|<m$, separating the $t=r$ factor w and the $t=-r$ factor $2r+w$ (or both $t=0$ factors w^2 at $r=0$) from (2.19), and writing the remaining prefactor $(w+r)=r(1+w/r)$ when $r\neq 0$, gives the exact degree-$2m$ factorization

$$\mathcal{E}_m((w+r)^2)=w(2r+w)\mathcal{E}_m'(r^2)\left(1+\frac{w}{r}\right)^{1-\delta_{r,0}}\prod_{\substack{-(m-1)\le t\le m-1\\ t\neq\pm r}}\left(1+\frac{w}{r-t}\right). \tag{2.20}$$

Write the leading quadratic factor in (2.20) as $w(2r+w)=2rw+w^2$, and note that the remaining product consists of $2m-2$ linear factors of the form $(1+\frac{w}{r-t})$ over the $(2m-2)$-element index set $J_{m,r}$ (namely $t\in\{-(m-1),\dots,m-1\}\setminus\{\pm r\}$ together with $(1+\frac{w}{r-0})$ when $1\le|r|<m$, or $t\in\{-(m-1),\dots,m-1\}\setminus\{0\}$ when $r=0$). Every non-zero difference satisfies $1\le|r-t|<2m$ and hence $v_p(r-t)\le\lfloor\frac{\log(2m)}{\log p}\rfloor$. Letting $\tilde{e}_k:=e_k(\{\frac{1}{r-t}:t\in J_{m,r}\})$ (with $\tilde{e}_k=0$ for $k<0$ or $k>2m-2$), we have $[w^0]\mathcal{E}_m((w+r)^2)=0$ and $[w^d]\mathcal{E}_m((w+r)^2)=\mathcal{E}_m'(r^2)(2r\,\tilde{e}_{d-1}+\tilde{e}_{d-2})$ for all $d\ge 1$. Since $2r\in\mathbb{Z}$ and $\min(v_p(\tilde{e}_{d-1}),v_p(\tilde{e}_{d-2}))\ge-(d-1)\lfloor\frac{\log(2m)}{\log p}\rfloor\ge-d\lfloor\frac{\log(2m)}{\log p}\rfloor$, the ultrametric inequality gives $v_p([w^d]\mathcal{E}_m((w+r)^2))\ge -v_p(1/\mathcal{E}_m'(r^2))-d\lfloor\frac{\log(2m)}{\log p}\rfloor$ for all $0\le|r|<m$ and $d\ge 0$.

Now suppose $|r|\ge m$ (with $m\ge 1$; for $m=0$, $\mathcal{E}_0\equiv 1$ satisfies (2.23) trivially). Since $(w+r)^2=(|r|+\operatorname{sgn}(r)w)^2$, replacing w by $\pm w$ preserves the p-adic valuation of $[w^d]\mathcal{E}_m((w+r)^2)$, so we may assume $r\ge m$. By Proposition 2.2 (2.3), the polynomial identity

$$\mathcal{E}_m((w+r)^2)=\binom{w+r+m}{2m}+\binom{w+r+m-1}{2m} \tag{2.21}$$

holds for all positive integers $w\in\mathbb{Z}_{\ge 1}$, and since both sides are polynomials of degree $2m$ in w, (2.21) holds identically in $\mathbb{Q}[w]$. For any integer $N_0\in\{r+m,r+m-1\}$, Vandermonde's

convolution identity in $\mathbb{Q}[w]$ gives

$$\binom{w+N_0}{2m} = \sum_{k=0}^{2m} \binom{N_0}{2m-k}\binom{w}{k}, \qquad \text{where} \qquad \binom{N_0}{2m-k} \in \mathbb{Z}. \tag{2.22}$$

For $d=0$, $[w^0]\mathcal{E}_m((w+r)^2) = \mathcal{E}_m(r^2) \in \mathbb{Z}$ has $v_p \geq 0$. For $d \geq 1$ and each $1 \leq k \leq 2m$, expanding $\binom{w}{k} = \frac{(-1)^{k-1}}{k}\, w \prod_{j=1}^{k-1}(1-\frac{w}{j})$ in powers of w yields

$$[w^d]\binom{w}{k} = \frac{(-1)^{k-d}}{k} \sum_{\substack{S\subset\{1,2,\ldots,k-1\}\\ |S|=d-1}} \prod_{j\in S} \frac{1}{j} \qquad (1 \leq d \leq k \leq 2m),$$

and $[w^d]\binom{w}{k} = 0$ for $d > k$. In every summand of $[w^d]\binom{w}{k}$, the denominator is a product of k and $d-1$ distinct integers $j \in \{1,\ldots,k-1\}$, which consists of at most $1+(d-1) = d$ integers in $\{1,2,\ldots,2m\}$, each having p-adic valuation at most $\lfloor \frac{\log(2m)}{\log p} \rfloor$. By additivity of v_p and the ultrametric inequality across (2.21)–(2.22), we obtain $v_p([w^d]\mathcal{E}_m((w+r)^2)) \geq -d\lfloor \frac{\log(2m)}{\log p} \rfloor$ for all $|r| \geq m$ and $d \geq 0$. Combining both ranges establishes the uniform valuation bound

$$v_p\Big([w^d]\mathcal{E}_m\big((w+r)^2\big)\Big) \geq -\mathbf{1}_{m>|r|} \cdot v_p\left(\frac{1}{\mathcal{E}_m'(r^2)}\right) - d\left\lfloor \frac{\log(2m)}{\log p} \right\rfloor \qquad (\forall\, d \geq 0,\ \forall\, r \in \mathbb{Z}). \tag{2.23}$$

□

Theorem 2.6 (Strict Total Positivity of $\mathcal{E}_r(x^2)$ via Lindström–Gessel–Viennot)**.** *For any positive integer $M \geq 1$, any strictly increasing sequence of integer degree indices $0 \leq r_1 < r_2 < \cdots < r_M$, and any strictly increasing sequence of integers $r_M < x_1 < x_2 < \cdots < x_M$, the evaluation determinant is strictly positive:*

$$\det_{1\leq i,j\leq M}\big(\mathcal{E}_{r_j}(x_i^2)\big) > 0. \tag{2.24}$$

Proof. By Proposition 2.2 (and $\mathcal{E}_0(x^2) = 1 = \frac{1}{2}(\binom{x}{0} + \binom{x-1}{0})$ at $r=0$), for every integer $x \in \{x_1,\ldots,x_M\}$ and every integer $r \in \{r_1,\ldots,r_M\}$, we have the finite matrix factorization

$$\mathcal{E}_r(x^2) = \frac{1}{1+\delta_{r,0}}\left(\binom{x+r}{2r} + \binom{x+r-1}{2r}\right) = \frac{1}{1+\delta_{r,0}} \sum_{y=0}^{x_M} B_{x,y}C_{y,r}, \tag{2.25}$$

where $B_{x,y} := \delta_{y,x} + \delta_{y,x-1}$ $(1 \leq x \leq x_M, 0 \leq y \leq x_M)$ is the lower bidiagonal matrix with 1's on $y=x$ and $y=x-1$, and $C_{y,r} := \binom{y+r}{2r}$ $(0 \leq y \leq x_M, 0 \leq r \leq r_M)$. Since $B_{x_i,y}$ is an $M \times (x_M+1)$ matrix and C_{y,r_j} is an $(x_M+1)\times M$ matrix with $x_M+1 > M$, the finite-dimensional Cauchy–Binet formula and multilinearity in column $j=1$ (where $\delta_{r_j,0} = 0$ for $j \geq 2$) give

$$\det_{1\leq i,j\leq M}\big(\mathcal{E}_{r_j}(x_i^2)\big) = \frac{1}{1+\delta_{r_1,0}} \sum_{0\leq y_1<y_2<\cdots<y_M\leq x_M} \det_{1\leq i,\ell\leq M}(B_{x_i,y_\ell}) \det_{1\leq \ell,j\leq M}(C_{y_\ell,r_j}). \tag{2.26}$$

We establish the non-negativity of every minor in (2.26) and the strict positivity of the term $(y_1,\ldots,y_M) = (x_1,\ldots,x_M)$:

(1) *Total non-negativity of B and unit diagonal minor:* Consider the planar directed acyclic bipartite graph with source vertices $U_x := (0,x)$ $(1 \leq x \leq x_M)$, sink vertices $V_y := (1,y)$ $(0 \leq y \leq x_M)$, horizontal edges $(0,x) \to (1,x)$, and diagonal edges $(0,x) \to (1,x-1)$ of weight 1. The number of directed paths from U_x to V_y is $B_{x,y} = \delta_{y,x} + \delta_{y,x-1}$. By planarity, two paths $U_{x_1} \to V_{y_{\pi(1)}}$ and $U_{x_2} \to V_{y_{\pi(2)}}$ with $x_1 < x_2$ and $y_{\pi(1)} > y_{\pi(2)}$ must cross, so only the identity permutation $\pi = \mathrm{id}$ can produce vertex-disjoint paths. Hence by the Lindström–Gessel–Viennot lemma [18, 12] (see also Karlin [13, Chapter 2]), for any $x_1 < \cdots < x_M$ and $y_1 < \cdots < y_M$, $\det_{1\leq i,\ell\leq M}(B_{x_i,y_\ell}) \in \{0,1\} \geq 0$. For the diagonal choice $y_\ell = x_\ell$ $(1 \leq \ell \leq M)$, whenever $i > \ell$ we have $x_\ell \leq x_i - 1 < x_i$, so $B_{x_i,x_\ell} = 0$ for all $i > \ell$ (upper triangularity) and $B_{x_i,x_i} = 1$, giving

$$\det_{1\leq i,\ell\leq M}(B_{x_i,x_\ell}) = 1 > 0. \tag{2.27}$$

(2) *Planar lattice-path representation and strict positivity of* $C_{y,r} = \binom{y+r}{2r}$*:* In the standard integer lattice $\mathbb{Z}^2$ equipped with East steps $(1,0)$ and North steps $(0,1)$, define for each integer $r \geq 0$ the source vertex $R_r := (-2r, r)$ on the line $u + 2v = 0$ in the left half-plane $u \leq 0$, and for each integer $y \geq 0$ the sink vertex $Q_y := (0, y)$ on the vertical axis $u = 0$. Any directed lattice path from $R_r = (-2r, r)$ to $Q_y = (0, y)$ consists of $2r$ East steps and $y - r$ North steps, so the total number of directed lattice paths from R_r to Q_y is $\binom{2r+(y-r)}{2r} = \binom{y+r}{2r} = C_{y,r}$. For any $0 \leq r_1 < r_2 < \cdots < r_M$ and $0 \leq y_1 < y_2 < \cdots < y_M$, the sources $R_{r_1}, \ldots, R_{r_M}$ are ordered along the ray $u = -2v \leq 0$ on the boundary of the region $\mathcal{D} := \{(u, v) \in \mathbb{R}^2 : u \leq 0, u + 2v \geq 0\}$, and the sinks $Q_{y_1}, \ldots, Q_{y_M}$ are ordered along the boundary $u = 0$ of $\mathcal{D}$. By the Jordan curve theorem in $\mathcal{D}$, any path from $R_{r_{j_1}}$ to $Q_{y_{\ell_1}}$ and any path from $R_{r_{j_2}}$ to $Q_{y_{\ell_2}}$ with $j_1 < j_2$ and $\ell_1 > \ell_2$ must intersect at a lattice vertex. Therefore, by the Lindström–Gessel–Viennot theorem [12, Theorem 1], $\det_{1 \leq \ell, j \leq M}(C_{y_\ell, r_j}) \geq 0$ equals the number of M-tuples of pairwise vertex-disjoint directed paths from R_{r_j} to Q_{y_j}. Moreover, whenever $0 \leq r_1 < \cdots < r_M \leq y_1 < \cdots < y_M$, we have $y_j \geq y_1 \geq r_M \geq r_j$ for every $1 \leq j \leq M$, and we explicitly construct an M-tuple of pairwise vertex-disjoint paths $(\gamma_1^*, \ldots, \gamma_M^*)$: let γ_j^* start at $R_{r_j} = (-2r_j, r_j)$, take $y_j - r_j \geq 0$ North steps vertically along $u = -2r_j$ to $(-2r_j, y_j)$, and then take $2r_j$ East steps horizontally along $v = y_j$ to $Q_{y_j} = (0, y_j)$. For any $1 \leq j_1 < j_2 \leq M$, we have $-2r_{j_2} < -2r_{j_1} \leq 0$ and $r_{j_1} < r_{j_2} \leq y_{j_1} < y_{j_2}$; thus the vertical segment of $\gamma_{j_2}^*$ lies strictly to the left of all vertices of $\gamma_{j_1}^*$, and the horizontal segment of $\gamma_{j_2}^*$ lies strictly above all vertices of $\gamma_{j_1}^*$. Hence $(\gamma_1^*, \ldots, \gamma_M^*)$ is pairwise vertex-disjoint, proving

$$\det_{1 \leq \ell, j \leq M}(C_{y_\ell, r_j}) \geq 1 > 0 \qquad \text{whenever } 0 \leq r_1 < \cdots < r_M \leq y_1 < \cdots < y_M. \tag{2.28}$$

Since every summand in (2.26) is non-negative and the term $(y_1, \ldots, y_M) = (x_1, \ldots, x_M)$ satisfies $r_M < x_1$, (2.27) and (2.28) give $\det_{1 \leq i, j \leq M}(\mathcal{E}_{r_j}(x_i^2)) \geq \frac{1}{2} > 0$, completing the proof. □

3. The Windowed Multi-Pole Kernel, Reflection Parity, and Positive-Series Representation

To confine all odd-zeta linear forms $S_3(P), S_5(P), \ldots, S_{2n+1}(P)$ to a narrow inner window of poles $|k - AK| \leq A_0 K$ while preserving a wide outer pole barrier $\{-2AK, \ldots, 0\}$ for archimedean decay and p-adic factorial cancellation, we construct the windowed multi-pole rational kernel $\mathcal{F}_{2n+1,2}^{\mathrm{odd},[A_0,A]}(z; P)$.

Definition 3.1 (Windowed Multi-Pole Rational Kernel)**.** Fix an integer $n \geq 1$, inner pole multiplicity $L_{\mathrm{in}} := 2n + 1$, outer pole multiplicity $L_{\mathrm{out}} := 2$, and rational scaling parameters $0 < A_0 < A$ such that $A_0 K, AK \in \mathbb{Z}_{\geq 1}$. For any polynomial $P(u) \in \mathbb{Q}[u]$ satisfying the degree bound

$$2 \deg_u P \leq 4AK + 2(2n-1)A_0 K - 2, \tag{3.1}$$

define the windowed rational function

$$\mathcal{F}_{2n+1,2}^{\mathrm{odd},[A_0,A]}(z; P) := \frac{((2AK)!)^2 ((2A_0K)!)^{2n-1}}{\left(\prod_{k=0}^{2AK}(z+k)^2\right)\left(\prod_{|k-AK| \leq A_0 K}(z+k)^{2n-1}\right)} P((z + AK)^2). \tag{3.2}$$

Denote by

$$L_k := \begin{cases} 2n+1, & |k - AK| \leq A_0 K, \\ 2, & A_0 K < |k - AK| \leq AK, \end{cases} \tag{3.3}$$

the pole multiplicity of $\mathcal{F}_{2n+1,2}^{\mathrm{odd},[A_0,A]}(z; 1)$ at $z = -k$ $(0 \leq k \leq 2AK)$, and write the partial fraction decomposition of $\mathcal{F}_{2n+1,2}^{\mathrm{odd},[A_0,A]}(z; P)$ as

$$\mathcal{F}_{2n+1,2}^{\mathrm{odd},[A_0,A]}(z; P) = \sum_{k=0}^{2AK} \sum_{t=1}^{L_k} \frac{A_{t,k}(P)}{(z+k)^t}, \qquad S_t(P) := \sum_{k=0}^{2AK} A_{t,k}(P) \quad (1 \leq t \leq 2n+1), \tag{3.4}$$

where $A_{t,k}(P) := 0$ whenever $t > L_k$.

Before establishing the global symmetry and support compression of $S_t(P)$, we derive the local Laurent expansion of the kernel at each pole $z = -k$.

Lemma 3.2 (Local Laurent Expansion of the Windowed Kernel)**.** *For each pole $k \in \{0, 1, \dots, 2AK\}$, set $r := AK - k$ and $w := z + k$, so that $u = (z + AK)^2 = (w + r)^2 = r^2 + 2rw + w^2$. Let*

$$g_k(w) := w^{L_k} \mathcal{F}_{2n+1,2}^{\mathrm{odd},[A_0,A]}(-k + w; 1) = \sum_{q=0}^{\infty} g_{k,q} w^q \tag{3.5}$$

be the Taylor expansion of the smooth kernel factor at $w = 0$. Then the leading coefficient $g_{k,0} \neq 0$ is given for every inner pole $|k - AK| \leq A_0 K$ and every outer pole $A_0 K < |k - AK| \leq AK$ by

$$g_{k,0} = \begin{cases} (-1)^{A_0K - r} \dbinom{2AK}{k}^2 \dbinom{2A_0K}{A_0K + r}^{2n-1}, & |r| \leq A_0 K, \\ \operatorname{sgn}(r)^{2n-1} \dbinom{2AK}{k}^2 \left((|r| - A_0K) \dbinom{|r| + A_0K}{2A_0K} \right)^{-(2n-1)}, & A_0K < |r| \leq AK, \end{cases} \tag{3.6}$$

and the higher Taylor coefficients $g_{k,q}$ $(1 \leq q < L_k)$ satisfy Newton's recurrence

$$g_{k,q} = \frac{1}{q} \sum_{d=1}^{q} p_{k,d}\, g_{k,q-d}, \qquad p_{k,d} := (-1)^d \sum_{\substack{0 \leq j \leq 2AK \\ j \neq k}} \frac{L_j}{(j - k)^d}. \tag{3.7}$$

For any polynomial $P(u) \in \mathbb{Q}[u]$, the Laurent coefficient $A_{t,k}(P)$ $(1 \leq t \leq L_k)$ is given by the finite convolution

$$A_{t,k}(P) = \sum_{q=0}^{L_k - t} g_{k, L_k - t - q} \sum_{d=\lceil q/2 \rceil}^{q} \binom{d}{q - d} (2r)^{2d-q} \frac{P^{(d)}(r^2)}{d!}. \tag{3.8}$$

Proof. At $z = -k + w$, the factor $(z + k)^{L_k} = w^{L_k}$ cancels against w^{L_k} in (3.5), leaving

$$g_k(w) = \big((2AK)!\big)^2 \big((2A_0K)!\big)^{2n-1} \prod_{\substack{0 \leq j \leq 2AK \\ j \neq k}} (w + j - k)^{-L_j}.$$

Evaluating at $w = 0$ for an inner pole $|r| \leq A_0 K$, we have

$$\prod_{\substack{0 \leq j \leq 2AK \\ j \neq k}} (j - k)^2 = \big((-1)^k k!(2AK - k)!\big)^2 = \big(k!(2AK - k)!\big)^2$$

and

$$\prod_{\substack{|j - AK| \leq A_0K \\ j \neq k}} (j - k)^{2n-1} = \big((-1)^{A_0K - r}(A_0K + r)!(A_0K - r)!\big)^{2n-1}.$$

Since $2n - 1$ is odd, $(-1)^{(2n-1)(A_0K - r)} = (-1)^{A_0K - r}$, so dividing these two factorial products into $((2AK)!)^2$ and $((2A_0K)!)^{2n-1}$ yields the first line of (3.6). Similarly, evaluating $g_k(0)$ at an outer pole $k = AK - r$ with $A_0K < |r| \leq AK$ (where $L_k = 2$, so $(z + k)^2 = w^2$ cancels from the outer denominator product), the outer product over $0 \leq j \leq 2AK, j \neq k$ again equals $(k!(2AK - k)!)^2$, which divides into $((2AK)!)^2$ to give $\binom{2AK}{k}^2$. In the inner denominator product over $j = AK - s$ $(-A_0K \leq s \leq A_0K)$, the difference $j - k = r - s$ has constant sign $\operatorname{sgn}(r)$ for all $|s| \leq A_0K < |r|$, with $|j - k| = |r| - \operatorname{sgn}(r)s$ running over the $2A_0K + 1$ consecutive positive integers from $|r| - A_0K$ to $|r| + A_0K$. Hence

$$\begin{aligned} \prod_{|j - AK| \leq A_0K} (j - k)^{2n-1} &= \operatorname{sgn}(r)^{(2n-1)(2A_0K+1)} \left(\frac{(|r| + A_0K)!}{(|r| - A_0K - 1)!} \right)^{2n-1} \\ &= \operatorname{sgn}(r)^{2n-1} \left((2A_0K)!\, (|r| - A_0K) \binom{|r| + A_0K}{2A_0K} \right)^{2n-1}. \end{aligned}$$

Dividing this product into $((2A_0K)!)^{2n-1}$ yields the second line of (3.6). Taking the logarithmic derivative of $g_k(w)$ gives

$$\frac{g_k'(w)}{g_k(w)} = -\sum_{\substack{0\le j\le 2AK\\ j\neq k}} \frac{L_j}{w+j-k} = -\sum_{\substack{0\le j\le 2AK\\ j\neq k}} \frac{L_j}{j-k}\sum_{d=1}^{\infty}\left(-\frac{w}{j-k}\right)^{d-1} = \sum_{d=1}^{\infty} p_{k,d}w^{d-1},$$

and equating coefficients of w^{q-1} in $g_k'(w) = g_k(w)\sum_{d\ge1} p_{k,d}w^{d-1}$ gives $qg_{k,q} = \sum_{d=1}^{q} p_{k,d}g_{k,q-d}$, proving (3.7).

To derive (3.8), let $D := \deg_u P$. By Taylor's formula for the polynomial $P(u)$ at $u = r^2$, substituting $u - r^2 = (w+r)^2 - r^2 = w(2r+w)$ and expanding $(2r+w)^d = \sum_{a=0}^{d}\binom{d}{a}(2r)^{d-a}w^a$ gives

$$P((w+r)^2) = \sum_{d=0}^{D}\frac{P^{(d)}(r^2)}{d!}w^d(2r+w)^d = \sum_{d=0}^{D}\sum_{a=0}^{d}\binom{d}{a}(2r)^{d-a}\frac{P^{(d)}(r^2)}{d!}w^{d+a}.$$

Grouping terms by the total exponent $q := d+a \in \{0,1,\dots,2D\}$, the constraints $0 \le a = q-d \le d$ are equivalent to $\lceil q/2\rceil \le d \le q$, so the coefficient of w^q in $P((w+r)^2)$ is

$$[w^q]P((w+r)^2) = \sum_{d=\lceil q/2\rceil}^{q}\binom{d}{q-d}(2r)^{2d-q}\frac{P^{(d)}(r^2)}{d!}.$$

Since $A_{t,k}(P)$ is the coefficient of $(z+k)^{-t} = w^{-t}$ in $\mathcal{F}(-k+w;P) = w^{-L_k}g_k(w)P((w+r)^2)$, it equals the coefficient of w^{L_k-t} in the Cauchy product of $g_k(w) = \sum_{m=0}^{\infty} g_{k,m}w^m$ and $P((w+r)^2)$, which is $\sum_{q=0}^{L_k-t} g_{k,L_k-t-q}[w^q]P((w+r)^2)$, proving (3.8). □

Theorem 3.3 (Odd Reflection Parity and Inner-Window Support Compression)**.** *For every integer $n \ge 1$ and every polynomial $P(u) \in \mathbb{Q}[u]$ satisfying the degree condition (3.1), the windowed kernel $\mathcal{F}(z;P) := \mathcal{F}_{2n+1,2}^{\mathrm{odd},[A_0,A]}(z;P)$ satisfies:*

(i) Odd Reflection Symmetry: *Under the involution $z \mapsto -2AK - z$,*

$$\mathcal{F}(-2AK-z;P) = -\mathcal{F}(z;P), \qquad \text{and hence} \qquad A_{t,2AK-k}(P) = (-1)^{t-1}A_{t,k}(P) \tag{3.9}$$

for all $0 \le k \le 2AK$ and $1 \le t \le 2n+1$. Consequently, all even-weight pole sums and the simple-pole residue sum vanish identically:

$$S_{2j}(P) = 0 \quad (1 \le j \le n), \qquad \text{and} \qquad S_1(P) = \sum_{k=0}^{2AK}\operatorname{Res}_{z=-k}\mathcal{F}(z;P) = 0. \tag{3.10}$$

(ii) Vanishing of Outer Poles in $S_{2j+1}(P)$: *For every outer pole k with $A_0K < |k-AK| \le AK$, the pole multiplicity of $\mathcal{F}(z;P)$ at $z = -k$ is $L_k = L_{\mathrm{out}} = 2 < 3$, so*

$$A_{t,k}(P) = 0 \qquad \text{for all } 3 \le t \le 2n+1 \text{ and all } A_0K < |k-AK| \le AK. \tag{3.11}$$

Thus every odd-weight pole sum $S_{2j+1}(P)$ $(1 \le j \le n)$ is supported exclusively on the inner window $|k-AK| \le A_0K$:

$$S_{2j+1}(P) = \sum_{|k-AK|\le A_0K} A_{2j+1,k}(P) \qquad (1 \le j \le n). \tag{3.12}$$

Proof. To prove (i), substitute $z \mapsto -2AK - z$ into each factor of (3.2):

- The variable $u = (-2AK - z + AK)^2 = (-z - AK)^2 = (z + AK)^2$ is invariant, so $P((-2AK - z + AK)^2) = P((z+AK)^2)$.
- In the outer denominator product, changing the index to $k' = 2AK - k$ gives

$$\prod_{k=0}^{2AK}(-2AK - z + k)^2 = \prod_{k'=0}^{2AK}(-z-k')^2 = \prod_{k'=0}^{2AK}(z+k')^2,$$

which is invariant because the exponent $2(2AK+1)$ is even.

- In the inner denominator product over $AK-A_0K \le k \le AK+A_0K$, setting $k' = 2AK-k$ (which runs over the same symmetric interval $|k'-AK| \le A_0K$ of odd cardinality $2A_0K+1$) gives

$$\prod_{|k-AK|\le A_0K} (-2AK - z + k)^{2n-1} = (-1)^{(2n-1)(2A_0K+1)} \prod_{|k'-AK|\le A_0K} (z + k')^{2n-1} = - \prod_{|k'-AK|\le A_0K} (z + k')^{2n-1},$$

since $(2n-1)(2A_0K+1)$ is a product of two odd integers and is therefore odd.

This proves $\mathcal{F}(-2AK - z; P) = -\mathcal{F}(z; P)$.

To determine the relation between the Laurent coefficients at $z = -k$ and $z = -(2AK-k)$, set $z = -(2AK-k)+w$, so that $-2AK - z = -k - w$. Expanding $\mathcal{F}(-2AK - z; P) = \mathcal{F}(-k - w; P)$ around $w = 0$ gives

$$\mathcal{F}(-2AK - z; P) = \sum_{t=1}^{L_k} \frac{A_{t,k}(P)}{(-w)^t} + O(1) = \sum_{t=1}^{L_k} \frac{(-1)^t A_{t,k}(P)}{\big(z + (2AK-k)\big)^t} + O(1).$$

On the other hand, expanding $-\mathcal{F}(z; P)$ around $z = -(2AK-k)+w$ gives

$$-\mathcal{F}(z; P) = - \sum_{t=1}^{L_{2AK-k}} \frac{A_{t,2AK-k}(P)}{\big(z + (2AK-k)\big)^t} + O(1).$$

By uniqueness of the Laurent principal part at $z = -(2AK-k)$, equating the coefficients of $(z + (2AK-k))^{-t}$ yields

$$A_{t,2AK-k}(P) = -(-1)^t A_{t,k}(P) = (-1)^{t-1} A_{t,k}(P), \tag{3.13}$$

proving (3.9). When $t = 2j$ is even, $(-1)^{2j-1} = -1$, so $A_{2j,2AK-k}(P) = -A_{2j,k}(P)$ is antisymmetric under $k \leftrightarrow 2AK-k$, and summing over $0 \le k \le 2AK$ gives $S_{2j}(P) = \frac{1}{2}\sum_{k=0}^{2AK}(A_{2j,k}(P) + A_{2j,2AK-k}(P)) = 0$, whereas when $t = 2j+1$ is odd, $(-1)^{(2j+1)-1} = +1$, so $A_{2j+1,2AK-k}(P) = A_{2j+1,k}(P)$ is symmetric under $k \leftrightarrow 2AK-k$. Finally, by (3.1), the numerator of $\mathcal{F}(z; P)$ has degree in z equal to $2\deg_u P \le 4AK + 2(2n-1)A_0K - 2$, while the denominator has degree $2(2AK+1)+(2n-1)(2A_0K+1) = 4AK+2(2n-1)A_0K+2n+1$, so $\deg_z \mathcal{F}(z; P) \le -(2n+3) \le -5 \le -2$. By Cauchy's residue theorem on the Riemann sphere $\mathbb{P}^1(\mathbb{C})$, the sum of all finite residues $S_1(P) = \sum_{k=0}^{2AK} \operatorname{Res}_{z=-k} \mathcal{F}(z; P) = -\operatorname{Res}_{z=\infty} \mathcal{F}(z; P) = 0$ vanishes identically, proving (i).

Part (ii) follows directly from Definition 3.1: for every outer pole k with $A_0K < |k-AK| \le AK$, the denominator of (3.2) has pole order $L_k = L_{\text{out}} = 2$, so the principal part at $z = -k$ contains only $(z+k)^{-1}$ and $(z+k)^{-2}$. Hence $A_{t,k}(P) = 0$ for all $t \ge 3$, which restricts the sum in (3.12) to $|k - AK| \le A_0K$. □

Theorem 3.4 (Positive-Series Moment Representation)**.** *For every integer $n \ge 1$ and every polynomial $P(u) \in \mathbb{Q}[u]$ satisfying (3.1), define the rational harmonic remainder*

$$\mathcal{R}_0(P) := - \sum_{k=0}^{2AK} \sum_{t=1}^{L_k} A_{t,k}(P) H_k^{(t)}, \qquad H_k^{(t)} := \sum_{\ell=1}^{k} \frac{1}{\ell^t} \quad (H_0^{(t)} := 0), \tag{3.14}$$

and the strictly positive discrete weights

$$W_K(\nu) := \mathcal{F}_{2n+1,2}^{\text{odd},[A_0,A]}(\nu; 1) = \frac{\big((2AK)!\big)^2 \big((2A_0K)!\big)^{2n-1}}{\Big(\prod_{k=0}^{2AK}(\nu+k)^2\Big)\Big(\prod_{|k-AK|\le A_0K}(\nu+k)^{2n-1}\Big)} > 0 \qquad (\nu \ge 1). \tag{3.15}$$

Then the linear form $\mathcal{A}(P) := \sum_{j=1}^{n} S_{2j+1}(P)\zeta(2j+1) + \mathcal{R}_0(P)$ converges absolutely and satisfies the identity

$$\mathcal{A}(P) = \sum_{j=1}^{n} S_{2j+1}(P)\zeta(2j+1) + \mathcal{R}_0(P) = \sum_{\nu=1}^{\infty} W_K(\nu) P\big((\nu+AK)^2\big). \tag{3.16}$$

Proof. By the degree bound (3.1), as $\nu \to \infty$ we have $\mathcal{F}_{2n+1,2}^{\mathrm{odd},[A_0,A]}(\nu;P) = W_K(\nu)P((\nu+AK)^2) = O(\nu^{-5})$, so the infinite series $\sum_{\nu=1}^{\infty} W_K(\nu)P((\nu+AK)^2)$ converges absolutely. For any finite truncation integer $M \geq 1$, substituting the finite partial fraction decomposition (3.4) into each term of the finite sum $\sum_{\nu=1}^{M} \mathcal{F}_{2n+1,2}^{\mathrm{odd},[A_0,A]}(\nu;P)$ gives

$$\sum_{\nu=1}^{M} W_K(\nu)P((\nu+AK)^2) = \sum_{k=0}^{2AK} A_{1,k}(P) \sum_{\nu=1}^{M} \frac{1}{\nu+k} + \sum_{t=2}^{2n+1} \sum_{k=0}^{2AK} A_{t,k}(P) \sum_{\nu=1}^{M} \frac{1}{(\nu+k)^t}. \tag{3.17}$$

For the $t=1$ sum in (3.17), shifting the summation index to $\ell = \nu + k$ gives

$$\sum_{\nu=1}^{M} \frac{1}{\nu+k} = H_{M+k}^{(1)} - H_k^{(1)} = H_M^{(1)} - H_k^{(1)} + \sum_{\ell=M+1}^{M+k} \frac{1}{\ell}.$$

Since $0 \leq \sum_{\ell=M+1}^{M+k} \frac{1}{\ell} \leq \frac{k}{M} \leq \frac{2AK}{M}$ for all $0 \leq k \leq 2AK$, multiplying by $A_{1,k}(P)$, summing over $0 \leq k \leq 2AK$, and using $\sum_{k=0}^{2AK} A_{1,k}(P) = S_1(P) = 0$ from Theorem 3.3(i), the term $H_M^{(1)} S_1(P) = 0$ vanishes identically for every $M \geq 1$, and

$$\left| \sum_{k=0}^{2AK} A_{1,k}(P) \sum_{\nu=1}^{M} \frac{1}{\nu+k} - \left(- \sum_{k=0}^{2AK} A_{1,k}(P) H_k^{(1)} \right) \right| \leq \frac{2AK}{M} \sum_{k=0}^{2AK} |A_{1,k}(P)| \xrightarrow[M\to\infty]{} 0,$$

which proves

$$\lim_{M\to\infty} \sum_{k=0}^{2AK} A_{1,k}(P) \sum_{\nu=1}^{M} \frac{1}{\nu+k} = - \sum_{k=0}^{2AK} A_{1,k}(P) H_k^{(1)}. \tag{3.18}$$

For each $2 \leq t \leq 2n+1$, the series $\sum_{\nu=1}^{\infty} (\nu+k)^{-t} = \sum_{\ell=k+1}^{\infty} \ell^{-t} = \zeta(t) - H_k^{(t)}$ converges absolutely, so taking $M \to \infty$ in the finite sum over $2 \leq t \leq 2n+1$ and $0 \leq k \leq 2AK$ in (3.17) gives

$$\begin{aligned} \lim_{M\to\infty} \sum_{t=2}^{2n+1} \sum_{k=0}^{2AK} A_{t,k}(P) \sum_{\nu=1}^{M} \frac{1}{(\nu+k)^t} &= \sum_{t=2}^{2n+1} S_t(P)\zeta(t) - \sum_{t=2}^{2n+1} \sum_{k=0}^{2AK} A_{t,k}(P) H_k^{(t)} \\ &= \sum_{j=1}^{n} S_{2j+1}(P)\zeta(2j+1) - \sum_{t=2}^{2n+1} \sum_{k=0}^{2AK} A_{t,k}(P) H_k^{(t)}, \end{aligned} \tag{3.19}$$

where the second equality uses $S_{2j}(P) = 0$ $(1 \leq j \leq n)$ from Theorem 3.3(i). Adding (3.18) and (3.19) and comparing with (3.14) establishes (3.16). □

Theorem 3.5 (Reflection-Symmetric Rising-Factorial Zero Barrier)**.** *Let $BK \in \mathbb{Z}_{\geq 0}$ and $M_0 \in \{0,1,2\}$ satisfy $2M_0BK + 2\deg_u P \leq 4AK + 2(2n-1)A_0K - 2$, and define for $BK \geq 1$ the reflection-invariant polynomial in $u = (z+AK)^2$:*

$$\Omega_B(u) := \frac{1}{((BK)!)^2} \prod_{j=1}^{BK} (u - (AK+j)^2) = (-1)^{BK} \binom{z-1}{BK} \binom{-2AK-z-1}{BK} \qquad (\Omega_0(u) := 1). \tag{3.20}$$

Then the zero-buffered windowed kernel $\mathcal{F}_{2n+1,2}^{\mathrm{odd},[A_0,A,B]}(z;P) := \mathcal{F}_{2n+1,2}^{\mathrm{odd},[A_0,A]}(z;\Omega_B(u)^{M_0}P(u))$ satisfies:

(i) *Because $\Omega_B(u)^{M_0}$ is a polynomial in $u = (z+AK)^2$ that is non-zero at every pole $u = r^2$ $(0 \leq r \leq AK < AK+1)$, Theorems 3.3 and 3.4 hold verbatim.*

(ii) *For every integer $1 \leq \nu \leq BK$, $\Omega_B((\nu+AK)^2) = 0$, so the positive series (3.16) starts at $\nu = BK+1$:*

$$\begin{aligned} \mathcal{A}(\Omega_B^{M_0}P) &= \sum_{\nu=BK+1}^{\infty} W_{K,B}(\nu)P((\nu+AK)^2), \\ W_{K,B}(\nu) &:= W_K(\nu) \binom{\nu-1}{BK}^{M_0} \binom{\nu+2AK+BK}{BK}^{M_0} > 0. \end{aligned} \tag{3.21}$$

(iii) *At every pole* $z = -k + w$ *(*$0 \le k \le 2AK$*,* $r = AK - k$*),*

$$\Omega_B((w+r)^2) = (-1)^{BK}\binom{k+BK}{BK}\binom{2AK-k+BK}{BK}\prod_{j=1}^{BK}\left(1-\frac{w}{k+j}\right)\left(1+\frac{w}{2AK-k+j}\right), \tag{3.22}$$

which adds the non-negative binomial valuation $+M_0 v_p(\binom{k+BK}{BK}\binom{2AK-k+BK}{BK}) \ge 0$ *to* $V_p(k)$ *at every pole* $0 \le k \le 2AK$*.*

Proof. At $u = (z + AK)^2$, each factor of (3.20) satisfies $u - (AK + j)^2 = (z - j)(z + 2AK + j) = -(z - j)(-2AK - z - j)$. Taking the product over $1 \le j \le BK$ and dividing by $((BK)!)^2$ gives $(-1)^{BK}\binom{z-1}{BK}\binom{-2AK-z-1}{BK}$, establishing (3.20). At $z = -k+w$ ($r = AK-k$), we have $(w+r)^2-(AK+j)^2 = (w-k-j)(w+2AK-k+j) = -(k+j)(2AK-k+j)(1-\frac{w}{k+j})(1+\frac{w}{2AK-k+j})$. Taking the product over $1 \le j \le BK$ (where the BK factors of -1 multiply to $(-1)^{BK}$) and dividing by $((BK)!)^2$ using $\prod_{j=1}^{BK}(k+j) = (BK)!\binom{k+BK}{BK}$ and $\prod_{j=1}^{BK}(2AK-k+j) = (BK)!\binom{2AK-k+BK}{BK}$ yields (3.22). Since $v_p((-1)^{BK}) = 0$, raising (3.22) to the power M_0 adds $+M_0 v_p(\binom{k+BK}{BK}\binom{2AK-k+BK}{BK}) \ge 0$ to $V_p(k)$. Finally, at $z = \nu \ge 1$, $(\nu + AK)^2 - (AK + j)^2 = (\nu - j)(\nu + 2AK + j)$, which vanishes for $1 \le \nu \le BK$ at $j = \nu$ and equals $((BK)!)^2\binom{\nu-1}{BK}\binom{\nu+2AK+BK}{BK} > 0$ for $\nu \ge BK + 1$, establishing (i), (ii), and (iii). □

4. Local Leibniz Jet Structure and Degree-Cutoff Jet Separation

Having confined all odd zeta sums $S_{2j+1}(P)$ $(1 \le j \le n)$ to the inner window $|k - AK| \le A_0K$, we now analyze their local derivative jet structure at the $A_0K + 1$ inner square nodes $u = r^2$ $(0 \le r \le A_0K)$.

Proposition 4.1 (Inner Jet Representation of $S_{2j+1}(P)$ for General $n \ge 2$)**.** *Fix an integer* $n \ge 2$ *and inner multiplicity* $L_{\text{in}} = 2n + 1$*. Define the inner node weights*

$$w_{2n+1,0} := g_{AK,0} \neq 0, \qquad w_{2n+1,r} := 2g_{AK-r,0} \neq 0 \quad (1 \le r \le A_0K). \tag{4.1}$$

Then for every polynomial $P(u) \in \mathbb{Q}[u]$ *and every* $1 \le j \le n$*:*

(i) *The highest odd-zeta functional* $S_{2n+1}(P)$ *depends exclusively on the values* $P(r^2)$ *at the* $A_0K + 1$ *inner nodes* $0 \le r \le A_0K$*:*

$$S_{2n+1}(P) = \sum_{r=0}^{A_0K} w_{2n+1,r} P(r^2). \tag{4.2}$$

(ii) *Each lower odd-zeta functional* $S_{2j+1}(P)$ *(*$1 \le j \le n-1$*) admits the local jet expansion*

$$S_{2j+1}(P) = \sum_{q=0}^{n-j} w_{2j+1,0}^{(q)} \frac{P^{(q)}(0)}{q!} + \sum_{r=1}^{A_0K}\sum_{q=0}^{2(n-j)} w_{2j+1,r}^{(q)} \frac{P^{(q)}(r^2)}{q!}, \tag{4.3}$$

where the rational jet weights $w_{2j+1,r}^{(q)}$ *are given in terms of the Laurent coefficients* $g_{AK-r,m}$ *of Lemma 3.2 by*

$$w_{2j+1,0}^{(q)} = g_{AK,2(n-j-q)}, \qquad w_{2j+1,r}^{(q)} = 2\sum_{a=0}^{\min(q,2(n-j)-q)} g_{AK-r,2(n-j)-q-a}\binom{q}{a}(2r)^{q-a}, \tag{4.4}$$

and the leading derivative weights at each node $0 \le r \le A_0K$ *are non-zero:*

$$w_{2j+1,0}^{(n-j)} = w_{2n+1,0} \neq 0, \qquad w_{2j+1,r}^{(2(n-j))} = (2r)^{2(n-j)} w_{2n+1,r} \neq 0 \quad (1 \le r \le A_0K). \tag{4.5}$$

(iii) *Consequently, for any row polynomial* $A(u) \in \mathbb{Q}[u]$ *and any tail column basis polynomial* $\mathcal{E}_m(u)$ *with* $m > A_0K$*, we have* $\mathcal{E}_m(r^2) = 0$ *for all* $0 \le r \le A_0K$*, and hence*

$$S_{2n+1}(A\mathcal{E}_m) = 0 \qquad (\forall\, m > A_0K). \tag{4.6}$$

Proof. By Theorem 3.3(i)–(ii), for any odd weight $t = 2j+1$ ($1 \le j \le n$), the Laurent coefficients satisfy $A_{2j+1,2AK-k}(P) = +A_{2j+1,k}(P)$ and vanish for $|k - AK| > A_0K$. Pairing each pole $k = AK - r$ with its reflection $2AK - k = AK + r$ for $1 \le r \le A_0K$ gives

$$S_{2j+1}(P) = A_{2j+1,AK}(P) + 2\sum_{r=1}^{A_0K} A_{2j+1,AK-r}(P). \tag{4.7}$$

At every inner pole $k = AK - r$ ($0 \le r \le A_0K$), the pole multiplicity is $L_k = 2n+1$. Setting $t = 2n+1$ in Lemma 3.2 (3.8) restricts the sum to $q = 0$ ($d = 0$), giving $A_{2n+1,AK-r}(P) = g_{AK-r,0}P(r^2)$. Substituting into (4.7) yields (4.2) with $w_{2n+1,0} = g_{AK,0} \neq 0$ and $w_{2n+1,r} = 2g_{AK-r,0} \neq 0$, proving (i).

For $1 \le j \le n-1$, setting $t = 2j+1$ and $L_k = 2n+1$ in (3.8) gives $L_k - t = 2(n-j)$. At the central pole $k = AK$ ($r = 0$), the factor $(2r)^{2d-q}$ in (3.8) vanishes unless $2d = q$, which forces $q = 2d$ with $0 \le d \le n-j$. Thus

$$A_{2j+1,AK}(P) = \sum_{d=0}^{n-j} g_{AK,2(n-j-d)}\frac{P^{(d)}(0)}{d!},$$

which gives the $r = 0$ sum in (4.3) with leading $d = n-j$ coefficient $w_{2j+1,0}^{(n-j)} = g_{AK,0} = w_{2n+1,0} \neq 0$. For $1 \le r \le A_0K$, denoting the outer summation index in (3.8) by $m \in \{0, \dots, 2(n-j)\}$ and the inner derivative order by $q \in \{\lceil m/2 \rceil, \dots, m\}$ gives

$$2A_{2j+1,AK-r}(P) = 2\sum_{m=0}^{2(n-j)} g_{AK-r,2(n-j)-m} \sum_{q=\lceil m/2 \rceil}^{m} \binom{q}{m-q}(2r)^{2q-m}\frac{P^{(q)}(r^2)}{q!}.$$

In this double sum, the index pairs (m, q) satisfy $0 \le q \le m \le \min(2q, 2(n-j))$. Interchanging the finite sums so that q runs from 0 to $2(n-j)$ and substituting $m = q+a$ (so $a = m-q$ runs from 0 to $\min(q, 2(n-j)-q)$, with $2q-m = q-a$ and $2(n-j)-m = 2(n-j)-q-a$) yields (4.3)–(4.4). At the highest derivative order $q = 2(n-j)$, the upper limit $\min(q, 2(n-j)-q) = 0$ restricts the sum over a to the single term $a = 0$, giving $w_{2j+1,r}^{(2(n-j))} = 2g_{AK-r,0}\binom{2(n-j)}{0}(2r)^{2(n-j)} = (2r)^{2(n-j)}w_{2n+1,r} \neq 0$, establishing (4.5) and proving (ii). Finally, if $m > A_0K$, then by Proposition 2.2(i) we have $\mathcal{E}_m(r^2) = 0$ for all $0 \le r \le A_0K$. Hence $(A\mathcal{E}_m)(r^2) = A(r^2)\mathcal{E}_m(r^2) = 0$ for every $0 \le r \le A_0K$, and substituting $P = A\mathcal{E}_m$ into (4.2) proves (4.6). □

Lemma 4.2 (Explicit Hermite Interpolation Basis on $\mathbb{Q}[u]_{\le 2A_0K}$)**.** *Let $A_0K \in \mathbb{Z}_{\ge 1}$ and consider the A_0K+1 distinct real nodes $u_0 := 0$ and $u_r := r^2$ ($1 \le r \le A_0K$). The linear evaluation map $\Pi_{\mathrm{Herm}} : \mathbb{Q}[u]_{\le 2A_0K} \to \mathbb{Q}^{2A_0K+1}$ defined by*

$$\Pi_{\mathrm{Herm}}(A) := \Big(A(0);\ A(1^2), A'(1^2);\ A(2^2), A'(2^2);\ \dots;\ A((A_0K)^2), A'((A_0K)^2)\Big) \tag{4.8}$$

is a vector space isomorphism. Explicitly, setting

$$\Omega(u) := \prod_{r=1}^{A_0K}(u-r^2)^2 \in \mathbb{Q}[u]_{\le 2A_0K}, \qquad \ell_r(u) := \frac{u}{r^2}\prod_{\substack{1\le t\le A_0K\\ t\neq r}}\left(\frac{u-t^2}{r^2-t^2}\right)^2 \quad (1 \le r \le A_0K), \tag{4.9}$$

where $r^2 - t^2 \neq 0$ for all $t \neq r$ and $\ell_r'(r^2) = \frac{1}{r^2} + \sum_{1\le t\le A_0K, t\neq r}\frac{2}{r^2-t^2}$, the fundamental Hermite basis polynomials

$$H_0(u) := \frac{\Omega(u)}{\Omega(0)}, \qquad H_{r,0}(u) := (1 - \ell_r'(r^2)(u-r^2))\ell_r(u), \qquad H_{r,1}(u) := (u-r^2)\ell_r(u) \tag{4.10}$$

all lie in $\mathbb{Q}[u]_{\le 2A_0K}$ and satisfy:

(i) *$H_0(0) = 1$, and $H_0(r^2) = H_0'(r^2) = 0$ for all $1 \le r \le A_0K$;*
(ii) *$H_{r,0}(0) = 0$, $H_{r,0}(t^2) = \delta_{r,t}$, and $H_{r,0}'(t^2) = 0$ for all $1 \le r, t \le A_0K$;*
(iii) *$H_{r,1}(0) = 0$, $H_{r,1}(t^2) = 0$, and $H_{r,1}'(t^2) = \delta_{r,t}$ for all $1 \le r, t \le A_0K$.*

Proof. Each polynomial $\ell_r(u)$ has degree $1+2(A_0K-1)=2A_0K-1$, so $H_{r,0}(u)$ and $H_{r,1}(u)$ have degree $2A_0K$ and $H_0(u)$ has degree $2A_0K$; thus all $2A_0K+1$ polynomials in (4.10) belong to $\mathbb{Q}[u]_{\le 2A_0K}$. For $H_0(u)$, $\Omega(0)=\prod_{r=1}^{A_0K} r^4=((A_0K)!)^4\neq 0$, so $H_0(0)=1$, and for each $1\le r\le A_0K$, $(u-r^2)^2 \mid H_0(u)$ implies $H_0(r^2)=H_0'(r^2)=0$, proving (i). For $1\le r\le A_0K$, the factor u in $\ell_r(u)$ gives $\ell_r(0)=0$, and for every $t\neq r$ the factor $(u-t^2)^2$ in $\ell_r(u)$ gives $\ell_r(t^2)=\ell_r'(t^2)=0$, while at $t=r$ we have $\ell_r(r^2)=1$. Differentiating $H_{r,1}(u)=(u-r^2)\ell_r(u)$ at $u=r^2$ gives $H_{r,1}(r^2)=0$ and $H_{r,1}'(r^2)=\ell_r(r^2)=1$, proving (iii). Taking the logarithmic derivative of $\ell_r(u)=\frac{u}{r^2}\prod_{t\neq r}(\frac{u-t^2}{r^2-t^2})^2$ gives $\frac{\ell_r'(u)}{\ell_r(u)}=\frac{1}{u}+2\sum_{1\le t\le A_0K,t\neq r}\frac{1}{u-t^2}$, and evaluating at $u=r^2$ yields $\ell_r'(r^2)=\frac{1}{r^2}+2\sum_{1\le t\le A_0K,t\neq r}\frac{1}{r^2-t^2}$. Differentiating $H_{r,0}(u)=(1-\ell_r'(r^2)(u-r^2))\ell_r(u)$ at $u=r^2$ gives $H_{r,0}(r^2)=\ell_r(r^2)=1$ and $H_{r,0}'(r^2)=-\ell_r'(r^2)\ell_r(r^2)+\ell_r'(r^2)=0$, proving (ii). Since the $2A_0K+1$ polynomials (4.10) map to the standard basis of $\mathbb{Q}^{2A_0K+1}$ under Π_{Herm} and $\dim_{\mathbb{Q}}\mathbb{Q}[u]_{\le 2A_0K}=2A_0K+1$, Π_{Herm} is an isomorphism. □

Fix $n\ge 2$ and $L_{\mathrm{in}}=2n+1$, and define the matrix dimensions

$$N:=A_0K+1,\qquad R_{\mathrm{each}}:=2A_0K+1,\qquad R:=(n-1)R_{\mathrm{each}},\qquad M:=N+R. \tag{4.11}$$

Partition the $M=N+R$ column indices $\{0,1,\dots,N+R-1\}$ into the N head columns and R tail columns:

$$J_{\mathrm{head}}:=\{0,1,\dots,N-1\},\qquad J_{\mathrm{tail}}:=\{N,N+1,\dots,N+R-1\}. \tag{4.12}$$

On the R tail columns $m\in J_{\mathrm{tail}}$, every basis polynomial $\mathcal{E}_m(u)=\frac{2}{(2m)!}\prod_{t=0}^{m-1}(u-t^2)$ $(N\le m\le N+R-1)$ contains the inner-window root polynomial

$$W_0(u):=\prod_{r=0}^{A_0K}(u-r^2)=\frac{(2N)!}{2}\,\mathcal{E}_N(u)\qquad(\deg_u W_0=A_0K+1=N) \tag{4.13}$$

as a factor, and since $\deg_u \mathcal{E}_m=m$ ranges over $N,N+1,\dots,N+R-1$, the tail basis $\{\mathcal{E}_N(u),\dots,\mathcal{E}_{N+R-1}(u)\}$ forms a $\mathbb{Q}$-vector space basis of $W_0(u)\mathbb{Q}[u]_{\le R-1}$ of dimension R.

Theorem 4.3 (Degree-Cutoff Jet Separation)**.** *Fix an integer $n\ge 2$ and $L_{\mathrm{in}}=2n+1$, and let N,R_{each},R,M be defined by (4.11). Take the row multipliers of each of the $n-1$ top elimination blocks $(1\le j\le n-1)$ to be $\mathcal{E}_0(u),\dots,\mathcal{E}_{R_{\mathrm{each}}-1}(u)$ of $\mathbb{Q}[u]_{\le 2A_0K}$, the bottom N row multipliers to be $\mathcal{E}_0(u),\dots,\mathcal{E}_{N-1}(u)$ of $\mathbb{Q}[u]_{\le A_0K}$, and the M column multipliers to be $\mathcal{E}_0(u),\dots,\mathcal{E}_{M-1}(u)$. For any tuple $(a_3,a_5,\dots,a_{2n+1})\in\mathbb{Q}^n$ with $a_{2n+1}\neq 0$, define the $R\times M$ stacked top operator matrix L_{top} and the $N\times M$ bottom $\zeta(2n+1)$-coefficient block $S_{2n+1,\mathrm{bot}}$ by*

$$L_{\mathrm{top}}:=\begin{pmatrix} L_{\mathrm{top}}^{(3)}\\ L_{\mathrm{top}}^{(5)}\\ \vdots\\ L_{\mathrm{top}}^{(2n-1)}\end{pmatrix},\qquad (L_{\mathrm{top}}^{(2j+1)})_{i,m}:=a_{2n+1}S_{2j+1}(\mathcal{E}_i\mathcal{E}_m)-a_{2j+1}S_{2n+1}(\mathcal{E}_i\mathcal{E}_m), \tag{4.14}$$

for $1\le j\le n-1, 0\le i<R_{\mathrm{each}}, 0\le m<M$, and $(S_{2n+1,\mathrm{bot}})_{k,m}:=S_{2n+1}(\mathcal{E}_k\mathcal{E}_m)$ $(0\le k<N, 0\le m<M)$. Then:

(i) Full Row Rank of L_{top} and Invertibility of the Tail Submatrix: $\mathrm{rank}(L_{\mathrm{top}})=R$. *Specifically, on the R tail columns $J_{\mathrm{tail}}=\{N,\dots,N+R-1\}$ we have $S_{2n+1,\mathrm{top}}[:,J_{\mathrm{tail}}]=0_{R\times R}$ identically, and the $R\times R$ tail submatrix*

$$L_{\mathrm{top}}[:,J_{\mathrm{tail}}]=a_{2n+1}S_{\mathrm{odd,top}}[:,J_{\mathrm{tail}}] \tag{4.15}$$

(where $S_{\mathrm{odd,top}}$ stacks $S_{3,\mathrm{top}},S_{5,\mathrm{top}},\dots,S_{2n-1,\mathrm{top}}$) is invertible over $\mathbb{Q}$.

(ii) Full Row Rank of $S_{2n+1,\mathrm{bot}}$ and Head Determinant: $\mathrm{rank}(S_{2n+1,\mathrm{bot}})=N$. *Specifically, on the N head columns $J_{\mathrm{head}}=\{0,\dots,N-1\}$, the $N\times N$ submatrix $S_{2n+1,\mathrm{bot}}[:,J_{\mathrm{head}}]$ is invertible with determinant*

$$\det(S_{2n+1,\mathrm{bot}}[:,J_{\mathrm{head}}])=\prod_{r=0}^{A_0K}w_{2n+1,r}\neq 0. \tag{4.16}$$

(iii) Invertibility of the Stacked Matrix and Zero Free Contour Rows: *With respect to the column partition* $\{0,\dots,N+R-1\} = J_{\text{head}} \sqcup J_{\text{tail}}$, *the* $(N+R)\times(N+R)$ *stacked matrix is block-triangular:*

$$\begin{pmatrix} L_{\text{top}} \\ S_{2n+1,\text{bot}} \end{pmatrix} = \begin{pmatrix} L_{\text{top}}[:, J_{\text{head}}] & a_{2n+1} S_{\text{odd,top}}[:, J_{\text{tail}}] \\ S_{2n+1,\text{bot}}[:, J_{\text{head}}] & 0_{N\times R} \end{pmatrix}, \tag{4.17}$$

and its determinant is non-zero:

$$\det \begin{pmatrix} L_{\text{top}} \\ S_{2n+1,\text{bot}} \end{pmatrix} = (-1)^{RN} a_{2n+1}^{R} \det(S_{2n+1,\text{bot}}[:, J_{\text{head}}]) \det(S_{\text{odd,top}}[:, J_{\text{tail}}]) \neq 0. \tag{4.18}$$

In particular, $\operatorname{row}(L_{\text{top}}) \cap \operatorname{row}(S_{2n+1,\text{bot}}) = \{0\}$ *and* $N_0 := N - \operatorname{rank}(S_{2n+1,\text{bot}} \bmod \operatorname{row}(L_{\text{top}})) = 0$ *identically.*

Proof. We establish the block-triangular structure (4.17) and then prove parts (ii), (i), and (iii) in turn.

Step 1: Block-triangular decomposition and proof of (ii). By Proposition 4.1(iii) (4.6), for every tail column index $m \in J_{\text{tail}} = \{N,\dots,N+R-1\}$ $(m \geq N = A_0K+1 > A_0K)$ and every row polynomial $\mathcal{E}_i(u)$, we have $S_{2n+1}(\mathcal{E}_i\mathcal{E}_m) = 0$. Therefore, on the R tail columns J_{tail}, the bottom block $S_{2n+1,\text{bot}}[:, J_{\text{tail}}] = 0_{N\times R}$ vanishes identically and the $-a_{2j+1}S_{2n+1}(\mathcal{E}_i\mathcal{E}_m)$ term in (4.14) drops out of $L_{\text{top}}[:, J_{\text{tail}}]$, giving (4.15) and (4.17). Expanding the determinant of (4.17) by block column exchange (exchanging N head columns past R tail columns with sign $(-1)^{RN}$) yields (4.18).

To evaluate $\det(S_{2n+1,\text{bot}}[:, J_{\text{head}}])$ in (4.16), apply Proposition 4.1(i) (4.2) to $P(u) = \mathcal{E}_k(u)\mathcal{E}_d(u)$ for $0 \leq k, d \leq A_0K = N-1$:

$$(S_{2n+1,\text{bot}}[:, J_{\text{head}}])_{k,d} = \sum_{r=0}^{A_0K} \mathcal{E}_k(r^2) w_{2n+1,r} \mathcal{E}_d(r^2) = (E_{\text{head}} W_{2n+1} E_{\text{head}}^T)_{k,d},$$

where $W_{2n+1} := \operatorname{diag}(w_{2n+1,0}, w_{2n+1,1}, \dots, w_{2n+1,A_0K})$ and $(E_{\text{head}})_{k,r} := \mathcal{E}_k(r^2)$ $(0 \leq k, r \leq A_0K)$. By Proposition 2.2(i), $\mathcal{E}_k(r^2) = 0$ whenever $r < k$ and $\mathcal{E}_k(k^2) = 1$ for every $0 \leq k \leq A_0K$. Thus E_{head} is an $N \times N$ upper unitriangular matrix with $\det E_{\text{head}} = 1$, and hence $\det(S_{2n+1,\text{bot}}[:, J_{\text{head}}]) = (\det E_{\text{head}})^2 \det W_{2n+1} = \prod_{r=0}^{A_0K} w_{2n+1,r} \neq 0$, proving (ii).

Step 2: Proof of (i) and (iii) by backward Hermite induction. To prove that the $R \times R$ matrix $S_{\text{odd,top}}[:, J_{\text{tail}}]$ is invertible, it suffices to show that its right kernel is trivial. Since $\{\mathcal{E}_N(u),\dots,\mathcal{E}_{N+R-1}(u)\}$ is a basis of $W_0(u)\mathbb{Q}[u]_{\leq R-1}$ and $\{\mathcal{E}_0(u),\dots,\mathcal{E}_{2A_0K}(u)\}$ is a basis of $\mathbb{Q}[u]_{\leq 2A_0K}$, any vector $\mathbf{c} = (c_N,\dots,c_{N+R-1})^T \in \ker(S_{\text{odd,top}}[:, J_{\text{tail}}])$ corresponds uniquely to a tail polynomial

$$B(u) := \sum_{m=N}^{N+R-1} c_m \mathcal{E}_m(u) = W_0(u)Q(u), \qquad Q(u) \in \mathbb{Q}[u]_{\leq R-1}, \tag{4.19}$$

such that

$$S_{2j+1}(AB) = 0 \qquad \text{for all } A(u) \in \mathbb{Q}[u]_{\leq 2A_0K} \text{ and all } 1 \leq j \leq n-1. \tag{4.20}$$

Set $m := n-j \in \{1,2,\dots,n-1\}$, so that $2j+1 = 2n-2m+1$. We prove by induction on $m \in \{1,2,\dots,n-1\}$ that $Q(u)$ satisfies the divisibility relation

$$u^m \prod_{r=1}^{A_0K} (u-r^2)^{2m} \,\Big|\, Q(u) \qquad \text{in } \mathbb{Q}[u]. \tag{4.21}$$

Fix $m \in \{1,2,\dots,n-1\}$, and assume that (4.21) holds with $m-1$ in place of m (for $m=1$, the divisor is 1, which divides $Q(u)$ trivially). Multiplying the induction hypothesis $u^{m-1}\prod_{r=1}^{A_0K}(u-r^2)^{2m-2} \mid Q(u)$ by $W_0(u) = u\prod_{r=1}^{A_0K}(u-r^2)$ shows that $B(u) = W_0(u)Q(u)$ is divisible by $u^m\prod_{r=1}^{A_0K}(u-r^2)^{2m-1}$. Equivalently,

$$B^{(t)}(0) = 0 \quad (0 \leq t \leq m-1), \qquad B^{(t)}(r^2) = 0 \quad (0 \leq t \leq 2m-2,\ 1 \leq r \leq A_0K). \tag{4.22}$$

Now consider the equation $S_{2n-2m+1}(AB) = 0$ from (4.20). By Proposition 4.1(ii) (4.3) and Leibniz's rule $\frac{(AB)^{(q)}(u_r)}{q!} = \sum_{a+b=q} \frac{A^{(a)}(u_r)}{a!}\frac{B^{(b)}(u_r)}{b!}$, substituting (4.22) leaves at $u_0 = 0$ only

$(q,a,b) = (m,0,m)$ and at each $u_r = r^2$ $(1 \le r \le A_0K)$ only $(q,a,b) \in \{(2m-1,0,2m-1),(2m,0,2m),(2m,1,2m-1)\}$:

$$\begin{aligned} 0 = S_{2n-2m+1}(AB) = w^{(m)}_{2n-2m+1,0} A(0)\frac{B^{(m)}(0)}{m!} \\ + \sum_{r=1}^{A_0K} A(r^2)\left(w^{(2m-1)}_{2n-2m+1,r}\frac{B^{(2m-1)}(r^2)}{(2m-1)!} + w^{(2m)}_{2n-2m+1,r}\frac{B^{(2m)}(r^2)}{(2m)!}\right) \\ + \sum_{r=1}^{A_0K} A'(r^2) w^{(2m)}_{2n-2m+1,r}\frac{B^{(2m-1)}(r^2)}{(2m-1)!}. \end{aligned} \tag{4.23}$$

We evaluate (4.23) on the Hermite basis polynomials $H_{r_0,1}(u), H_{r_0,0}(u), H_0(u) \in \mathbb{Q}[u]_{\le 2A_0K}$ of Lemma 4.2:

(1) Fix $r_0 \in \{1,\dots,A_0K\}$ and substitute $A(u) = H_{r_0,1}(u)$ into (4.23). By Lemma 4.2(iii), $A(0) = 0$, $A(r^2) = 0$ for all $1 \le r \le A_0K$, and $A'(r^2) = \delta_{r,r_0}$. Thus (4.23) reduces to $w^{(2m)}_{2n-2m+1,r_0}\frac{B^{(2m-1)}(r_0^2)}{(2m-1)!} = 0$. By (4.5), $w^{(2m)}_{2n-2m+1,r_0} = (2r_0)^{2m} w_{2n+1,r_0} \ne 0$, forcing $B^{(2m-1)}(r_0^2) = 0$ for every $1 \le r_0 \le A_0K$.

(2) Substitute $B^{(2m-1)}(r^2) = 0$ $(1 \le r \le A_0K)$ back into (4.23), fix $r_0 \in \{1,\dots,A_0K\}$, and choose $A(u) = H_{r_0,0}(u)$. By Lemma 4.2(ii), $A(0) = 0$, $A(r^2) = \delta_{r,r_0}$, and $A'(r^2) = 0$, so (4.23) reduces to $w^{(2m)}_{2n-2m+1,r_0}\frac{B^{(2m)}(r_0^2)}{(2m)!} = 0$, forcing $B^{(2m)}(r_0^2) = 0$ for every $1 \le r_0 \le A_0K$.

(3) Finally, choose $A(u) = H_0(u)$ in (4.23). By Lemma 4.2(i), $A(0) = 1$ and $A(r^2) = A'(r^2) = 0$ for all $1 \le r \le A_0K$, so (4.23) reduces to $w^{(m)}_{2n-2m+1,0}\frac{B^{(m)}(0)}{m!} = 0$. By (4.5), $w^{(m)}_{2n-2m+1,0} = w_{2n+1,0} \ne 0$, forcing $B^{(m)}(0) = 0$.

Combining $B^{(m)}(0) = 0$ and $B^{(2m-1)}(r^2) = B^{(2m)}(r^2) = 0$ $(1 \le r \le A_0K)$ with (4.22) shows that $B(u) = W_0(u)Q(u)$ vanishes to order at least $m+1$ at $u = 0$ and to order at least $2m+1$ at each $u = r^2$ $(1 \le r \le A_0K)$. Dividing by $W_0(u) = u\prod_{r=1}^{A_0K}(u-r^2)$ proves (4.21) for m, completing the induction.

Evaluating (4.21) at the terminal step $m = n-1$ shows that $Q(u)$ is divisible by $u^{n-1}\prod_{r=1}^{A_0K}(u-r^2)^{2(n-1)}$, which is a monic polynomial of degree $(n-1)+2(n-1)A_0K = (n-1)(2A_0K+1) = R$. However, by (4.19), $\deg_u Q \le R-1 < R$. Thus $Q(u) \equiv 0$, $B(u) \equiv 0$, and $\mathbf{c} = \mathbf{0}$, proving (i) and (iii). □

Theorem 4.4 (Determinant of the Tail Submatrix)**.** *Fix an integer $n \ge 2$, $A_0K \ge 1$, and $AK > A_0K$, and let $N = A_0K+1$, $R_{\text{each}} = 2A_0K+1$, and $R = (n-1)R_{\text{each}}$. Then the $R \times R$ rational matrix $S_{\text{odd,top}}[:, J_{\text{tail}}]$ of Theorem 4.3(i) satisfies the closed-form identity*

$$\begin{aligned} \left|\det\left(S_{\text{odd,top}}[:,J_{\text{tail}}]\right)\right| = \left(\prod_{i=1}^{2A_0K}\frac{2}{(2i)!}\right)^{n-1}\left(\prod_{m=N}^{N+R-1}\frac{2}{(2m)!}\right)|w_{2n+1,0}|^{n-1} \\ \times\left(\prod_{r=1}^{A_0K}|w_{2n+1,r}|^{2(n-1)}(2r)^{2n(n-1)}r^{(n-1)(4n+6)}\right) \\ \times\prod_{1\le s<r\le A_0K}(r^2-s^2)^{4(n^2-1)} \ne 0. \end{aligned} \tag{4.24}$$

In particular, after factoring out the row and column central factorial prefactors, the remaining monic-basis tail determinant is a non-zero integer divisible by $|w_{2n+1,0}|^{n-1}$ and by $\prod_{r=1}^{A_0K}|w_{2n+1,r}|^{2(n-1)}$.

Proof. Recall the classical confluent Vandermonde determinant identity: for any $K+1$ distinct nodes $u_0 < u_1 < \cdots < u_K$ with positive integer multiplicities $m_0, m_1, \dots, m_K \ge 1$ summing to $D := \sum_{r=0}^{K} m_r$, the $D \times D$ confluent Vandermonde matrix $V_{\text{conf}}(u_0^{(m_0)},\dots,u_K^{(m_K)})$ whose rows evaluate $\delta_{u_r,q}(P) := \frac{1}{q!}P^{(q)}(u_r)$ $(0 \le r \le K, 0 \le q < m_r)$ on the monic power basis

$\{1, u, u^2, \dots, u^{D-1}\}$ satisfies

$$\det V_{\mathrm{conf}}(u_0^{(m_0)}, u_1^{(m_1)}, \dots, u_K^{(m_K)}) = \prod_{0 \le s < r \le K} (u_r - u_s)^{m_r m_s}. \tag{4.25}$$

(Indeed, viewing $\det V_{\mathrm{conf}}(u_0^{(m_0)}, \dots, u_K^{(m_K)})$ as a polynomial in the variables $u_0, \dots, u_K$, for each pair $0 \le s < r \le K$ the $m_r + m_s$ rows at u_r and u_s span the Taylor jets of orders $< m_r$ at u_r and $< m_s$ at u_s, so $(u_r - u_s)^{m_r m_s}$ divides $\det V_{\mathrm{conf}}$ in $\mathbb{Q}[u_0, \dots, u_K]$. Since $\sum_{0 \le s < r \le K} m_r m_s + \sum_{r=0}^{K} \binom{m_r}{2} = \binom{D}{2}$ equals the total u-degree of the columns $\{1, u, \dots, u^{D-1}\}$ minus the total differentiation order $\sum_{r=0}^{K} \binom{m_r}{2}$ of the rows, and the diagonal product has coefficient $+1$, (4.25) holds identically.) Specializing (4.25) to the $A_0K + 1$ nodes $u_0 = 0$ and $u_r = r^2$ $(1 \le r \le A_0K)$ gives:

- For multiplicities $(m_0; m_1, \dots, m_{A_0K}) = (1; 2, \dots, 2)$ of sum $D = 2A_0K + 1 = R_{\mathrm{each}}$:

$$\det V_{\mathrm{Herm}}^{\mathrm{monic}} = \prod_{r=1}^{A_0K} (r^2 - 0)^{2 \cdot 1} \prod_{1 \le s < r \le A_0K} (r^2 - s^2)^{2 \cdot 2} = \left(\prod_{r=1}^{A_0K} r^4 \right) \prod_{1 \le s < r \le A_0K} (r^2 - s^2)^4. \tag{4.26}$$

- For multiplicities $(m_0; m_1, \dots, m_{A_0K}) = (n-1; 2(n-1), \dots, 2(n-1))$ of sum $D = (n-1)(2A_0K + 1) = R$:

$$\det V_Q^{\mathrm{monic}} = \left(\prod_{r=1}^{A_0K} r^{4(n-1)^2} \right) \prod_{1 \le s < r \le A_0K} (r^2 - s^2)^{4(n-1)^2}. \tag{4.27}$$

We factor $S_{\mathrm{odd,top}}[:, J_{\mathrm{tail}}]$ into four triangular/invertible transitions:

(1) *Row transition from $\{\mathcal{E}_0, \dots, \mathcal{E}_{2A_0K}\}$ to the Hermite basis $\{H_0, H_{r,0}, H_{r,1}\}$:* In each of the $n-1$ row blocks $S_{2j+1,\mathrm{top}}$ $(1 \le j \le n-1)$, $\mathcal{E}_0(u) = 1$ and $\mathcal{E}_i(u) = \frac{2}{(2i)!} u^i + (\deg < i)$ $(1 \le i \le 2A_0K)$, so the transition matrix T_{row} from $\{1, u, \dots, u^{2A_0K}\}$ to $\{\mathcal{E}_0, \dots, \mathcal{E}_{2A_0K}\}$ is upper triangular with $\det T_{\mathrm{row}} = \prod_{i=1}^{2A_0K} \frac{2}{(2i)!}$. Since Π_{Herm} maps $\{H_0, H_{r,0}, H_{r,1}\}$ to the standard basis of $\mathbb{Q}^{2A_0K+1}$ by Lemma 4.2, the transition matrix from $\{H_0, H_{r,0}, H_{r,1}\}$ to $\{\mathcal{E}_0, \dots, \mathcal{E}_{2A_0K}\}$ has determinant $(\prod_{i=1}^{2A_0K} \frac{2}{(2i)!}) \det V_{\mathrm{Herm}}^{\mathrm{monic}}$. Across all $n-1$ row blocks, this contributes $(\prod_{i=1}^{2A_0K} \frac{2}{(2i)!})^{n-1} (\det V_{\mathrm{Herm}}^{\mathrm{monic}})^{n-1}$.

(2) *Column transition to $W_0(u) \cdot \{1, u, \dots, u^{R-1}\}$:* For each $m = N + d \in J_{\mathrm{tail}}$ $(0 \le d < R)$, $\mathcal{E}_{N+d}(u) = \frac{2}{(2N+2d)!} W_0(u) Q_d^{\mathrm{monic}}(u)$, where $W_0(u) = u \prod_{r=1}^{A_0K} (u - r^2)$ and $Q_d^{\mathrm{monic}}(u) = u^d + (\deg < d)$. The transition matrix from $\{W_0(u) u^d\}_{d=0}^{R-1}$ to $\{\mathcal{E}_m(u)\}_{m=N}^{N+R-1}$ is upper triangular with determinant $\prod_{m=N}^{N+R-1} \frac{2}{(2m)!}$.

(3) *Block-triangular Hermite jet evaluation on $\{W_0(u) Q(u) : Q \in \mathbb{Q}[u]_{\le R-1}\}$:* Order the $n-1$ row blocks by $m = n - j \in \{1, 2, \dots, n-1\}$ (so block m is $S_{2n-2m+1}$), and order the R confluent jet functionals $\delta_{u_r, q}(Q)$ on $Q \in \mathbb{Q}[u]_{\le R-1}$ by step $m \in \{1, \dots, n-1\}$, where step m consists of the $2A_0K + 1$ functionals $\delta_{0,m-1}(Q) = \frac{Q^{(m-1)}(0)}{(m-1)!}$, $\delta_{r^2,2m-2}(Q) = \frac{Q^{(2m-2)}(r^2)}{(2m-2)!}$, and $\delta_{r^2,2m-1}(Q) = \frac{Q^{(2m-1)}(r^2)}{(2m-1)!}$. For any polynomial $B(u) = W_0(u) Q(u)$ whose quotient $Q(u)$ has vanishing jets at all steps $1 \le \tilde{m} < m$, differentiating $B(u) = (u - u_r) \widetilde{W}_{0,r}(u) Q(u)$ (where $\widetilde{W}_{0,r}(u_r) = W_0'(u_r)$) by Leibniz's rule at $u_0 = 0$ and $u_r = r^2$ gives

$$\frac{B^{(m)}(0)}{m!} = W_0'(0) \delta_{0,m-1}(Q), \qquad \frac{B^{(2m-1)}(r^2)}{(2m-1)!} = W_0'(r^2) \delta_{r^2,2m-2}(Q),$$

and $\frac{B^{(2m)}(r^2)}{(2m)!} = W_0'(r^2) \delta_{r^2,2m-1}(Q) + \frac{1}{2} W_0''(r^2) \delta_{r^2,2m-2}(Q)$. Substituting these into (4.23) and evaluating on $(H_0; H_{r,1}, H_{r,0})$ at step m yields at $u_0 = 0$ the 1×1 multiplier $|w_{2n-2m+1,0}^{(m)} W_0'(0)| = |w_{2n+1,0}| \prod_{r=1}^{A_0K} r^2$, and at each $u_r = r^2$ $(1 \le r \le A_0K)$ the 2×2

lower-triangular matrix

$$\begin{pmatrix} S_{2n-2m+1}(H_{r,1}W_0Q) \\ S_{2n-2m+1}(H_{r,0}W_0Q) \end{pmatrix} = \begin{pmatrix} w^{(2m)}_{2n-2m+1,r}W_0'(r^2) & 0 \\ w^{(2m-1)}_{2n-2m+1,r}W_0'(r^2) + \frac{1}{2}w^{(2m)}_{2n-2m+1,r}W_0''(r^2) & w^{(2m)}_{2n-2m+1,r}W_0'(r^2) \end{pmatrix} \times \begin{pmatrix} \delta_{r^2,2m-2}(Q) \\ \delta_{r^2,2m-1}(Q) \end{pmatrix},$$

whose determinant is $(w^{(2m)}_{2n-2m+1,r})^2 W_0'(r^2)^2$. Since $W_0'(r^2) = r^2\prod_{s\neq r}(r^2-s^2)$ and $w^{(2m)}_{2n-2m+1,r} = (2r)^{2m}w_{2n+1,r}$ by (4.5), the product of these 2×2 block determinants over $1\le r\le A_0K$ at step m equals

$$\prod_{r=1}^{A_0K}(w^{(2m)}_{2n-2m+1,r})^2 W_0'(r^2)^2 = \left(\prod_{r=1}^{A_0K} w^2_{2n+1,r}(2r)^{4m}r^4\right)\prod_{1\le s<r\le A_0K}(r^2-s^2)^4.$$

Combining the $u_0=0$ factor $|w_{2n+1,0}|\prod_{r=1}^{A_0K}r^2$ with the $1\le r\le A_0K$ product gives $r^2\cdot r^4 = r^6$ at each step $m\in\{1,\dots,n-1\}$. Multiplying across all $n-1$ steps $1\le m\le n-1$ (where $\sum_{m=1}^{n-1}4m = 2n(n-1)$) yields the combined step-(3) product

$$|w_{2n+1,0}|^{n-1}\left(\prod_{r=1}^{A_0K}|w_{2n+1,r}|^{2(n-1)}(2r)^{2n(n-1)}r^{6(n-1)}\right)\prod_{1\le s<r\le A_0K}(r^2-s^2)^{4(n-1)}.$$

(4) *Confluent Vandermonde evaluation of $Q\in\{1,u,\dots,u^{R-1}\}$:* Evaluating the R functionals $\delta_{0,q}$ $(0\le q<n-1)$ and $\delta_{r^2,q}$ $(0\le q<2(n-1), 1\le r\le A_0K)$ on $\{1,u,\dots,u^{R-1}\}$ gives $\det V_Q^{\mathrm{monic}}$ in (4.27).

Multiplying the four determinants together and summing the exponents of r $(4(n-1)+6(n-1)+4(n-1)^2 = (n-1)(4n+6))$ and of (r^2-s^2) $(4(n-1)+4(n-1)+4(n-1)^2 = 4(n^2-1))$ establishes (4.24). □

Remark 4.5 (Uniform Odd Inner Multiplicity $L_{\mathrm{in}} = 2n+1$)**.** Throughout this paper, we work exclusively with the odd inner pole multiplicity $L_{\mathrm{in}} = 2n+1$ for every $n\ge 1$ (in particular, $L_{\mathrm{in}} = 5$ for $n=2$). Because $(2n-1)(2A_0K+1)$ is odd, the odd inner multiplicity $L_{\mathrm{in}} = 2n+1$ guarantees the odd reflection symmetry $\mathcal{F}(-2AK-z;P) = -\mathcal{F}(z;P)$ directly from the denominator product in (3.2) without auxiliary odd linear factors $(z+AK)$.

5. The Schur-Complement Moment Determinant, Block-Triangular Reduction, and Reversed-Laplace Asymptotics

Fix an integer $n\ge 2$, and assume for induction on n that $1,\zeta(3),\dots,\zeta(2n-1)$ are linearly independent over $\mathbb{Q}$ (for the base case $n=1$, 1 and $\zeta(3)$ are linearly independent over $\mathbb{Q}$ by Apéry's theorem [1]). Suppose for contradiction that there exists a nontrivial rational linear relation among $1,\zeta(3),\zeta(5),\dots,\zeta(2n+1)$. Clearing rational denominators, we obtain a nontrivial integer linear relation

$$a_3\zeta(3)+a_5\zeta(5)+\cdots+a_{2n+1}\zeta(2n+1)+c=0 \qquad (a_3,a_5,\dots,a_{2n+1},c\in\mathbb{Z}\text{ not all zero}). \tag{5.1}$$

If $a_{2n+1}=0$, then (5.1) reduces to a nontrivial integer linear relation among $1,\zeta(3),\dots,\zeta(2n-1)$, contradicting the induction hypothesis. Thus $a_{2n+1}\neq 0$, and multiplying (5.1) by -1 if necessary, we may assume without loss of generality that $a_{2n+1}>0$.

Definition 5.1 (The $(N+R)\times(N+R)$ Schur-Complement Matrix and Determinant)**.** Fix $n\ge 2$, $L_{\mathrm{in}} = 2n+1$, and scaling parameters $0<A_0<A$ with $A_0K, AK\in\mathbb{Z}_{\ge 1}$. Let $N = A_0K+1 = CK+1$, $R_{\mathrm{each}} = 2A_0K+1 = R_{c,\mathrm{each}}K+1$, and $R = (n-1)R_{\mathrm{each}} = R_cK+n-1$ (where $C := A_0$, $R_{c,\mathrm{each}} := 2A_0$, and $R_c := 2(n-1)A_0$) be as in Theorem 4.3. Define the $(N+R)\times(N+R)$ rational matrix $M_K(\mathbf{a},c)$ and its determinant Δ_{N+R} by

$$M_K(\mathbf{a},c) := \begin{pmatrix} L_{\mathrm{top}} \\ a_{2n+1}\mathcal{R}_{0,\mathrm{bot}} - cS_{2n+1,\mathrm{bot}} \end{pmatrix} \in \mathbb{Q}^{(N+R)\times(N+R)}, \qquad \Delta_{N+R} := \det M_K(\mathbf{a},c)\in\mathbb{Q}, \tag{5.2}$$

where L_{top} is the $R \times (N+R)$ stacked top matrix of (4.14) and the bottom $N \times (N+R)$ block has entries

$$\left(a_{2n+1}\mathcal{R}_{0,\mathrm{bot}} - cS_{2n+1,\mathrm{bot}}\right)_{k,m} := a_{2n+1}\mathcal{R}_0(\mathcal{E}_k\mathcal{E}_m) - cS_{2n+1}(\mathcal{E}_k\mathcal{E}_m) \quad (0 \le k < N,\ 0 \le m < N+R).$$

Proposition 5.2 (Schur-Complement Moment Identity)**.** *Under the linear relation (5.1), the rational determinant Δ_{N+R} of Definition 5.1 satisfies the identity*

$$\Delta_{N+R} = \det\begin{pmatrix} L_{\mathrm{top}} \\ a_{2n+1}\mathcal{A}_{\mathrm{bot}} \end{pmatrix}, \tag{5.3}$$

where $(\mathcal{A}_{\mathrm{bot}})_{k,m} := \mathcal{A}(\mathcal{E}_k\mathcal{E}_m) = \sum_{\nu=1}^{\infty} W_K(\nu)\mathcal{E}_k((\nu+AK)^2)\mathcal{E}_m((\nu+AK)^2)$ $(0 \le k < N, 0 \le m < N+R)$ is the positive-series moment matrix of Theorem 3.4.

Proof. Substituting $c = -\sum_{j=1}^{n} a_{2j+1}\zeta(2j+1)$ from (5.1) into row k $(0 \le k < N)$ of the bottom block of $M_K(\mathbf{a}, c)$ and adding and subtracting $\sum_{j=1}^{n-1} a_{2n+1}\zeta(2j+1)S_{2j+1,\mathrm{bot}}[k,\bullet]$ gives

$$\begin{aligned} a_{2n+1}\mathcal{R}_{0,\mathrm{bot}}[k,\bullet] - cS_{2n+1,\mathrm{bot}}[k,\bullet] &= a_{2n+1}\mathcal{R}_{0,\mathrm{bot}}[k,\bullet] + \sum_{j=1}^{n} a_{2j+1}\zeta(2j+1)S_{2n+1,\mathrm{bot}}[k,\bullet] \\ &= a_{2n+1}\left(\sum_{j=1}^{n}\zeta(2j+1)S_{2j+1,\mathrm{bot}}[k,\bullet] + \mathcal{R}_{0,\mathrm{bot}}[k,\bullet]\right) \\ &\quad - \sum_{j=1}^{n-1}\zeta(2j+1)\left(a_{2n+1}S_{2j+1,\mathrm{bot}}[k,\bullet] - a_{2j+1}S_{2n+1,\mathrm{bot}}[k,\bullet]\right) \\ &= a_{2n+1}\mathcal{A}_{\mathrm{bot}}[k,\bullet] - \sum_{j=1}^{n-1}\zeta(2j+1)L_{\mathrm{top}}^{(2j+1)}[k,\bullet]. \end{aligned}$$

Since $0 \le k < N = A_0K + 1 < 2A_0K + 1 = R_{\mathrm{each}}$, the vector $L_{\mathrm{top}}^{(2j+1)}[k,\bullet]$ is row k of the j-th top block $L_{\mathrm{top}}^{(2j+1)}$ in $M_K(\mathbf{a}, c)$. Adding $\sum_{j=1}^{n-1}\zeta(2j+1)L_{\mathrm{top}}^{(2j+1)}[k,\bullet]$ from the top blocks to row k of the bottom block for each $0 \le k < N$ is a sequence of elementary row operations that leaves the determinant invariant and transforms $M_K(\mathbf{a}, c)$ into $\left(\begin{smallmatrix} L_{\mathrm{top}} \\ a_{2n+1}\mathcal{A}_{\mathrm{bot}} \end{smallmatrix}\right)$, proving (5.3). □

Theorem 5.3 (Block-Triangular Column Reduction, Homogeneous Scalar Cancellation, and Osculatory Decoupling)**.** *For every finite $K \ge 1$ with $A_0K, AK \in \mathbb{Z}_{\ge 1}$, define the $R \times N$ rational transition matrix*

$$\Gamma := \left(L_{\mathrm{top}}[:, J_{\mathrm{tail}}]\right)^{-1} L_{\mathrm{top}}[:, J_{\mathrm{head}}] \in \mathbb{Q}^{R\times N}, \tag{5.4}$$

and the N reduced kernel basis polynomials

$$Q_d(u) := \mathcal{E}_d(u) - \sum_{m=N}^{N+R-1} \Gamma_{m-N,d}\mathcal{E}_m(u) \in \ker(L_{\mathrm{top}}) \qquad (0 \le d < N). \tag{5.5}$$

Then:

(i) Finite-K Block-Triangular Determinant Reduction: *For every admissible $K \ge 1$, the Schur-complement determinant Δ_{N+R} factors algebraically as*

$$\begin{aligned} \Delta_{N+R} &= (-1)^{RN}\det\left(L_{\mathrm{top}}[:, J_{\mathrm{tail}}]\right) \cdot \det_{0\le k,d<N}\left(a_{2n+1}\mathcal{R}_0(\mathcal{E}_kQ_d) - cS_{2n+1}(\mathcal{E}_k\mathcal{E}_d)\right) \\ &= (-1)^{RN}a_{2n+1}^{R}\det\left(S_{\mathrm{odd,top}}[:, J_{\mathrm{tail}}]\right) \cdot \det_{0\le k,d<N}\left(a_{2n+1}\mathcal{A}(\mathcal{E}_kQ_d)\right). \end{aligned} \tag{5.6}$$

(ii) Homogeneous Scalar Cancellation and Outer-Barrier Convergence of Γ: *Every entry of L_{top} is homogeneous of degree 1 in the inner-pole weights $(w_{2n+1,0}, \dots, w_{2n+1,A_0K})$, so the global factorial prefactor $((2AK)!)^2((2A_0K)!)^{2n-1}$ and the central weight $w_{2n+1,0}$ cancel identically in $\Gamma = \left(L_{\mathrm{top}}[:, J_{\mathrm{tail}}]\right)^{-1}L_{\mathrm{top}}[:, J_{\mathrm{head}}]$ for every $K \ge 1$. If A_0K is held fixed while $AK \to \infty$, each weight ratio $w_{2n+1,r}/w_{2n+1,0}$ converges to $2(-1)^r\left(\binom{2A_0K}{A_0K+r}/\binom{2A_0K}{A_0K}\right)^{2n-1}$ with error $O((AK)^{-1})$, and $\Gamma(AK)$ converges entrywise and in operator norm to a finite*

rational limit with error $O((AK)^{-1})$. Furthermore, if the row multipliers are chosen from the dual basis of the order-$(2n-1)$ osculatory Hermite map $\Pi_{\mathrm{osc}} : \mathbb{Q}[u]_{\le R-1} \to \mathbb{Q}^R$,

$$\Pi_{\mathrm{osc}}(A) := \left(\left(\frac{A^{(q)}(0)}{q!} \right)_{0 \le q < n-1} ; \left(\frac{A^{(q)}(r^2)}{q!} \right)_{1 \le r \le A_0 K,\, 0 \le q < 2(n-1)} \right), \tag{5.7}$$

then both $L_{\mathrm{top}}[:, J_{\mathrm{head}}]$ and $L_{\mathrm{top}}[:, J_{\mathrm{tail}}]$ factor on the left through $D_{\mathrm{wt}} := \mathrm{diag}(w_{2n+1,0} I_{n-1}, w_{2n+1,1} I_{2(n-1)}, \dots, w_{2n+1,A_0K} I_{2(n-1)})$ at every finite $K \ge 1$.

Proof. To prove (i), fix any admissible $K \ge 1$ and multiply $M_K(\mathbf{a}, c)$ (or $\binom{L_{\mathrm{top}}}{a_{2n+1}\mathcal{A}_{\mathrm{bot}}}$) on the right by the upper unitriangular block column-operation matrix

$$U_\Gamma := \begin{pmatrix} I_N & 0_{N \times R} \\ -\Gamma & I_R \end{pmatrix} \in \mathbb{Q}^{(N+R)\times(N+R)}, \qquad \det U_\Gamma = 1.$$

In the top R rows, the head N columns become $L_{\mathrm{top}}[:, J_{\mathrm{head}}] - L_{\mathrm{top}}[:, J_{\mathrm{tail}}]\Gamma = 0_{R \times N}$ by (5.4), while the tail R columns remain $L_{\mathrm{top}}[:, J_{\mathrm{tail}}] = a_{2n+1} S_{\mathrm{odd,top}}[:, J_{\mathrm{tail}}]$. In the bottom N rows, column $d \in J_{\mathrm{head}}$ $(0 \le d < N)$ is evaluated on $\mathcal{E}_d(u) - \sum_{m=N}^{N+R-1} \Gamma_{m-N,d} \mathcal{E}_m(u) = Q_d(u)$, where $S_{2n+1}(\mathcal{E}_k Q_d) = S_{2n+1}(\mathcal{E}_k \mathcal{E}_d)$ since $S_{2n+1}(\mathcal{E}_k \mathcal{E}_m) = 0$ for all $m \ge N$. Taking the determinant of the resulting block anti-triangular matrix proves (5.6).

To prove (ii), observe from Lemma 3.2 (3.7) and Proposition 4.1(ii) (4.4) that at each inner node $r \in \{0, 1, \dots, A_0K\}$, every Laurent coefficient $g_{AK-r,q}$ $(0 \le q \le 2n-2)$ is a rational multiple of $g_{AK-r,0}$ by a polynomial of degree at most $q \le 2n-2$ in the logarithmic power sums $p_{AK-r,d}$ $(1 \le d \le 2n-2)$. Dividing both $L_{\mathrm{top}}[:, J_{\mathrm{tail}}]$ and $L_{\mathrm{top}}[:, J_{\mathrm{head}}]$ by the non-zero scalar $w_{2n+1,0} = g_{AK,0}$ cancels $w_{2n+1,0}$ (and hence $((2AK)!)^2((2A_0K)!)^{2n-1}$) identically in $\Gamma = (L_{\mathrm{top}}[:, J_{\mathrm{tail}}])^{-1} L_{\mathrm{top}}[:, J_{\mathrm{head}}]$. When $A_0K \in \mathbb{Z}_{\ge 1}$ is held fixed while $AK \to \infty$, for each $1 \le r \le A_0K$ we have

$$\begin{aligned} \frac{w_{2n+1,r}}{w_{2n+1,0}} &= 2(-1)^r \left(\frac{\binom{2A_0K}{A_0K+r}}{\binom{2A_0K}{A_0K}} \right)^{2n-1} \prod_{j=1}^{r} \left(\frac{AK-j+1}{AK+j} \right)^2 \\ &= 2(-1)^r \left(\frac{\binom{2A_0K}{A_0K+r}}{\binom{2A_0K}{A_0K}} \right)^{2n-1} (1 + O((AK)^{-1})), \end{aligned}$$

and each power sum $p_{AK-r,d}$ $(0 \le r \le A_0K, 1 \le d \le 2n-2)$ splits into an AK-independent inner sum over $|j - AK| \le A_0K$ and the symmetric outer-barrier sum over $s = |j - AK| \in \{A_0K+1, \dots, AK\}$ ($j = AK \mp s$, where $L_j = 2$ and $j - (AK - r) = r \mp s$):

$$p^{\mathrm{out}}_{AK-r,d}(AK) = 2 \sum_{s=A_0K+1}^{AK} \left(\frac{1}{(s-r)^d} + \frac{(-1)^d}{(s+r)^d} \right). \tag{5.8}$$

For $d = 1$, $(-1)^1 = -1$ and the symmetric outer poles cancel at leading order $O(s^{-1})$ via

$$\frac{1}{s-r} - \frac{1}{s+r} = \frac{2r}{s^2 - r^2} \qquad (s \ge A_0K + 1 > r \ge 0), \tag{5.9}$$

whose tail beyond $s = AK$ is bounded by $\sum_{s=AK+1}^{\infty} \frac{4A_0K}{(s-A_0K)^2} \le \frac{4A_0K}{AK - A_0K} = O((AK)^{-1})$. For all $2 \le d \le 2n-2$, $2|(s-r)^{-d} + (-1)^d (s+r)^{-d}| \le 4(s - A_0K)^{-2}$, whose tail beyond $s = AK$ is bounded by $\frac{4}{AK - A_0K} = O((AK)^{-1})$. Since $S_{\mathrm{odd,top}}[:, J_{\mathrm{tail}}]/w_{2n+1,0}$ converges to an invertible limit by Theorem 4.4 (4.24), Cramer's rule gives $\|\Gamma(AK) - \Gamma(\infty)\|_{\mathrm{op}} = O((AK)^{-1})$. Finally, for Π_{osc} in (5.7), setting $W_{\mathrm{osc},0}(u) := \prod_{r=1}^{A_0K} (u - r^2)^{2(n-1)}$ and $W_{\mathrm{osc},r_0}(u) := u^{n-1} \prod_{1 \le r \le A_0K, r \ne r_0} (u - r^2)^{2(n-1)}$ $(1 \le r_0 \le A_0K)$, the R dual basis polynomials in $\mathbb{Q}[u]_{\le R-1}$ are given explicitly for $0 \le q_0 < n-1$ at $u_0 = 0$ and for $0 \le q_0 < 2(n-1)$ at $1 \le r_0 \le A_0K$ by

$$H^{\mathrm{osc}}_{0,q_0}(u) := u^{q_0} W_{\mathrm{osc},0}(u) \sum_{s=0}^{n-2-q_0} \frac{1}{s!} \frac{d^s}{du^s} (W_{\mathrm{osc},0}(u)^{-1}) \Big|_{u=0} u^s,$$

$$H^{\mathrm{osc}}_{r_0,q_0}(u) := (u - r_0^2)^{q_0} W_{\mathrm{osc},r_0}(u) \sum_{s=0}^{2n-3-q_0} \frac{1}{s!} \frac{d^s}{du^s} (W_{\mathrm{osc},r_0}(u)^{-1}) \Big|_{u=r_0^2} (u - r_0^2)^s,$$

so $W_{\mathrm{osc},0}(u) \mid H^{\mathrm{osc}}_{0,q_0}(u)$ and $W_{\mathrm{osc},r_0}(u) \mid H^{\mathrm{osc}}_{r_0,q_0}(u)$, so for every column polynomial $\mathcal{E}_m(u)$ $(0 \le m < N+R)$ the product $H^{\mathrm{osc}}_{r_0,q_0}(u)\mathcal{E}_m(u)$ has vanishing derivatives up to order $n-1$ at $u=0$ and up to order $2(n-1)$ at every $r^2 \neq r_0^2$, decoupling node r_0 and factoring out D_{wt} on the left at every finite $K \ge 1$. □

Theorem 5.4 (Cauchy–Binet Series Expansion and Asymptotic Archimedean Rate)**.** *For $\nu \ge 1$, set $x_\nu := \nu + AK$ and $v_\nu := x_\nu^2 = (\nu + AK)^2$. Then:*

(i) Convergent Cauchy–Binet Series Expansion: *The determinant (5.3) admits the absolutely convergent series expansion*

$$\Delta_{N+R} = a_{2n+1}^N \sum_{1\le\nu_1<\nu_2<\cdots<\nu_N<\infty} \left(\prod_{j=1}^N W_K(\nu_j)\right) \det_{\substack{0\le k<N\\ 1\le j\le N}} (\mathcal{E}_k(v_{\nu_j}))\, \det V_{N+R}(L_{\mathrm{top}}; v_{\nu_1}, \dots, v_{\nu_N}), \tag{5.10}$$

where $V_{N+R}(L_{\mathrm{top}}; v_1, \dots, v_N)$ denotes the $(N+R)\times(N+R)$ matrix whose top R rows are L_{top} and whose bottom N rows are $(\mathcal{E}_m(v_j))_{1\le j\le N, 0\le m<N+R}$.

(ii) Asymptotic Archimedean Rate: *With $N = CK+1$ $(C = A_0)$, $R_{\mathrm{each}} = R_{c,\mathrm{each}}K + 1$, and $R = (n-1)R_{\mathrm{each}} = R_cK + n - 1$, the archimedean exponential rate $\Lambda^{(N+R)}_{\mathrm{arch}} := \limsup_{K\to\infty} \frac{1}{K^2}\log|\Delta_{N+R}|$ satisfies the variational upper bound*

$$\begin{aligned}\Lambda^{(N+R)}_{\mathrm{arch}} \le \sup_{\mu_c}\Big\{ & C^2 \iint K_{cc}(y_1,y_2)\, d\mu_c(y_1) d\mu_c(y_2) + C\int V_{\mathrm{ext}}(y)\, d\mu_c(y) \\ & + CR_c \int_0^{R_c}\int K_{pc}(\tau, y)\, \frac{d\tau}{R_c} d\mu_c(y)\Big\} \\ & + 2n(n-1)A_0^2 \iint_{[0,A_0]^2} K_{pp}(\alpha_1,\alpha_2)\, \frac{d\alpha_1}{A_0}\frac{d\alpha_2}{A_0} + R_c\int_0^{A_0} w_p(\alpha)\, \frac{d\alpha}{A_0} + C_{\mathrm{fac}},\end{aligned} \tag{5.11}$$

where the supremum is taken over probability measures μ_c on $[0,\infty)$,

$$\begin{aligned}C_{\mathrm{fac}} := -\Big[& C^2\log(2C) - \tfrac{3}{2}C^2 + (n-1)(R_{c,\mathrm{each}}^2 \log(2R_{c,\mathrm{each}}) - \tfrac{3}{2}R_{c,\mathrm{each}}^2) \\ & + (C+R_c)^2\log(2(C+R_c)) - \tfrac{3}{2}(C+R_c)^2\Big]\end{aligned} \tag{5.12}$$

is the normalized central factorial prefactor rate under the rescaling $u = K^2\tilde{u}$, and the five kernel and potential functions are defined for $y, y_1, y_2 \ge 0$, $\tau \in [0, R_c]$, and $\alpha, \alpha_1, \alpha_2 \in [0, A_0]$ by

$$K_{cc}(y_1,y_2) := \frac{1}{2}\log|y_1^2 - y_2^2| + \frac{1}{4}\log\Big((4A^2 + (y_1+y_2)^2)(4A^2 + (y_1-y_2)^2)\Big), \tag{5.13}$$

$$K_{pc}(\tau, y) := \frac{1}{2}\log\Big(((A-\tau)^2 + y^2)((A+\tau)^2 + y^2)\Big), \qquad K_{pp}(\alpha_1,\alpha_2) := \log|\alpha_1^2 - \alpha_2^2|, \tag{5.14}$$

$$V_{\mathrm{ext}}(y) := G^{[A_0,A]}_{2n+1,2}(y) - 2\pi y, \qquad \Phi_{\log}(u,y) := u\log(u^2+y^2) - 2u + 2y\arctan(u/y), \tag{5.15}$$

$$\begin{aligned}G^{[A_0,A]}_{2n+1,2}(y) := {} & 4A(\log(2A) - 1) + 2(2n-1)A_0(\log(2A_0) - 1) \\ & - \Phi_{\log}(2A, y) - \frac{2n-1}{2}(\Phi_{\log}(A+A_0, y) - \Phi_{\log}(A-A_0, y)),\end{aligned} \tag{5.16}$$

$$\begin{aligned}w_p(\alpha) := {} & 2(2A\log(2A) - (A-\alpha)\log(A-\alpha) - (A+\alpha)\log(A+\alpha)) \\ & + (2n-1)(2A_0\log(2A_0) - (A_0-\alpha)\log(A_0-\alpha) - (A_0+\alpha)\log(A_0+\alpha)).\end{aligned} \tag{5.17}$$

Proof. By Proposition 5.2 (5.3), each bottom-row entry equals $a_{2n+1}\sum_{\nu=1}^\infty W_K(\nu)\mathcal{E}_k(v_\nu)\mathcal{E}_m(v_\nu)$. Expanding Δ_{N+R} by multilinearity in the bottom N rows over all tuples $(\nu_1,\dots,\nu_N) \in \mathbb{Z}_{\ge 1}^N$ and grouping the $N!$ permutations of each strictly increasing N-tuple $1 \le \nu_1 < \cdots < \nu_N < \infty$ by skew-symmetry of the determinant yields (5.10).

To establish (5.11)–(5.17), expand $\det V_{N+R}(L_{\mathrm{top}}; v_{\nu_1}, \dots, v_{\nu_N})$ along the top R rows L_{top} by Laplace expansion:

$$\det V_{N+R}(L_{\mathrm{top}}; v_{\nu_1}, \dots, v_{\nu_N}) = \sum_{\substack{J^c \subset \{0,\dots,N+R-1\} \\ |J^c|=R}} (-1)^{\sigma(J^c)} \det L_{\mathrm{top}}[:, J^c] \cdot \det_{\substack{m\in J \\ 1\le j\le N}} \big(\mathcal{E}_m(v_{\nu_j})\big), \quad (5.18)$$

where $J = \{0, \dots, N+R-1\} \setminus J^c$. Applying the triangle inequality across the $\binom{N+R}{R} \le 2^{N+R} = e^{O(K)}$ summands gives

$$|\Delta_{N+R}| \le \binom{N+R}{R} \max_{|J^c|=R} \Big(\big|\det L_{\mathrm{top}}[:, J^c]\big| \cdot \mathcal{S}_K(J) \Big), \qquad \frac{1}{K^2} \log \binom{N+R}{R} = O(K^{-1}) \to 0. \quad (5.19)$$

Rescaling all variables by K via $u = K^2\tilde{u}$ and $t = K\tau$ in $\mathcal{E}_m(u) = \frac{2K^{2m}}{(2m)!} \prod_{t=0}^{m-1}(\tilde{u} - (t/K)^2)$, we extract the rescaled central factorial prefactors $\prod_{k=0}^{N-1} \frac{2K^{2k}}{(2k)!}$ from the bottom N rows, $(\prod_{i=0}^{R_{\mathrm{each}}-1} \frac{2K^{2i}}{(2i)!})^{n-1}$ from the $n-1$ top row blocks, and $\prod_{m=0}^{N+R-1} \frac{2K^{2m}}{(2m)!} = (\prod_{m\in J^c} \frac{2K^{2m}}{(2m)!})(\prod_{m\in J} \frac{2K^{2m}}{(2m)!})$ from the $N+R$ columns. By Stirling's formula $\log((2m)!) = 2m\log(2m) - 2m + O(\log m)$, for any scaling constant $D > 0$ and $m = \mu K$:

$$\lim_{K\to\infty} \frac{1}{K^2} \log \prod_{m=0}^{\lfloor DK \rfloor - 1} \frac{2K^{2m}}{(2m)!} = -\int_0^D 2\mu(\log(2\mu) - 1)\, d\mu = -\left(D^2 \log(2D) - \frac{3}{2} D^2 \right),$$

which sums across $D \in \{C, R_{c,\mathrm{each}}, C+R_c\}$ to C_{fac} in (5.12), leaving the rescaled monic polynomials $\widetilde{\mathcal{E}}_m^{\mathrm{resc}}(\tilde{u}) := \prod_{t=0}^{m-1}(\tilde{u} - (t/K)^2)$ inside $\widetilde{L}_{\mathrm{top}}[:, J^c]$ and $\widetilde{\mathcal{S}}_K(J)$. We now evaluate each term of (5.11):

(1) *Top-block inner-pole weight $w_p(\alpha)$ and exact unitriangular confluent Vandermonde rate $K_{pp}(\alpha_1, \alpha_2)$:* On the anchor partition $J_{\max} := \{R, R+1, \dots, R+N-1\}$ and $J^c_{\max} := \{0, 1, \dots, R-1\}$, the R column polynomials $\widetilde{\mathcal{E}}_0^{\mathrm{resc}}(\tilde{u}), \widetilde{\mathcal{E}}_1^{\mathrm{resc}}(\tilde{u}), \dots, \widetilde{\mathcal{E}}_{R-1}^{\mathrm{resc}}(\tilde{u})$ are monic polynomials of degrees $0, 1, \dots, R-1$, so the change-of-basis matrix from the monic power basis $\{1, \tilde{u}, \dots, \tilde{u}^{R-1}\}$ to $\{\widetilde{\mathcal{E}}_m^{\mathrm{resc}}(\tilde{u})\}_{m=0}^{R-1}$ is upper unitriangular with determinant 1. Likewise, in each of the $n-1$ top row blocks $L_{\mathrm{top}}^{(2j+1)}$ $(1 \le j \le n-1)$, the rescaled row polynomials $\{\widetilde{\mathcal{E}}_i^{\mathrm{resc}}(\tilde{u})\}_{i=0}^{2A_0K}$ are monic of degrees $0, 1, \dots, 2A_0K$, so the row transition from the Hermite basis $\{H_0, H_{r,0}, H_{r,1}\}$ of Lemma 4.2 to $\{\widetilde{\mathcal{E}}_i^{\mathrm{resc}}(\tilde{u})\}_{i=0}^{2A_0K}$ has determinant $(\det V_{\mathrm{Herm}}^{\mathrm{monic}})^{n-1}$ across all $n-1$ blocks, where by (4.26) $\det V_{\mathrm{Herm}}^{\mathrm{monic}} = (\prod_{r=1}^{A_0K} (r/K)^4) \prod_{1\le s<r\le A_0K} ((r/K)^2 - (s/K)^2)^4$. Ordering the row blocks by $m = n - j \in \{1, \dots, n-1\}$ and eliminating lower derivative jets triangularly as in Step (3) of Theorem 4.4 projects the R rows onto the R confluent jet functionals $\delta_{0,q}$ $(0 \le q < n-1)$ and $\delta_{(r/K)^2,q}$ $(0 \le q < 2(n-1), 1 \le r \le A_0K)$ with diagonal pole-weight product $|w_{2n+1,0}|^{n-1} \prod_{r=1}^{A_0K} |w_{2n+1,r}|^{2(n-1)} (2r/K)^{2n(n-1)}$, while evaluating those R confluent jet functionals on the unitriangular column basis $\{\widetilde{\mathcal{E}}_m^{\mathrm{resc}}(\tilde{u})\}_{m=0}^{R-1}$ gives $\det V_Q^{\mathrm{monic}} = (\prod_{r=1}^{A_0K} (r/K)^{4(n-1)^2}) \prod_{1\le s<r\le A_0K} ((r/K)^2 - (s/K)^2)^{4(n-1)^2}$ by (4.27). Multiplying the row transition factor $(\det V_{\mathrm{Herm}}^{\mathrm{monic}})^{n-1}$ and the column confluent Vandermonde factor $\det V_Q^{\mathrm{monic}}$ gives on each unordered pair $1 \le s < r \le A_0K$ the exact exponent

$$4(n-1) + 4(n-1)^2 = 4n(n-1).$$

Converting the unordered pair sum $\sum_{1\le s<r\le A_0K} 4n(n-1)\log|(r/K)^2 - (s/K)^2|$ (and the diagonal r-power sum $\sum_{r=1}^{A_0K} (4(n-1) + 2n(n-1) + 4(n-1)^2)\log(r/K) = \sum_{r=1}^{A_0K} 2n(n-1)\log((r/K)^2)$) by Euler–Maclaurin summation to the ordered double integral over $[0, A_0]^2$ introduces the symmetry factor $\frac{1}{2}$, yielding $\frac{1}{2} \cdot 4n(n-1)A_0^2 = 2n(n-1)A_0^2$ multiplying $\iint_{[0,A_0]^2} K_{pp}(\alpha_1, \alpha_2) \frac{d\alpha_1}{A_0} \frac{d\alpha_2}{A_0}$. Moreover, for any other R-element subset $J^c = \{m_1 < \cdots < m_R\} \ne J^c_{\max}$ of total degree shift $D := \sum_{s=1}^{R} (m_s - (s-1)) \ge 1$, Schur's bialternant formula expresses the generalized confluent Vandermonde determinant on J^c as $\det V_Q^{\mathrm{monic}}$ times the confluent Schur polynomial $s_\lambda(\tilde{u}_1, \dots, \tilde{u}_R)$ of degree D, which consists of at most $\binom{R+D-1}{D}$ monomials each bounded on $[0, A_0^2]$ by A_0^{2D}, while in $\widetilde{\mathcal{S}}_K(J)$ those D degree steps remove D

factors bounded below on $\tilde{u}_\nu \geq A^2$ by $(A^2-c_0^2)^D$; since $\frac{A_0^2}{A^2-c_0^2} \leq \frac{1}{52.25} < 1$, every $J^c \neq J^c_{\max}$ has strictly smaller exponential rate than $J^c_{\max}$. Finally, Stirling's formula applied to $|g_{AK-r,0}| = \binom{2AK}{AK-r}^2 \binom{2A_0K}{A_0K+r}^{2n-1}$ at $r = \alpha K$ yields $\frac{1}{K}\log|g_{AK-r,0}| = w_p(\alpha)+O(K^{-1}\log K)$ as in (5.17).

(2) *Contour/Laplace representation of $\widetilde{\mathcal{S}}_K(J_{\max})$ and external potential $V_{\text{ext}}(y)$:* On the vertical line $\Gamma_K := \{z = \frac{1}{2} + iyK : y \in \mathbb{R}\}$ (which passes strictly to the right of all poles $\{-2AK, \dots, 0\}$ with distance $\frac{1}{2}$ from both $\{-2AK, \dots, 0\}$ and $\mathbb{Z}_{\geq 1}$), Cauchy's residue theorem and Andréief's multi-integral identity express $\widetilde{\mathcal{S}}_K(J_{\max})$ as an N-fold integral over $y_1, \dots, y_N \in \mathbb{R}$ with $\tilde{u}_j = (A + \frac{1}{2K} + iy_j)^2$. On $z = \frac{1}{2} + iyK$, $|\pi\cot(\pi(\frac{1}{2} + iyK))| = \pi|\tanh(\pi yK)| \leq \pi$, and by Euler–Maclaurin summation and Stirling's formula on $((2AK)!)^2((2A_0K)!)^{2n-1}$, $\frac{1}{K}\log|\mathcal{F}^{\text{odd},[A_0,A]}_{2n+1,2}(\frac{1}{2} + iyK; 1)| = G^{[A_0,A]}_{2n+1,2}(|y|) + O(K^{-1}\log K)$, while the residue series decay $e^{-2\pi|y|K}$ combines with $G^{[A_0,A]}_{2n+1,2}(|y|)$ to give $V_{\text{ext}}(|y|) = G^{[A_0,A]}_{2n+1,2}(|y|) - 2\pi|y|$ in (5.15)–(5.16) (and on the positive real series $x = (\nu + AK)/K \geq A$, the real logarithmic potential is bounded above by the same endpoint $G^{[A_0,A]}_{2n+1,2}(0) = V_{\text{ext}}(0)$).

(3) *Contour-contour kernel $K_{cc}(y_1, y_2)$ and pole-contour kernel $K_{pc}(\tau, y)$:* For $J_{\max} = \{R, \dots, R+N-1\}$, factoring the common prefix $W^{\text{resc}}_R(\tilde{u}_j) = \prod_{t=0}^{R-1}(\tilde{u}_j - (t/K)^2)$ from each row $1 \leq j \leq N$ of $\det_{0\leq d<N}(\widetilde{\mathcal{E}}^{\text{resc}}_{R+d}(\tilde{u}_j))$ leaves the Vandermonde determinant $\det_{0\leq d<N}(\tilde{u}_j^d) = \prod_{1\leq a<b\leq N}(\tilde{u}_b - \tilde{u}_a)$. Pairing $\det_{0\leq k<N}(\tilde{u}_j^k)$ and $\det_{0\leq d<N}(\tilde{u}_j^d)$ gives $\frac{1}{2}\log|(A+iy_1)^2-(A+iy_2)^2|+\frac{1}{2}\log|(A+iy_1)^2-(A-iy_2)^2| = K_{cc}(y_1, y_2)$ in (5.13), while the $R = R_cK + n - 1$ roots $\tau = t/K \in [0, R_c]$ of $W^{\text{resc}}_R(\tilde{u}_j)$ give $\log|(A+iy)^2 - \tau^2| = K_{pc}(\tau, y)$ in (5.14).

By weak-$*$ compactness of $\mathcal{P}([0, Y_{\max}])$ and the upper-envelope theorem for logarithmic potentials [21, Theorem I.6.8], (5.11) follows. □

6. p-Adic Valuations, Harmonic Centering, and Net Exponent Bounds

We now analyze the p-adic valuation of the entries and determinant of $M_K(\mathbf{a}, c)$ across all primes p and construct the determinant-clearing rational multiplier $D^{\text{eff}}_{N+R} \in \mathbb{Q}_{>0}$. Throughout this section, for any logical condition $\mathcal{P}$, we write $\mathbf{1}_{\mathcal{P}} := 1$ if $\mathcal{P}$ holds and $\mathbf{1}_{\mathcal{P}} := 0$ otherwise.

Lemma 6.1 (Entry-Wise p-Adic Valuation Lower Bound)**.** *Let p be a prime, and let $0 \leq m < N+R$ with $2m \leq 2AK$. For each pole $k \in \{0, 1, \dots, 2AK\}$ of multiplicity $L_k \in \{2, 2n+1\}$ with $r = |AK - k|$, define the base kernel factorial valuation at k:*

$$V_p(k) := \begin{cases} 2v_p\left(\binom{2AK}{k}\right) + (2n-1)v_p\left(\binom{2A_0K}{A_0K+r}\right), & |k - AK| \leq A_0K, \\ 2v_p\left(\binom{2AK}{k}\right) - (2n-1)v_p\left((r - A_0K)\binom{r+A_0K}{2A_0K}\right), & A_0K < |k - AK| \leq AK. \end{cases} \tag{6.1}$$

Then the p-adic valuation of every entry in row ℓ (with row multiplier $\mathcal{E}_i$) and column m (with column multiplier $\mathcal{E}_m$) of $M_K(\mathbf{a}, c)$ satisfies

$$v_p(M_K(\mathbf{a}, c)_{\ell,m}) \geq \min_{0\leq k\leq 2AK} \Bigg\{ V_p(k) - L_k \left\lfloor \frac{\log(2AK)}{\log p} \right\rfloor - \mathbf{1}_{i>r}\, v_p\left(\frac{2i(2i-1)}{1+\delta_{r,0}}\binom{2i-2}{i-1+r}\right) - \mathbf{1}_{m>r}\, v_p\left(\frac{2m(2m-1)}{1+\delta_{r,0}}\binom{2m-2}{m-1+r}\right) \Bigg\}, \tag{6.2}$$

where on the top R rows L_{top} the minimum in (6.2) runs only over the inner poles $|k-AK| \leq A_0K$.

Proof. Fix a row ℓ (with row polynomial $\mathcal{E}_i, 0 \leq i \leq 2A_0K$) and a column m (with column polynomial $\mathcal{E}_m, 0 \leq m < N+R$). Since $a_3, \dots, a_{2n+1}, c \in \mathbb{Z}$ have $v_p(a_{2j+1}) \geq 0$ and $v_p(c) \geq 0$, every entry $M_K(\mathbf{a}, c)_{\ell,m}$ is a $\mathbb{Z}$-linear combination of $A_{t,k}(\mathcal{E}_i\mathcal{E}_m)$ ($|k - AK| \leq A_0K, t \in \{2j+1, 2n+1\}$)

on the top R rows, and of $A_{t,k}(\mathcal{E}_i\mathcal{E}_m)H_k^{(t)}$ $(0 \le k \le 2AK, 1 \le t \le L_k)$ and $A_{2n+1,k}(\mathcal{E}_i\mathcal{E}_m)$ on the bottom N rows.

Fix a pole $k \in \{0, 1, \dots, 2AK\}$ with $r = |AK - k|$ and $w = z + k$, and fix $1 \le t \le L_k$. By Definition 3.1 and Lemma 3.2, $A_{t,k}(\mathcal{E}_i\mathcal{E}_m)$ is the coefficient of $w^{L_k - t}$ in the Cauchy product of $g_k(w)$, $\mathcal{E}_i((w + AK - k)^2)$, and $\mathcal{E}_m((w + AK - k)^2)$:

$$A_{t,k}(\mathcal{E}_i\mathcal{E}_m)H_k^{(t)} = \sum_{q_0+d_1+d_2=L_k-t} g_{k,q_0}\Big([w^{d_1}]\mathcal{E}_i((w + AK - k)^2)\Big)\Big([w^{d_2}]\mathcal{E}_m((w + AK - k)^2)\Big)H_k^{(t)}. \tag{6.3}$$

We bound the p-adic valuation of each of the four factors in every summand $q_0 + d_1 + d_2 = L_k - t$ of (6.3):

(1) *Kernel Taylor factor g_{k,q_0}:* By Lemma 3.2 (3.6), the leading term $g_{k,0}$ has valuation $v_p(g_{k,0}) = V_p(k)$ as defined in (6.1). Since $g_k(w) = g_{k,0}\prod_{j\neq k}(1 + \frac{w}{j-k})^{-L_j}$ and every pole difference satisfies $1 \le |j - k| \le 2AK$ (so $v_p(j - k) \le \lfloor \frac{\log(2AK)}{\log p} \rfloor$) while the negative binomial coefficients $\binom{L_j+a-1}{a}$ are integers, the coefficient $g_{k,q_0} = [w^{q_0}]g_k(w)$ satisfies $v_p(g_{k,q_0}) \ge V_p(k) - q_0\lfloor \frac{\log(2AK)}{\log p} \rfloor$.

(2) *Row and column polynomial factors $[w^{d_1}]\mathcal{E}_i$ and $[w^{d_2}]\mathcal{E}_m$:* By Theorem 2.5 (2.23) and Proposition 2.3 (2.8), since $2i < 2m \le 2AK$, for every $d_1, d_2 \ge 0$ we have

$$v_p\Big([w^{d_1}]\mathcal{E}_i((w + AK - k)^2)\Big) \ge -\mathbf{1}_{i>r}\, v_p\left(\frac{2i(2i-1)}{1+\delta_{r,0}}\binom{2i-2}{i-1+r}\right) - d_1\left\lfloor \frac{\log(2AK)}{\log p}\right\rfloor,$$

$$v_p\Big([w^{d_2}]\mathcal{E}_m((w + AK - k)^2)\Big) \ge -\mathbf{1}_{m>r}\, v_p\left(\frac{2m(2m-1)}{1+\delta_{r,0}}\binom{2m-2}{m-1+r}\right) - d_2\left\lfloor \frac{\log(2AK)}{\log p}\right\rfloor.$$

(3) *Harmonic factor $H_k^{(t)}$:* For $H_k^{(t)} = \sum_{\ell=1}^{k} \ell^{-t}$ $(k \le 2AK)$, every denominator $\ell \in \{1, \dots, k\}$ satisfies $v_p(\ell) \le \lfloor \frac{\log(2AK)}{\log p} \rfloor$, so $v_p(H_k^{(t)}) \ge -t\lfloor \frac{\log(2AK)}{\log p} \rfloor$.

Summing the four valuations for any (q_0, d_1, d_2) with $q_0 + d_1 + d_2 = L_k - t$, the coefficients of $\lfloor \frac{\log(2AK)}{\log p} \rfloor$ add to $-(q_0 + d_1 + d_2 + t) = -L_k$ identically. Applying the ultrametric inequality across all summands and all poles $0 \le k \le 2AK$ (restricted to $|k - AK| \le A_0K$ on L_{top} by Theorem 3.3(ii)) establishes (6.2). □

Proposition 6.2 (Inner-Window p-Adic Integrality)**.** *Let p be a prime, and let $k \in [AK - A_0K, AK + A_0K]$ be an inner pole with $r = AK - k$ $(|r| \le A_0K)$. Decompose the smooth kernel factor $g_k(w) = w^{2n+1}\mathcal{F}_{2n+1,2}^{\text{odd},[A_0,A]}(-k + w; 1)$ of (3.5) as the product*

$$g_k(w) = U_{\text{out},k}(w) \cdot U_{\text{in},k}(w), \tag{6.4}$$

where

$$U_{\text{out},k}(w) := \frac{((2AK)!)^2}{\prod_{\substack{0\le j\le 2AK\\ j\neq k}}(w + j - k)^2}, \qquad U_{\text{in},k}(w) := \frac{((2A_0K)!)^{2n-1}}{\prod_{\substack{|j-AK|\le A_0K\\ j\neq k}}(w + j - k)^{2n-1}}. \tag{6.5}$$

Suppose that k is the unique inner pole in its residue class modulo p, meaning that

$$\left\lfloor \frac{A_0K - r}{p}\right\rfloor + \left\lfloor \frac{A_0K + r}{p}\right\rfloor = 0 \tag{6.6}$$

(which holds for all $|r| \le A_0K$ whenever $p > 2A_0K$, and for $|r| < p - A_0K$ whenever $A_0K < p \le 2A_0K$). Then for every integer $q \ge 0$, the q-th Taylor coefficient $[w^q]U_{\text{in},k}(w) = \frac{1}{q!}U_{\text{in},k}^{(q)}(0)$ satisfies

$$v_p([w^q]U_{\text{in},k}(w)) \ge (2n-1)v_p\left(\binom{2A_0K}{A_0K + r}\right) = (2n-1)\left\lfloor \frac{2A_0K}{p}\right\rfloor \ge 0 \qquad (\forall\, q \ge 0). \tag{6.7}$$

Consequently, whenever (6.6) holds, the Taylor coefficients $g_{k,q}$ inherit their p-adic differentiation denominators solely from the outer factor $U_{\text{out},k}(w)$ of multiplicity $L_{\text{out}} = 2$.

Proof. Factor out $\prod_{|j-AK|\le A_0K, j\neq k}(j-k)^{2n-1} = ((-1)^{A_0K-r}(A_0K+r)!(A_0K-r)!)^{2n-1}$ from the denominator of $U_{\mathrm{in},k}(w)$ in (6.5) to write

$$U_{\mathrm{in},k}(w) = (-1)^{A_0K-r}\binom{2A_0K}{A_0K+r}^{2n-1} \prod_{\substack{|j-AK|\le A_0K \\ j\neq k}} \left(1+\frac{w}{j-k}\right)^{-(2n-1)}. \tag{6.8}$$

As j ranges over $[AK-A_0K, AK+A_0K]\setminus\{k\}$, the difference $s := j-k = r-(AK-j)$ ranges over the non-zero integers in $[-(A_0K-r), A_0K+r]\setminus\{0\}$. The number of non-zero multiples of p in this interval is $\lfloor\frac{A_0K-r}{p}\rfloor + \lfloor\frac{A_0K+r}{p}\rfloor = 0$ by (6.6). Thus $p\nmid(j-k)$ and $(j-k)^{-1}\in\mathbb{Z}_{(p)}^{\times}$ is a p-adic unit for every inner pole $j\neq k$. Expanding each factor in (6.8) by the negative binomial series gives

$$\left(1+\frac{w}{j-k}\right)^{-(2n-1)} = \sum_{d=0}^{\infty}(-1)^d\binom{2n-2+d}{d}\frac{w^d}{(j-k)^d} \in \mathbb{Z}_{(p)}[[w]].$$

Because $\mathbb{Z}_{(p)}[[w]]$ is closed under Cauchy multiplication, the product over $j\neq k$ in (6.8) belongs to $\mathbb{Z}_{(p)}[[w]]$. Moreover, by (6.6) we have $A_0K-r<p$ and $A_0K+r<p$ (so $2A_0K<2p\le p^2$), whence $v_p(\binom{2A_0K}{A_0K+r}) = \lfloor\frac{2A_0K}{p}\rfloor$, proving (6.7). □

Theorem 6.3 (Midpoint Harmonic Centering)**.** *Fix $n\ge 1$, $A_0K\ge 1$, and $AK>A_0K$. For each $1\le t\le 2n+1$ and $0\le k\le 2AK$, define the midpoint-centered harmonic number*

$$\widetilde{H}_{k,AK}^{(t)} := H_k^{(t)} - H_{AK}^{(t)} = \begin{cases} +\sum_{\ell=AK+1}^{k}\frac{1}{\ell^t}, & k>AK, \\ 0, & k=AK, \\ -\sum_{\ell=k+1}^{AK}\frac{1}{\ell^t}, & k<AK, \end{cases} \tag{6.9}$$

and the centered harmonic remainder

$$\mathcal{R}_0^{\mathrm{cent}}(P) := -\sum_{k=0}^{2AK}\sum_{t=1}^{L_k} A_{t,k}(P)\,\widetilde{H}_{k,AK}^{(t)}. \tag{6.10}$$

Then:

(i) Polynomial and Determinant Identity: *For every polynomial $P(u)\in\mathbb{Q}[u]$ satisfying (3.1),*

$$\mathcal{R}_0(P) = \mathcal{R}_0^{\mathrm{cent}}(P) - \sum_{j=1}^{n} H_{AK}^{(2j+1)} S_{2j+1}(P). \tag{6.11}$$

Setting $c_{\mathrm{cent}} := c + \sum_{j=1}^{n} a_{2j+1}H_{AK}^{(2j+1)}$, adding $\sum_{j=1}^{n-1} H_{AK}^{(2j+1)} L_{\mathrm{top}}^{(2j+1)}[k,\bullet]$ from the top R rows to row k of the bottom N rows of $M_K(\mathbf{a},c)$ $(0\le k<N)$ is an elementary row operation over $\mathbb{Q}$ that leaves $\Delta_{N+R} = \det M_K(\mathbf{a},c)$ invariant and replaces $a_{2n+1}\mathcal{R}_{0,\mathrm{bot}} - cS_{2n+1,\mathrm{bot}}$ by $a_{2n+1}\mathcal{R}_{0,\mathrm{bot}}^{\mathrm{cent}} - c_{\mathrm{cent}}S_{2n+1,\mathrm{bot}}$.

(ii) Support of Inner Centered Harmonic Numbers: *For every inner pole $k\in[AK-A_0K, AK+A_0K]$ and every $1\le t\le 2n+1$, $\widetilde{H}_{k,AK}^{(t)}$ sums exclusively over $\ell\in[AK-A_0K+1, AK+A_0K]$ (which does not involve any integer $\ell\in[1, AK-A_0K]$). In particular, for every prime $p>2A_0K$, the interval $[AK-A_0K+1, AK+A_0K]$ of length $2A_0K<p$ contains at most a single multiple of p, and contains zero multiples of p whenever $\lfloor\frac{(A+A_0)K}{p}\rfloor = \lfloor\frac{(A-A_0)K}{p}\rfloor$ (which holds for all $p>(A+A_0)K$ and on every subinterval $p/K\in(\frac{A+A_0}{m}, \frac{A-A_0}{m-1}]$ for $m\ge 2$).*

Proof. Substituting $H_k^{(t)} = H_{AK}^{(t)} + \widetilde{H}_{k,AK}^{(t)}$ into (3.14) and interchanging the finite sums over k and t gives $\mathcal{R}_0(P) = \mathcal{R}_0^{\mathrm{cent}}(P) - \sum_{t=1}^{2n+1} H_{AK}^{(t)} S_t(P)$. By Theorem 3.3(i) (3.10), $S_1(P)=0$ and $S_{2j}(P)=0$ for all $1\le j\le n$, leaving only $t=2j+1$ $(1\le j\le n)$, which proves (6.11). Substituting (6.11) into row k $(0\le k<N)$ of the bottom block of $M_K(\mathbf{a},c)$ and using $a_{2n+1}S_{2j+1} = L_{\mathrm{top}}^{(2j+1)} + a_{2j+1}S_{2n+1}$ proves (i). Part (ii) follows directly from (6.9), since for $k\in[AK-A_0K, AK+A_0K]$ every index

ℓ between $AK+1$ and k (or $k+1$ and AK) lies in $[AK-A_0K+1, AK+A_0K]$, which contains no multiple of p whenever $\lfloor\frac{(A+A_0)K}{p}\rfloor = \lfloor\frac{(A-A_0)K}{p}\rfloor$. □

Proposition 6.4 (Outer-Pole Binomial Divisibility, Boundary Cancellation, and Elimination of Tail Primes)**.** *In the continuum prime scaling $p = xK$, $k = (\frac{A}{x}+w)p$ ($r = |AK-k| = p|w|$), and $m = \mu K$:*

(i) Outer-Pole Binomial Divisibility and $o(K^2)$ Boundary-Layer Error: *For every outer pole $k \in \{0,\dots,2AK\}$ with $r = |AK-k| \in \{A_0K+1,\dots,AK\}$, the inner-window denominator factor in (3.6) and (6.1) satisfies the identities*

$$(r-A_0K)\binom{r+A_0K}{2A_0K} = (2A_0K+1)\binom{r+A_0K}{2A_0K+1} \;\Big|\; \operatorname{lcm}(1,2,\dots,(A+A_0)K), \tag{6.12}$$

and is linked to the first tail polynomial $\mathcal{E}_{A_0K+1}(u)$ at $u = r^2$ by

$$\mathcal{E}_{A_0K+1}(r^2) = \frac{2r}{(2A_0K+1)(2A_0K+2)}\cdot(r-A_0K)\binom{r+A_0K}{2A_0K} \in \mathbb{Z}_{\geq 1}. \tag{6.13}$$

For any real $\alpha > \beta \geq 0$, define the inner-window boundary floor function

$$\Omega_{\mathrm{bdry}}(\alpha,\beta) := \lfloor\alpha+\beta\rfloor - \lfloor\alpha-\beta\rfloor - \lfloor 2\beta\rfloor \in \{0,1\}. \tag{6.14}$$

At any outer pole $A_0K < r \leq AK$, Legendre's formula applied to (6.1) expands $v_p(\binom{r+A_0K}{2A_0K}) = \sum_{\ell\geq 1}\Omega_{\mathrm{bdry}}(\frac{r}{p^\ell},\frac{A_0K}{p^\ell})$. At level $\ell = 1$, the boundary loss $-(2n-1)\Omega_{\mathrm{bdry}}(\frac{r}{p},\frac{A_0K}{p}) \geq -(2n-1)$ combines with $-L_{\mathrm{out}} = -2$ to give at least $-(2n+1) = -L_{\mathrm{in}}$ (and equals 0 whenever $\lfloor\frac{r+A_0K}{p}\rfloor = \lfloor\frac{r-A_0K}{p}\rfloor$, in particular for all $p > (A+A_0)K$). Across all $N+R$ columns and all primes $p \leq 2AK$, the contribution to $\log D^{\mathrm{eff}}_{N+R}$ from $v_p(r-A_0K)$, from prime powers $p^\ell \leq 2AK$ ($\ell \geq 2$) of the kernel binomial factors, and from the floor-jump boundary layers $xw = A_0, xw = A$, and $\frac{A\pm xw}{x}, \frac{A_0\pm xw}{x} \in \mathbb{Z}$ is bounded by

$$E_{\mathrm{bdry}}(K) \leq C_1(n;A,c_0)K^{3/2}\log K + C_2(n;A,c_0)K(\log K)^2 = o(K^2) \qquad (K\to\infty), \tag{6.15}$$

with $C_1(n;A,c_0) := 2(c_0+0.01)\sqrt{2A}\,(2n+3)$ and $C_2(n;A,c_0) := 40A(2n+3)$.

(ii) Elimination of Tail Primes $p > 2AK$ and Cancellation of Central Factorial Prefactors: *Whenever $A > C+R_c = (2n-1)A_0$, every row index $0 \leq i < R_{\mathrm{each}}$ and every column index $0 \leq m < N+R = (C+R_c)K+n$ satisfies $2i < 2m \leq 2AK$ for all sufficiently large K. Consequently, for every prime $p > 2AK$, we have $p > 2m$ and $p > 2AK$, so every binomial and harmonic valuation in (6.2) is zero, giving $v_p(M_K(\mathbf{a},c)_{\ell,m}) \geq 0$ for all ℓ, m and all $p > 2AK$. Furthermore, for every integer $1 \leq m < N+R$, every prime p dividing $\frac{(2m)!}{2}$ satisfies $p \leq 2m \leq 2AK$, so $\prod_{p\leq 2AK} p^{v_p((2m)!/2)} = \frac{(2m)!}{2}$; under the rescaling $u = K^2\tilde{u}$ (which pairs the K^{2m} rescaling factor with the small-prime Vandermonde counterterm), the p-adic prefactor denominators over $p \leq 2AK$ and the archimedean prefactor rate $+C_{\mathrm{fac}}(n)$ in (5.11) cancel identically.*

Proof. To prove (i), write $(r-A_0K)\binom{r+A_0K}{2A_0K} = \frac{(r+A_0K)!}{(2A_0K)!(r-A_0K-1)!} = (2A_0K+1)\binom{r+A_0K}{2A_0K+1}$, and since $\mathcal{E}_{A_0K+1}(r^2) = \frac{2}{(2A_0K+2)!}r^2\prod_{s=1}^{A_0K}(r^2-s^2) = \frac{2r}{(2A_0K+1)(2A_0K+2)}(r-A_0K)\binom{r+A_0K}{2A_0K}$, we obtain (6.13). Moreover, for any integers $1 \leq m_1 \leq M_1$, in $m_1\binom{M_1}{m_1} = \frac{M_1(M_1-1)\cdots(M_1-m_1+1)}{(m_1-1)!}$, no factor in the numerator is divisible by p^ℓ for $\ell > \lfloor\log M_1/\log p\rfloor$, while for each $1 \leq s \leq \lfloor\log M_1/\log p\rfloor$ the number of multiples of p^s among the m_1 consecutive integers $\{M_1-m_1+1,\dots,M_1\}$ exceeds $\lfloor(m_1-1)/p^s\rfloor$ by at most 1; hence $v_p(m_1\binom{M_1}{m_1}) \leq \lfloor\log M_1/\log p\rfloor = v_p(\operatorname{lcm}(1,\dots,M_1))$, proving (6.12) at $m_1 = 2A_0K+1$ and $M_1 = r+A_0K \leq (A+A_0)K$. Setting $X := \alpha-\beta > 0$ and $Y := 2\beta \geq 0$, the floor inequality $0 \leq \lfloor X+Y\rfloor - \lfloor X\rfloor - \lfloor Y\rfloor \leq 1$ gives $\Omega_{\mathrm{bdry}}(\alpha,\beta) \in \{0,1\}$. The bounds $E^{(1)}_{\mathrm{bdry}}(K) \leq C_1(n;A,c_0)K^{3/2}\log K$ for prime powers $p^\ell \leq 2AK$ ($\ell \geq 2$) and $E^{(2)}_{\mathrm{bdry}}(K) \leq C_2(n;A,c_0)K(\log K)^2$ for the $O(AK/p)$ jump points summed via Mertens' bound

$\sum_{p\le 2AK}\frac{\log p}{p}\le 2\log K$ establish (6.15). Part (ii) follows immediately from $2(N+R-1)\le 2AK$ for $A>C+R_c$ and the unique prime factorization $\frac{(2m)!}{2}=\prod_{p\le 2AK}p^{v_p((2m)!/2)}$. □

Theorem 6.5 (Effective Determinant-Clearing Multiplier and Uniform Negativity Across All $n\ge 1$)**.** *Factor out the scalar central factorial prefactors* $P_{\mathrm{fac}}:=\prod_{\ell=0}^{N+R-1}\frac{2}{(2i_\ell)!}\prod_{m=0}^{N+R-1}\frac{2}{(2m)!}$ *from the rows and columns of* $M_K(\mathbf{a},c)$ *by multilinearity of the determinant:*

$$\Delta_{N+R}=P_{\mathrm{fac}}\cdot\det M_K^{\mathrm{monic}}(\mathbf{a},c),\qquad P_{\mathrm{fac}}=\prod_{p\le 2AK}p^{v_p(P_{\mathrm{fac}})}\quad(\forall\,K\ \text{with}\ 2(N+R-1)\le 2AK). \tag{6.16}$$

For each prime $p\le 2AK$*, let* $-v_{p,\mathrm{occ}}(K)\in\mathbb{Z}$ *be the* p*-adic valuation lower bound on* $\det M_K^{\mathrm{monic}}(\mathbf{a},c)$ *obtained by residue-class* p*-adic Gaussian elimination over* $w\pmod p$*, and define the determinant-clearing rational multiplier*

$$D_{N+R}^{\mathrm{eff}}:=P_{\mathrm{fac}}^{-1}\prod_{p\le 2AK}p^{v_{p,\mathrm{occ}}(K)}=\prod_{p\le 2AK}p^{-v_p(P_{\mathrm{fac}})+v_{p,\mathrm{occ}}(K)}\in\mathbb{Q}_{>0}. \tag{6.17}$$

Then:

(i) Integrality of $D_{N+R}^{\mathrm{eff}}\cdot\Delta_{N+R}$: *For every admissible* $K\ge 1$ *with* $2(N+R-1)\le 2AK$*,*

$$D_{N+R}^{\mathrm{eff}}\cdot\Delta_{N+R}=\left(\prod_{p\le 2AK}p^{v_{p,\mathrm{occ}}(K)}\right)\det M_K^{\mathrm{monic}}(\mathbf{a},c)\in\mathbb{Z}, \tag{6.18}$$

and the effective denominator exponent $\Phi_{\mathrm{den,eff}}^{(N+R)}(n):=\limsup_{K\to\infty}\frac{1}{K^2}\log D_{N+R}^{\mathrm{eff}}=-C_{\mathrm{fac}}(n)-\mathcal{I}_{\mathrm{occ}}(n)$ *is strictly negative for every* $n\ge 1$*.*

(ii) Uniform Analytical Negativity of $\Lambda_{\mathrm{net,full}}^{(N+R)}(n)$ Across All $n\ge 1$: *Define the full-range net exponent*

$$\Lambda_{\mathrm{net,full}}^{(N+R)}(n):=\Lambda_{\mathrm{arch}}^{(N+R)}(n)+\Phi_{\mathrm{den,eff}}^{(N+R)}(n). \tag{6.19}$$

For every integer $n\ge 1$*, set* $L_{\mathrm{in}}=2n+1$*,* $c_0:=0.03$*, and choose the scaling parameters*

$$A_0(n)=C(n)=\frac{0.03}{2n-1},\qquad R_c(n)=2(n-1)A_0(n),\qquad A(n)=2.5\big(C(n)+R_c(n)\big)=0.0750. \tag{6.20}$$

Then $\Lambda_{\mathrm{arch}}^{(N+R)}(n)$ *satisfies the closed-form upper bounds* $\Lambda_{\mathrm{arch}}^{(N+R)}(n)\le\Lambda_{\mathrm{arch}}^{\mathrm{tight}}(n)\le\Lambda_{\mathrm{arch}}^{\mathrm{UB}}(n)$*, where*

$$\begin{aligned}\Lambda_{\mathrm{arch}}^{\mathrm{UB}}(n):=\;&C(n)^2\left(\log\frac{C(n)}{4}-\frac{3}{2}+\log(2A)\right)+C(n)G_{2n+1,2}^{[A_0,A]}(0)+2C(n)R_c(n)\log A\\&+2n(n-1)A_0(n)^2\big(2\log A_0(n)+2\log 2-3\big)\\&+R_c(n)\left(4A\log 2-\frac{2A^2}{A_0(n)}H\left(\frac{A_0(n)}{A}\right)+c_0\right)+C_{\mathrm{fac}}(n)\end{aligned} \tag{6.21}$$

(and $\Lambda_{\mathrm{arch}}^{\mathrm{tight}}(n)$ *replaces* $2C(n)R_c(n)\log A$ *by* $\mathcal{I}_{pc}^{(0)}(n):=C(n)\int_0^{R_c(n)}\log(A^2-\tau^2)\,d\tau=C(n)R_c(n)\big(2\log A+(1+\frac{R_c(n)}{A})\log(1+\frac{R_c(n)}{A})+(1-\frac{R_c(n)}{A})\log(1-\frac{R_c(n)}{A})-2\big)$*, with* $\mathcal{I}_{pc}^{(0)}(1):=0$*), while* $\mathcal{I}_{\mathrm{occ}}(n)$ *satisfies the closed-form analytic lower bound*

$$\begin{aligned}\mathcal{I}_{\mathrm{occ}}(n)\ge\mathcal{I}_{\mathrm{occ}}^{\mathrm{LB}}(n):=\;&L_{\mathrm{pos}}(n)\Big(2C(n)+2\min\big(R_c(n),4A_0(n)\big)\Big)\\&+\big(2A-2A_0(n)\big)\cdot 2A_0(n)\bar m_1(n)\big(\bar m_1(n)-1\big)\\&-2C(n)\big(2A-2A_0(n)-L_{\mathrm{pos}}(n)\big)-c_0^2\left(\frac{3}{2}+\frac{4}{(2n-1)^2}\right),\end{aligned} \tag{6.22}$$

where $\bar m_1(n):=\min(n-1,2)$ *and* $L_{\mathrm{pos}}(n):=\sum_{m=1}^{3}\max\big(0,\min(2(A-A_0(n)),\frac{2A}{2m-1})-\max(2A_0(n),\frac{A+A_0(n)}{m})\big)$ *is the combined length of the three disjoint harmonic-free positive-valuation subintervals* $I_1=(A,2(A-A_0(n))]$*,* $I_2=(\frac{A+A_0(n)}{2},\frac{2A}{3}]$*, and* $I_3=(\frac{A+A_0(n)}{3},\frac{2A}{5}]$*.*

Consequently, for every integer $n \geq 1$,

$$\Lambda_{\mathrm{net,full}}^{(N+R)}(n) \leq \Lambda_{\mathrm{arch}}^{\mathrm{tight}}(n) - C_{\mathrm{fac}}(n) - \mathcal{I}_{\mathrm{occ}}^{\mathrm{LB}}(n) \leq -0.000146 < 0 \qquad (\forall\, n \geq 1), \tag{6.23}$$

and $\Lambda_{\mathrm{arch}}^{\mathrm{UB}}(n) - C_{\mathrm{fac}}(n) - \mathcal{I}_{\mathrm{occ}}^{\mathrm{LB}}(n) \leq -0.000122 < 0$ *for all* $n \geq 1$.

Proof. Part (i): In the monic-rescaled matrix $M_K^{\mathrm{monic}}(\mathbf{a}, c)$, every row polynomial $\mathcal{E}_i^{\mathrm{monic}}(u) = \prod_{t=0}^{i-1}(u - t^2) \in \mathbb{Z}[u]$ and every column polynomial $\mathcal{E}_m^{\mathrm{monic}}(u) = \prod_{t=0}^{m-1}(u - t^2) \in \mathbb{Z}[u]$ has integer coefficients. Fix a prime $p \leq 2AK$, and partition the poles $k \in \{0, 1, \ldots, 2AK\}$ into symmetric residue classes $\mathcal{R}_w := \{k \in \{0, \ldots, 2AK\} : k \equiv AK \pm r_w \pmod p\}$ indexed by $w = r_w/p \in [0, 1/2]$. For any polynomial $P(u) \in \mathbb{Z}[u]$, expanding $P((z + AK)^2)$ around a representative $k_0 \in \mathcal{R}_w$ in powers of $\xi := z + k_0$ gives $P((z + AK)^2) = \sum_{d \geq 0} c_d(P)\xi^d$ with $c_d(P) \in \mathbb{Z}$. For every pole $k = k_0 + s_k p \in \mathcal{R}_w$ ($s_k \in \mathbb{Z}$), substituting $\xi = w_k + s_k p$ ($w_k := z + k$) shows that if $c_0(P) = c_1(P) = \cdots = c_{\nu-1}(P) = 0$, then at every pole $k \in \mathcal{R}_w$ and every derivative order $d \geq 0$, the local Taylor coefficient $[w_k^d]P((w_k + AK - k)^2) = \sum_{j \geq \max(d,\nu)} \binom{j}{d}(s_k p)^{j-d} c_j(P)$ is divisible by $p^{\max(0,\nu-d)}$.

In the Cauchy product (6.3) of Lemma 6.1, when $m_1(w)$ top rows and $m_2(w)$ bottom rows in residue class $w \pmod p$ are eliminated by Gaussian elimination over $\mathbb{Z}_{(p)}$, the j_1-th eliminated top row ($0 \leq j_1 < m_1(w)$) vanishes at $\xi = 0$ to order $\nu_1 = j_1$, the j_2-th eliminated bottom row ($0 \leq j_2 < m_2(w)$) vanishes at $\xi = 0$ to order $\nu_2 = j_2$, and the j_{col}-th eliminated column ($0 \leq j_{\mathrm{col}} < m_1(w) + m_2(w)$) vanishes at $\xi = 0$ to order $\nu_{\mathrm{col}} = j_{\mathrm{col}}$. Summing the row and column vanishing orders across the $m_1(w) + m_2(w)$ pivots in $\mathcal{R}_w$ gives the total p-adic vanishing gain

$$\begin{aligned}\sum_{j_1=0}^{m_1-1} j_1 + \sum_{j_2=0}^{m_2-1} j_2 + \sum_{j_{\mathrm{col}}=0}^{m_1+m_2-1} j_{\mathrm{col}} &= \frac{m_1(m_1-1)}{2} + \frac{m_2(m_2-1)}{2} + \frac{(m_1+m_2)(m_1+m_2-1)}{2} \\ &= m_1(m_1-1) + m_2(m_2-1) + m_1 m_2.\end{aligned}$$

Adding the base kernel valuation $(\Omega_{\mathrm{in}} - L_{\mathrm{eff}} + 2)m_1$ on the m_1 top rows (where $+2$ comes from the quadratic double root $w(2r + w)$ of $\mathcal{E}_m^{\mathrm{monic}}((w + r)^2)$ at inner poles $m > r$ in (2.20)) and $(\Omega_{\mathrm{in}} - s_{\mathrm{eff}})m_2$ on the m_2 bottom rows yields the local valuation increment $f_1(m_1(w)) + f_2(m_2(w)) + m_1(w)m_2(w)$ (up to $E_{\mathrm{bdry}}(K) = o(K^2)$ from Proposition 6.4(i)), where $f_1(m_1) := m_1(m_1 - 1) + (\Omega_{\mathrm{in}} - L_{\mathrm{eff}} + 2)m_1$ and $f_2(m_2) := m_2(m_2 - 1) + (\Omega_{\mathrm{in}} - s_{\mathrm{eff}})m_2$. Thus $v_p(\det M_K^{\mathrm{monic}}(\mathbf{a}, c)) \geq -v_{p,\mathrm{occ}}(K)$ for all $p \leq 2AK$, while for $p > 2AK$ Proposition 6.4(ii) gives $v_p(\det M_K^{\mathrm{monic}}(\mathbf{a}, c)) \geq 0$ and $\prod_{p \leq 2AK} p^{v_p(P_{\mathrm{fac}})} = P_{\mathrm{fac}}$, proving (6.18). By the Prime Number Theorem $\vartheta(y) = \sum_{p \leq y} \log p = y + O(ye^{-c\sqrt{\log y}})$, Stieltjes integration over $p = xK \in (0, 2AK]$ gives $\lim_{K\to\infty} \frac{1}{K^2} \sum_{p \leq 2AK} (\log p)\, v_{p,\mathrm{occ}}(K) = -\mathcal{I}_{\mathrm{occ}}(n)$.

Part (ii): Under (6.20), $(2n - 1)A_0(n) = c_0 = 0.03$, $C(n) + R_c(n) = c_0 = 0.03$, and $A(n) = 2.5c_0 = 0.0750$. We establish (6.21), (6.22), and the pure analytical proof of (6.23) in three steps:

(1) *Closed-Form Archimedean Upper Bounds (6.21) and* $\Lambda_{\mathrm{arch}}^{\mathrm{tight}}(n)$*:* In (5.11)–(5.17), decompose $K_{cc}(y_1, y_2) = \frac{1}{2}\log|y_1^2 - y_2^2| + \log(2A) + \frac{1}{4}\log((1 + \frac{(y_1+y_2)^2}{4A^2})(1 + \frac{(y_1-y_2)^2}{4A^2}))$. By concavity of $t \mapsto \log(1 + t)$ and $1 + a + b \leq (1 + a)(1 + b)$ for $a, b \geq 0$, the last term is bounded above by $\frac{1}{2}\log(1 + \frac{y_1^2}{2A^2}) + \frac{1}{2}\log(1 + \frac{y_2^2}{2A^2})$. Similarly, for all $\tau \in [0, R_c(n)]$ and $y \geq 0$, since $((A-\tau)^2+y^2)((A+\tau)^2+y^2) = (A^2-\tau^2+y^2)^2+4\tau^2y^2 \leq (A^2+y^2)^2$, we have both $K_{pc}(\tau, y) \leq 2\log A + \log(1 + \frac{y^2}{A^2})$ and the sharper decomposition $K_{pc}(\tau, y) \leq \log(A^2 - \tau^2) + \log(1 + \frac{y^2}{A^2-\tau^2})$ (which at $y = 0$ integrates over $\tau \in [0, R_c(n)]$ to $\mathcal{I}_{pc}^{(0)}(n)$). Thus the sum of all y-dependent terms in (5.11) is bounded above by

$$\mathcal{E}_{\mathrm{sym}}(\mu_c) + C(n)^2 \log(2A) + C(n)G_{2n+1,2}^{[A_0,A]}(0) + 2C(n)R_c(n)\log A + C(n)\int_0^\infty H_n(y)\, d\mu_c(y),$$

where $\mathcal{E}_{\mathrm{sym}}(\mu_c) := C^2 \iint \frac{1}{2}\log|y_1^2 - y_2^2|\, d\mu_c(y_1)d\mu_c(y_2) - 2\pi C \int_0^\infty y\, d\mu_c(y)$ and

$$H_n(y) := (G_{2n+1,2}^{[A_0,A]}(y) - G_{2n+1,2}^{[A_0,A]}(0)) + C(n)\log\left(1 + \frac{y^2}{2A^2}\right) + R_c(n)\log\left(1 + \frac{y^2}{A^2}\right).$$

Differentiating $H_n(y)$ for $y > 0$ using $\frac{\partial}{\partial y}\Phi_{\log}(v, y) = 2\arctan(v/y)$, $C + R_c = c_0 = 0.03$, $A = 2.5c_0 = 0.075$, and $\arctan(2/u) \geq \frac{2u}{u^2+4}$ ($u := y/A > 0$) gives

$$H_n'(y) \leq -\frac{4u}{u^2+4} + \frac{c_0}{A}\frac{2u}{u^2+1} = -\frac{u(3.2u^2+0.8)}{(u^2+4)(u^2+1)} < 0,$$

so $H_n(y) \leq H_n(0) = 0$ for all $y \geq 0$. By Saff–Totik [21, Example IV.1.17, p. 199], $\mathcal{E}_{\mathrm{sym}}(\mu_c) \leq C(n)^2(\log\frac{C(n)}{4} - \frac{3}{2})$. On the uniform inner-pole measure $\frac{d\alpha}{A_0(n)}$, rescaling $\alpha_i = A_0(n)u_i$ ($u_i \in [0,1]$) splits $\log|\alpha_1^2 - \alpha_2^2| = 2\log A_0(n) + \log|u_1 - u_2| + \log(u_1 + u_2)$, where by triangle symmetry on $0 \leq u_2 < u_1 \leq 1$, $\iint_{[0,1]^2}\log|u_1 - u_2|\,du_1du_2 = 2\int_0^1\int_0^{u_1}\log(u_1 - u_2)\,du_2du_1 = -\frac{3}{2}$ and $\iint_{[0,1]^2}\log(u_1 + u_2)\,du_1du_2 = 2\log 2 - \frac{3}{2}$, giving $\iint_{[0,A_0]^2}\log|\alpha_1^2 - \alpha_2^2|\frac{d\alpha_1}{A_0}\frac{d\alpha_2}{A_0} = 2\log A_0(n) + 2\log 2 - 3$. Similarly, substituting $\alpha = As$ ($s \in [0,u]$ with $u = A_0(n)/A$) into the outer part of $w_p(\alpha)$ in (5.17) and $\alpha = A_0(n)v$ ($v \in [0,1]$) into the inner part gives

$$\frac{1}{A_0(n)}\int_0^{A_0(n)} w_p(\alpha)\,d\alpha = 4A\log 2 - \frac{2A^2}{A_0(n)}H\left(\frac{A_0(n)}{A}\right) + c_0,$$

where $H(u) := \int_0^u((1-s)\log(1-s) + (1+s)\log(1+s))\,ds = \frac{1}{2}((1+u)^2\log(1+u) - (1-u)^2\log(1-u)) - u$ (with $H(1) = 2\log 2 - 1$), establishing (6.21) and $\Lambda_{\mathrm{arch}}^{\mathrm{tight}}(n)$.

(2) *Closed-Form Analytic Lower Bound* (6.22) *on* $\mathcal{I}_{\mathrm{occ}}(n)$*:* Decompose $(0, 2A]$ into $(0, A_0(n)]$, $(A_0(n), 2A_0(n)]$, and $(2A_0(n), 2A]$:

- On $x \in (0, A_0(n)]$: Since $m_1(m_1-1) \geq 0$, $m_2(m_2-1) \geq 0$, $m_1m_2 \geq 0$, $\Omega_{\mathrm{in}} \geq 0$, and $L_{\mathrm{eff}}, s_{\mathrm{eff}} \leq 2n+1$, we have $f_1(m_1) + f_2(m_2) + m_1m_2 \geq -(2n-1)(m_1+m_2) - 2m_2$. Integrating over $w \in [0, 1/2]$ (where $2x\int_0^{1/2}(m_1+m_2)\,dw \leq c_0$ and $2x\int_0^{1/2}m_2\,dw = C(n)$) and $x \in (0, A_0(n)]$ of width $A_0(n) = \frac{c_0}{2n-1}$ gives $\mathcal{I}_{(0,A_0(n)]}(n) \geq -c_0^2(1 + \frac{2}{(2n-1)^2})$.
- On $x \in (A_0(n), 2A_0(n)]$: For $w \in [0, 1 - A_0(n)/x)$, Proposition 6.2 holds with $L_{\mathrm{eff}} = s_{\mathrm{eff}} = 2$ and $\Omega_{\mathrm{in}} \geq 2n-1$, so $f_1(m_1) \geq 0$ and $f_2(m_2) \geq -2m_2$. On $w \in [1 - A_0(n)/x, 1/2]$ (which has relative fraction $\frac{A_0(n)/x - 1/2}{A_0(n)/x} = 1 - \frac{x}{2A_0(n)}$ of $[0, A_0(n)/x]$), since $\int_{A_0(n)}^{2A_0(n)}(1 - \frac{x}{2A_0(n)})\,dx = \frac{1}{2}A_0(n)$, integrating over $x \in (A_0(n), 2A_0(n)]$ gives $\mathcal{I}_{(A_0(n),2A_0(n)]}(n) \geq -\frac{1}{2}A_0(n)(2n-1)c_0 - 2C(n)A_0(n) = -c_0^2(\frac{1}{2} + \frac{2}{(2n-1)^2})$. Summing with $\mathcal{I}_{(0,A_0(n)]}(n)$ gives $-c_0^2(\frac{3}{2} + \frac{4}{(2n-1)^2})$.
- On $x \in (2A_0(n), 2A]$: By Theorem 3.3(ii) and Proposition 6.2, on the top R rows L_{top} (supported purely on $|k - AK| \leq A_0K$) we have $L_{\mathrm{eff}} = 2$ and $\Omega_{\mathrm{in}}(x,w) = 2(\lfloor 2A/x\rfloor - \lfloor A/x + w\rfloor - \lfloor A/x - w\rfloor) \geq 0$ everywhere. With constant top-row occupation $\bar{m}_1(n) = \min(n-1, 2) \in \{0,1,2\}$ on $w \in [0, A_0(n)/x]$, the quadratic term $m_1(m_1-1)$ contributes $2x\int_0^{A_0(n)/x}\bar{m}_1(\bar{m}_1-1)\,dw = 2A_0(n)\bar{m}_1(n)(\bar{m}_1(n)-1)$ across the entire interval $(2A_0(n), 2A]$ of length $2A - 2A_0(n)$. Furthermore, on the three disjoint subintervals $I_1 = (A, 2(A - A_0(n))]$, $I_2 = (\frac{A+A_0(n)}{2}, \frac{2A}{3}]$, and $I_3 = (\frac{A+A_0(n)}{3}, \frac{2A}{5}]$ (where for $n \geq 2$, $u = \frac{A_0(n)}{A} \leq \frac{2}{15}$ ensures both $(m - \frac{1}{2})(1-u) \geq m - 1 + \frac{3-2m}{30} > m-1$ at the upper endpoint $x = \frac{2A}{2m-1}$ and $m\frac{1-u}{1+u} \geq \frac{13m}{17} > m-1$ at the lower endpoint $x = \frac{A+A_0(n)}{m}$, so $m - 1 \leq \frac{A-A_0(n)}{x} < \frac{A+A_0(n)}{x} < m$ across all of I_m for each $m \in \{1,2,3\}$), every inner residue $w \in [0, A_0(n)/x]$ satisfies $\lfloor 2A/x\rfloor = 2m-1$ and $\lfloor A/x \pm w\rfloor = m-1$, so $\Omega_{\mathrm{in}}(x,w) = 2((2m-1) - 2(m-1)) = +2$ and $v_p(\widetilde{H}_{k,AK}^{(t)}) = 0$ by Theorem 6.3(ii), contributing $+L_{\mathrm{pos}}(n)(2C(n) + 2\min(R_c(n), 4A_0(n)))$, while the remaining portion of length $2A - 2A_0(n) - L_{\mathrm{pos}}(n)$ contributes at least $-2C(n)(2A - 2A_0(n) - L_{\mathrm{pos}}(n))$.

Summing the three subinterval bounds yields (6.22).

(3) *Pure Analytical Proof of Strict Negativity* (6.23) *Across All* $n \geq 1$*:* Set $t := \frac{1}{2n-1} \in (0, 1]$ and $u := \frac{A_0(n)}{A} = \frac{2t}{5} \in (0, \frac{2}{5}]$. Since $4A(\log(2A) - 1) - \Phi_{\log}(2A, 0) = 0$, expanding $\log(1 \pm u)$

in powers of u gives the exact identities

$$G^{[A_0,A]}_{2n+1,2}(0) = 2c_0\left(\log\frac{4t}{5}-1\right)+c_0\sum_{k=1}^{\infty}\frac{u^{2k}}{k(2k+1)}, \qquad \frac{2A^2}{A_0(n)}H(u) = 2A\sum_{k=1}^{\infty}\frac{u^{2k}}{(2k-1)(2k)(2k+1)}.$$

At $n=1$ ($t=1, R_c(1)=0, L_{\rm pos}(1)=\frac{1}{2}c_0, -\mathcal{I}^{\rm LB}_{\rm occ}(1)=\frac{19}{2}c_0^2$), direct substitution gives $\frac{1}{c_0^2}(\Lambda^{\rm UB}_{\rm arch}(1)-C_{\rm fac}(1)-\mathcal{I}^{\rm LB}_{\rm occ}(1)) = \log\frac{5c_0^2}{4}-\frac{3}{2}+2\log 2-\frac{7}{2}\log\frac{7}{2}+\frac{3}{2}\log\frac{3}{2}+\frac{19}{2} = 8+\log\frac{27\sqrt{21}}{1200500} = -1.18015 < -1$, so $\Lambda^{(N+R)}_{\rm net,full}(1)\le -0.001062 < 0$. At $n=2$ ($t=\frac{1}{3}, R_c(2)=\frac{2}{3}c_0, L_{\rm pos}(2)=\frac{77}{36}c_0$), we obtain $\frac{1}{c_0^2}(\Lambda^{\rm UB}_{\rm arch}(2)-C_{\rm fac}(2)-\mathcal{I}^{\rm LB}_{\rm occ}(2)) = -4.1535 < -4$, so $\Lambda^{(N+R)}_{\rm net,full}(2)\le -0.003738 < 0$ (and $\le -0.004124 < 0$ with $\Lambda^{\rm tight}_{\rm arch}(2)$).

Now let $n\ge 3$, so that $t=\frac{1}{2n-1}\in(0,\frac{1}{5}]$, $u=\frac{2t}{5}\in(0,\frac{2}{25}]$, $\min(R_c(n),4A_0(n)) = 4c_0t$, $\bar{m}_1(n)(\bar{m}_1(n)-1)=2$, $L_{\rm pos}(n)=(\frac{37}{12}-\frac{17}{6}t)c_0$, and $2n(n-1)A_0(n)^2=\frac{1-t^2}{2}c_0^2$. Using $\sum_{k=1}^{\infty}\frac{u^{2k}}{k(2k+1)}\le\frac{u^2}{3(1-(2/25)^2)}=\frac{25}{465.75}t^2$ and $\sum_{k=1}^{\infty}\frac{u^{2k}}{(2k-1)(2k)(2k+1)}\ge\frac{u^2}{6}=\frac{2}{75}t^2$, the $t^2\log t$ terms in $C(n)^2\log(C(n)/4)$ and $\frac{1-t^2}{2}c_0^2(2\log A_0(n))$ cancel identically ($t^2\log t+(1-t^2)\log t=\log t$), yielding for all $n\ge 3$ the elementary closed-form upper bound

$$\frac{1}{c_0^2}\Big(\Lambda^{\rm UB}_{\rm arch}(n)-C_{\rm fac}(n)-\mathcal{I}^{\rm LB}_{\rm occ}(n)\Big)\le f_{\rm upper}(t):=(1+2t)\log t+b_0+b_1t+b_2t^2+b_3t^3, \tag{6.24}$$

where the constants b_0,b_1,b_2,b_3 are given explicitly by

$$b_0:=1+\log\frac{1536}{25}, \qquad b_1:=-50+2\log\frac{3}{1600}, \qquad b_2:=\frac{626}{15}+\log\frac{10}{3}, \qquad b_3:=\frac{2984}{9315}.$$

Differentiating $f_{\rm upper}(t)$ twice on $t\in(0,\frac{1}{5}]$ gives

$$f'_{\rm upper}(t)=\frac{1}{t}+2\log t+(2+b_1)+2b_2t+3b_3t^2, \qquad f''_{\rm upper}(t)=\frac{-1+2t+2b_2t^2+6b_3t^3}{t^2}.$$

Since $2b_2<86$ and $6b_3<2$, the numerator $q(t):=-1+2t+2b_2t^2+6b_3t^3$ is strictly increasing on $t>0$ with $q(\frac{1}{11})<-1+\frac{2}{11}+\frac{86}{121}+\frac{2}{1331}=-\frac{141}{1331}<0$. Thus $f''_{\rm upper}(t)<0$ (strict concavity) on $(0,\frac{1}{11}]$, while on $[\frac{1}{11},\frac{1}{5}]$ the convexity of $f'_{\rm upper}$ beyond its unique inflection point in $(\frac{1}{11},\frac{1}{10})$ together with $f'_{\rm upper}(\frac{1}{11})<-41<0$ and $f'_{\rm upper}(\frac{1}{5})<-41<0$ implies $f'_{\rm upper}(t)<0$ across $[\frac{1}{11},\frac{1}{5}]$. On the strictly concave interval $(0,\frac{1}{11}]$, evaluating at the adjacent integer parameters $n=34$ ($t_0=\frac{1}{67}$) and $n=35$ ($t_1=\frac{1}{69}$) gives $f'_{\rm upper}(t_1)=+1.2183>0>-0.6858=f'_{\rm upper}(t_0)$, so by concavity the maximum of $f_{\rm upper}(t)$ over all integers $n\ge 3$ is attained at $n=34$ ($t_0=\frac{1}{67}$) with value $f_{\rm upper}(\frac{1}{67}) = -\frac{69}{67}\log 67+b_0+\frac{b_1}{67}+\frac{b_2}{67^2}+\frac{b_3}{67^3}=-0.136285<0$ (verified purely by rational arithmetic via $\log\frac{1+z}{1-z}=2\sum_{k=0}^{\infty}\frac{z^{2k+1}}{2k+1}$ applied to $2=\frac{1+1/3}{1-1/3}$, $3=2\cdot\frac{1+1/5}{1-1/5}$, $5=4\cdot\frac{1+1/9}{1-1/9}$, and $67=64\cdot\frac{1+3/131}{1-3/131}$, which give $\log 2\in(0.6931471,0.6931472)$, $\log 3\in(1.0986122,1.0986123)$, $\log 5\in(1.6094379,1.6094380)$, and $\log 67>6\log 2+\frac{6}{131}>4.204684$, whence $b_0<5.118062$, $b_1<-62.55829$, $b_2<42.93731$, $b_3<0.32035$, and $f_{\rm upper}(\frac{1}{67})<-0.1362<0$; moreover, over all real $t\in(0,\frac{1}{5}]$, tangent-line concavity gives $f_{\rm upper}(t)\le f_{\rm upper}(\frac{1}{67})+f'_{\rm upper}(\frac{1}{67})(\frac{1}{69}-\frac{1}{67})=-0.135988<0$). Multiplying by $c_0^2=0.0009$ proves analytically that $\Lambda^{\rm UB}_{\rm arch}(n)-C_{\rm fac}(n)-\mathcal{I}^{\rm LB}_{\rm occ}(n)\le c_0^2 f_{\rm upper}(\frac{1}{67})=-0.00012265<-0.000122<0$ (and $\Lambda^{\rm tight}_{\rm arch}(n)-C_{\rm fac}(n)-\mathcal{I}^{\rm LB}_{\rm occ}(n)\le -0.00014641<-0.000146<0$, attained at $n=35$) for every integer $n\ge 1$.

□

7. Asymptotic Non-Vanishing of Δ_{N+R} and Proof of Theorem 1.1

To complete the proof of Theorem 1.1, we now prove that for every integer $n\ge 2$ and every integer tuple $\mathbf{a}=(a_3,a_5,\dots,a_{2n+1})\in\mathbb{Z}^n$ with $a_{2n+1}>0$, the moment determinant $\Delta_{N+R}=\det\left({}^{L_{\rm top}}_{a_{2n+1}}\mathcal{A}_{\rm bot}\right)$ is non-zero for all admissible $K\ge K_0(n,\mathbf{a})$.

Theorem 7.1 (Asymptotic Non-Vanishing of Δ_{N+R})**.** *Fix an integer $n \geq 2$, $L_{\mathrm{in}} = 2n+1$, and the scaling parameters $A_0(n) = C(n) = \frac{0.03}{2n-1}$, $R_c(n) = 2(n-1)A_0(n)$, and $A(n) = 2.5(C(n)+R_c(n)) = 0.0750$ of Theorem 6.5. For any fixed integer tuple $\mathbf{a} = (a_3, a_5, \ldots, a_{2n+1}) \in \mathbb{Z}^n$ with $a_{2n+1} > 0$, define the Euler sine-square constant*

$$\begin{aligned}\widetilde{\Lambda}_{\mathrm{Euler}}(\mathbf{a}) &:= \sum_{j=1}^{n} a_{2j+1}(-1)^{j-1}\frac{2^{2j-1}\pi^{2(j-1)}}{(2j)!} \\ &= a_3 - \frac{\pi^2}{3}a_5 + \frac{2\pi^4}{45}a_7 - \cdots + (-1)^{n-1}\frac{2^{2n-1}\pi^{2(n-1)}}{(2n)!}a_{2n+1} \neq 0.\end{aligned} \tag{7.1}$$

Then there exists an explicit threshold $K_0(n, \mathbf{a}) \geq \lceil \frac{100n}{c_0} \rceil = \lceil \frac{10000n}{3} \rceil$ such that for every admissible scaling integer $K \geq K_0(n, \mathbf{a})$, setting $m_{\mathrm{core}}(n, K) := 2A_0K + 1$ for all $n \geq 2$ and defining the dominant Class I core

$$\mathcal{C}_{\mathrm{I}}^{\mathrm{core}} := \{J \subset \{2A_0K + 1, \ldots, N + R - 1\} : |J| = N\}, \tag{7.2}$$

the following properties hold:

(i) Hermite Block-Triangular Decomposition and Exact Sign Coherence on $\mathcal{C}_{\mathrm{I}}^{\mathrm{core}}$: *For every subset $J \in \mathcal{C}_{\mathrm{I}}^{\mathrm{core}}$, its complement $J^c = J_{\mathrm{head}} \sqcup (J_{\mathrm{tail}} \setminus J)$ contains $\{0, 1, \ldots, 2A_0K\}$ and satisfies*

$$(-1)^{\sigma(J^c)} \det L_{\mathrm{top}}[:, J^c] = (-1)^{\binom{R-N}{2}} \det M_{\mathrm{head}}(\mathbf{a}; K; J) \cdot \det \widetilde{S}_{\mathrm{der}}(J_{\mathrm{tail}} \setminus J), \tag{7.3}$$

where $\det M_{\mathrm{head}}(\mathbf{a}; K; J) = (\prod_{r=0}^{A_0K} w_{2n+1,r})(-\widetilde{\Lambda}_{\mathrm{Euler}}(\mathbf{a}))^N(1 + O(K^{-1})) \neq 0$ and $\det \widetilde{S}_{\mathrm{der}}(J_{\mathrm{tail}} \setminus J)$ has constant non-zero sign $\varepsilon_0 \in \{+1, -1\}$ across all $J \in \mathcal{C}_{\mathrm{I}}^{\mathrm{core}}$ by Cauchy–Vandermonde total positivity on $t^2 - r^2 > 0$ ($t \geq N > r \geq 0$). Consequently, every subset $J \in \mathcal{C}_{\mathrm{I}}^{\mathrm{core}}$ has the exact same sign $\varepsilon_{\mathrm{I}} \in \{+1, -1\}$ in the Laplace expansion of Δ_{N+R}, so all terms in $\mathcal{C}_{\mathrm{I}}^{\mathrm{core}}$ add constructively without cancellation:

$$\left| \sum_{J \in \mathcal{C}_{\mathrm{I}}^{\mathrm{core}}} (-1)^{\sigma(J^c)} \det L_{\mathrm{top}}[:, J^c] \cdot \mathcal{S}_K(J) \right| = \sum_{J \in \mathcal{C}_{\mathrm{I}}^{\mathrm{core}}} \left|\det L_{\mathrm{top}}[:, J^c]\right| \cdot \mathcal{S}_K(J) > 0. \tag{7.4}$$

(ii) Block-Triangular Non-Vanishing and Barrier Suppression: *By Theorem 5.3(i) (5.6), $\Delta_{N+R} = (-1)^{RN} a_{2n+1}^R \det(S_{\mathrm{odd,top}}[:, J_{\mathrm{tail}}]) \det_{0 \leq k,d < N}(a_{2n+1}\mathcal{A}(\mathcal{E}_k Q_d))$, where $\det(S_{\mathrm{odd,top}}[:, J_{\mathrm{tail}}]) \neq 0$ by Theorem 4.4 and the barrier-crossing complement $\mathcal{C}_{\mathrm{bar}} := \{J \subset \{0, \ldots, N + R - 1\} : |J| = N\} \setminus \mathcal{C}_{\mathrm{I}}^{\mathrm{core}}$ is exponentially suppressed relative to $\mathcal{C}_{\mathrm{I}}^{\mathrm{core}}$ across the A_0K-step inner-tail barrier $\{A_0K + 1, \ldots, 2A_0K\}$. Consequently, for all $K \geq K_0(n, \mathbf{a})$,*

$$\left|\Delta_{N+R}\right| > 0 \qquad (\forall\, K \geq K_0(n, \mathbf{a})). \tag{7.5}$$

Proof. Expanding $\det V_{N+R}(L_{\mathrm{top}}; v_{\nu_1}, \ldots, v_{\nu_N})$ in the Cauchy–Binet series (5.10) along the top R rows L_{top} gives

$$\Delta_{N+R} = \sum_{J \in \mathcal{C}_{\mathrm{I}}^{\mathrm{core}}} (-1)^{\sigma(J^c)} \det L_{\mathrm{top}}[:, J^c] \cdot \mathcal{S}_K(J) + \sum_{J \in \mathcal{C}_{\mathrm{bar}}} (-1)^{\sigma(J^c)} \det L_{\mathrm{top}}[:, J^c] \cdot \mathcal{S}_K(J), \tag{7.6}$$

where for each N-element subset $J \subset \{0, \ldots, N + R - 1\}$,

$$\mathcal{S}_K(J) := a_{2n+1}^N \sum_{1 \leq \nu_1 < \cdots < \nu_N < \infty} \left(\prod_{j=1}^{N} W_K(\nu_j) \right) \det_{\substack{0 \leq k < N \\ 1 \leq j \leq N}} (\mathcal{E}_k(x_{\nu_j}^2)) \det_{\substack{m \in J \\ 1 \leq j \leq N}} (\mathcal{E}_m(x_{\nu_j}^2)). \tag{7.7}$$

For every $K \geq K_0(n, \mathbf{a}) \geq \lceil \frac{100n}{c_0} \rceil$, every column index $m < N + R$ and every $\nu_j \geq 1$ satisfies $m < 1.01(C + R_c)K < 2.5(C + R_c)K + 1 = AK + 1 \leq x_{\nu_1} < \cdots < x_{\nu_N}$. By Theorem 2.6, both evaluation determinants in (7.7) are strictly positive, so $\mathcal{S}_K(J) > 0$ for every N-element subset J.

Step 1: Proof of (i). For any $J \in \mathcal{C}_{\mathrm{I}}^{\mathrm{core}}$, its complement is $J^c = J_{\mathrm{head}} \sqcup (J_{\mathrm{tail}} \setminus J)$ with $J_{\mathrm{tail}} \setminus J = \{m_1 < m_2 < \cdots < m_{R-N}\} \subset \{N, \ldots, N + R - 1\}$. Since every element of $J_{\mathrm{head}} = \{0, \ldots, N - 1\}$ precedes every element of $J \subset J_{\mathrm{tail}}$, no inversions occur between J_{head} and J. For

each $1 \le s \le R-N$, among $\{N, N+1, \dots, m_s\}$, s integers belong to $J_{\text{tail}} \setminus J$ and $(m_s - N) - (s-1)$ belong to J, giving

$$\sigma(J^c) = \sum_{s=1}^{R-N} ((m_s - N) - (s-1)) = \sum_{m \in J_{\text{tail}} \setminus J} (m-N) - \binom{R-N}{2}. \tag{7.8}$$

Writing $\mathcal{E}_m(u) = (-1)^{m-N} \frac{2}{(2m)!} W_0(u) \prod_{t=N}^{m-1} (t^2 - u)$ on $m \in J_{\text{tail}} \setminus J$ and factoring out $(-1)^{m-N}$ cancels $\sum_{m \in J_{\text{tail}} \setminus J} (m-N) \pmod 2$ against $(-1)^{\sigma(J^c)}$:

$$(-1)^{\sigma(J^c)} \prod_{m \in J_{\text{tail}} \setminus J} (-1)^{m-N} = (-1)^{\binom{R-N}{2}}. \tag{7.9}$$

Because $m_{\text{core}}(n, K) = 2A_0K + 1$, for every $J \in \mathcal{C}_{\text{I}}^{\text{core}}$ its complement J^c contains $\{0, 1, \dots, 2A_0K\}$. At each inner pole $k = AK - r$ ($0 \le r \le A_0K$), decompose $g_k(w)/g_{k,0} = \widetilde{U}_{\text{out},k}(w)\widetilde{U}_{\text{in},k}(w)$, where by (6.5) and Euler's infinite product for $\sin(\pi w)$,

$$\begin{aligned} \widetilde{U}_{\text{out},k}(w)^{-1} &:= \prod_{\substack{0 \le j \le 2AK \\ j \ne k}} \left(1 + \frac{w}{j-k}\right)^2 = \left(\frac{\sin(\pi w)}{\pi w}\right)^2 (1 + O((AK)^{-1})) \\ &= \sum_{m=0}^{\infty} c_m w^{2m} + O((AK)^{-1}), \end{aligned}$$

with $c_m = (-1)^m \frac{2^{2m+1}\pi^{2m}}{(2m+2)!}$. For any polynomial $P(u) \in \mathbb{Q}[u]$, multiplying $\frac{g_k(w)}{g_{k,0}} P((w+r)^2) = \sum_{q=0}^{2n} \frac{A_{2n+1-q,k}(P)}{g_{k,0}} w^q + O(w^{2n+1})$ by $\widetilde{U}_{\text{out},k}(w)^{-1}$ and extracting $[w^{2n-2}]$ via the Cauchy product gives

$$[w^{2n-2}]\Big(\widetilde{U}_{\text{in},k}(w) P((w+r)^2)\Big) = \sum_{j=1}^{n} c_{j-1} \frac{A_{2j+1,k}(P)}{g_{k,0}} + O((AK)^{-1}).$$

Taking the linear combination $\sum_{j=1}^{n-1} c_{j-1} L_{\text{top}}^{(2j+1)}(P) = a_{2n+1} \sum_{j=1}^{n} c_{j-1} S_{2j+1}(P) - \widetilde{\Lambda}_{\text{Euler}}(\mathbf{a}) S_{2n+1}(P)$ (where $\widetilde{\Lambda}_{\text{Euler}}(\mathbf{a}) = \sum_{j=1}^{n} a_{2j+1} c_{j-1}$ is given by (7.1)): on the $R - N = (2n-3)A_0K + n - 2$ tail columns $m \in J_{\text{tail}} \setminus J$ (which include the initial A_0K tail columns $\{N, \dots, 2A_0K\}$), we have $W_0(u) \mid \mathcal{E}_m(u)$ and hence $S_{2n+1}(\mathcal{E}_i \mathcal{E}_m) = 0$ identically, while the $(R-N) \times (R-N)$ derivative jet submatrix $\widetilde{S}_{\text{der}}(J_{\text{tail}} \setminus J)$ evaluating the confluent derivatives of order $1 \le q_0 \le n-2$ at $u = 0$ and $1 \le q \le 2(n-1)$ at $u = r^2$ ($1 \le r \le A_0K$) on $\{\mathcal{E}_m(u)\}_{m \in J_{\text{tail}} \setminus J}$ is invertible and has constant sign $\varepsilon_0 \in \{+1, -1\}$ across all $J \in \mathcal{C}_{\text{I}}^{\text{core}}$ because by (7.9) the sign-adjusted tail polynomials $(-1)^{m-N}\mathcal{E}_m(u) = \frac{2}{(2m)!} W_0(u) \prod_{t=N}^{m-1} (t^2 - u)$ have positive divided differences on $u_r = r^2 < N^2 \le t^2$ by Theorem 2.6. Performing block Gaussian elimination with $\widetilde{S}_{\text{der}}(J_{\text{tail}} \setminus J)$ annihilates all positive-order derivative terms $[w^q]P((w+r)^2)$ ($q \ge 1$) from the combined row block $\sum_{j=1}^{n-1} c_{j-1} L_{\text{top}}^{(2j+1)}$, leaving on the N head columns J_{head} only the $q = 0$ value $P(r^2)$ multiplied by $-w_{2n+1,r}(\widetilde{\Lambda}_{\text{Euler}}(\mathbf{a}) + O(K^{-1}))$. Since $\det_{0 \le i,d < N}(\mathcal{E}_i(r^2)\mathcal{E}_d(r^2)) = 1$ by unitriangularity of E_{head}, this gives $\det M_{\text{head}}(\mathbf{a}; K; J) = (\prod_{r=0}^{A_0K} w_{2n+1,r})(-\widetilde{\Lambda}_{\text{Euler}}(\mathbf{a}))^N (1 + O(K^{-1}))$. Because π is transcendental over $\mathbb{Q}$ by Lindemann's theorem [17], π^2 is transcendental over $\mathbb{Q}$, and since $\widetilde{P}_{\mathbf{a}}(X) := \sum_{j=1}^{n} a_{2j+1}(-1)^{j-1} \frac{2^{2j-1}}{(2j)!} X^{j-1} \in \mathbb{Q}[X]$ has non-zero leading coefficient $(-1)^{n-1} \frac{2^{2n-1}}{(2n)!} a_{2n+1} \ne 0$, we have $\widetilde{\Lambda}_{\text{Euler}}(\mathbf{a}) = \widetilde{P}_{\mathbf{a}}(\pi^2) \ne 0$, proving (7.3) and (7.4).

Step 2: Proof of (ii) (Normalized Step Ratios and Exponential Barrier Suppression of $\mathcal{C}_{\text{bar}}$). In the normalized basis $\mathcal{E}_\ell(u) = \frac{2}{(2\ell)!} \prod_{t=0}^{\ell-1} (u - t^2)$, we bound the single-step ratios across $\ell \in \{1, \dots, N + R - 1\}$:

- *Bottom-block normalized step ratio:* For any $\ell \in \{1, \dots, N + R - 1\}$ and any $\nu_j \ge 1$ ($x_{\nu_j} \ge AK + 1 > 2.5(C + R_c)K$), we have the exact identity

$$\frac{\mathcal{E}_{\ell-1}(x_{\nu_j}^2)}{\mathcal{E}_\ell(x_{\nu_j}^2)} = \frac{2\ell(2\ell-1)}{x_{\nu_j}^2 - (\ell-1)^2}. \tag{7.10}$$

For general $\ell < N+R < 1.01(C+R_c)K$, (7.10) is at most $\frac{4(1.01)^2}{2.5^2-(1.01)^2} < \frac{4}{5}$ (and asymptotically $\frac{4}{5.25} = \frac{16}{21} < 1$). Moreover, across the A_0K inner-tail barrier indices $\ell \in \{A_0K+1,\dots,2A_0K\}$, since $A = 2.5(2n-1)A_0 \geq 7.5A_0$ for all $n \geq 2$, we have $\ell \leq 2A_0K \leq \frac{2}{7.5}AK$, so on the barrier:

$$\frac{\mathcal{E}_{\ell-1}(x_{\nu_j}^2)}{\mathcal{E}_\ell(x_{\nu_j}^2)} \leq \frac{4(2A_0)^2}{(7.5A_0)^2-(2A_0)^2} = \frac{16}{52.25} = \frac{64}{209} < \frac{1}{3} \qquad (\forall\, \ell \in \{A_0K+1,\dots,2A_0K\}). \tag{7.11}$$

- *Top-block normalized step ratio:* Simultaneously, in $\mathcal{E}_\ell(u) = \frac{u-(\ell-1)^2}{2\ell(2\ell-1)}\mathcal{E}_{\ell-1}(u)$, at every inner-window node $u = r^2$ ($0 \leq r \leq A_0K < \ell$), advancing from $\ell-1$ to ℓ in J^c multiplies the inner-window factor by

$$\frac{\ell^2-r^2}{2\ell(2\ell-1)} \leq \frac{1}{4}(1+O(K^{-1})). \tag{7.12}$$

Multiplying the normalized top-block factor $\frac{1}{4}$ from (7.12) and the normalized bottom-block factor from (7.10)–(7.11), the factors of 4 and $\frac{1}{4}$ cancel identically, yielding the combined per-step product

$$\rho_{\mathrm{bar}} := \frac{1}{4}\cdot\frac{64}{209} = \frac{16}{209} < \frac{1}{13} \quad (\ell \in \{A_0K+1,\dots,2A_0K\}), \qquad \rho := \frac{1}{4}\cdot\frac{16}{21} = \frac{4}{21} \quad (\ell > 2A_0K). \tag{7.13}$$

Set $\Psi := (L_{\mathrm{top}}[:,:R])^{-1}L_{\mathrm{top}}[:,R:] \in \mathbb{Q}^{R\times N}$. By the Sylvester–Jacobi complementary minor identity, for any R-element subset $J^c \subset \{0,\dots,N+R-1\}$ with $I_{\mathrm{out}} := \{0,\dots,R-1\}\setminus J^c$ and $I_{\mathrm{in}} := \{m-R : m \in J^c, m \geq R\}$, we have $|\det L_{\mathrm{top}}[:,J^c]| = |\det L_{\mathrm{top}}[:,:R]|\cdot|\det \Psi[I_{\mathrm{out}},I_{\mathrm{in}}]|$. Every barrier-crossing subset $J \in \mathcal{C}_{\mathrm{bar}}$ replaces $r = |J\cap\{0,\dots,2A_0K\}| \in \{1,\dots,N\}$ indices $\geq 2A_0K+1$ of $J \in \mathcal{C}_{\mathrm{I}}^{\mathrm{core}}$ across the A_0K-step barrier $\{A_0K+1,\dots,2A_0K\}$ (contributing $\rho_{\mathrm{bar}}^{rA_0K}$). Since the number of index subsets with total intra-block displacement $D \geq 0$ is bounded by the partition number $p(D)$ and Euler's partition generating function satisfies $\sum_{D=0}^{\infty}p(D)\rho^D = \prod_{d=1}^{\infty}(1-(4/21)^d)^{-1} = 1.292825\cdots < 1.30$, summing over all $\binom{N}{r}$ choices ($1 \leq r \leq N = A_0K+1$) bounds the barrier sum by $1.30((1+(16/209)^{A_0K})^{A_0K+1}-1) \leq 0.2067 < \frac{1}{4}$ times the $\mathcal{C}_{\mathrm{I}}^{\mathrm{core}}$ sum, and the reverse triangle inequality on (7.6) establishes $|\Delta_{N+R}| > 0$ for all $K \geq K_0(n,\mathbf{a})$. □

Proof of Theorem 1.1. We proceed by induction on the positive integer $n \geq 1$. For the base case $n = 1$, the numbers 1 and $\zeta(3)$ are linearly independent over $\mathbb{Q}$ by Apéry's theorem [1].

Now let $n \geq 2$ and assume as the induction hypothesis that $1,\zeta(3),\zeta(5),\dots,\zeta(2n-1)$ are linearly independent over $\mathbb{Q}$. Suppose for contradiction that there exist rational numbers $q_0,q_1,\dots,q_n \in \mathbb{Q}$, not all zero, such that $q_0 + \sum_{j=1}^{n} q_j\zeta(2j+1) = 0$. Clearing rational denominators, we obtain integers $a_3,a_5,\dots,a_{2n+1},c \in \mathbb{Z}$, not all zero, satisfying

$$a_3\zeta(3) + a_5\zeta(5) + \cdots + a_{2n+1}\zeta(2n+1) + c = 0. \tag{7.14}$$

If $a_{2n+1} = 0$, then (7.14) is a nontrivial linear relation over $\mathbb{Q}$ among $1,\zeta(3),\dots,\zeta(2n-1)$, contradicting the induction hypothesis. Hence $a_{2n+1} \neq 0$, and replacing $(\mathbf{a},c)$ by $(-\mathbf{a},-c)$ if necessary, we have $a_{2n+1} > 0$.

Fix $L_{\mathrm{in}} = 2n+1$ and the rational scaling parameters $A_0(n) = C(n) = \frac{0.03}{2n-1}$, $R_c(n) = 2(n-1)A_0(n)$, and $A(n) = 2.5(C(n)+R_c(n)) = 0.0750$ of Theorem 6.5(ii). For every $0 \leq i < R_{\mathrm{each}} = 2A_0K+1$ and $0 \leq m < N+R = (2n-1)A_0K+n$, the polynomial $P(u) = \mathcal{E}_i(u)\mathcal{E}_m(u)$ satisfies

$$2\deg_u P = 2(i+m) \leq 4A_0K+2(2n-1)A_0K+2n-2 < 12(2n-1)A_0K-2 = 4AK+2(2n-1)A_0K-2 \tag{7.15}$$

for all $K \geq K_0(n,\mathbf{a})$ (since $(4AK+2(2n-1)A_0K-2)-2\deg_u P \geq (20n-14)A_0(n)K-2n \geq \frac{20n-14}{2n-1}(100n)-2n \geq 598n > 0$), confirming the kernel degree condition (3.1).

For each scaling integer $K \geq K_0(n,\mathbf{a})$ such that $A_0(n)K, A(n)K \in \mathbb{Z}_{\geq 1}$, form the $(N+R)\times(N+R)$ rational matrix $M_K(\mathbf{a},c)$ of Definition 5.1, its determinant $\Delta_{N+R} = \det M_K(\mathbf{a},c) \in \mathbb{Q}$,

and the determinant-clearing rational multiplier $D^{\mathrm{eff}}_{N+R} \in \mathbb{Q}_{>0}$ of Theorem 6.5(i) (6.17). By Theorem 6.5(i) (6.18),

$$D^{\mathrm{eff}}_{N+R} \cdot \Delta_{N+R} \in \mathbb{Z} \qquad \text{for every admissible } K \geq K_0(n, \mathbf{a}). \tag{7.16}$$

By Proposition 5.2 and Theorem 7.1, $\Delta_{N+R} \neq 0$ for every admissible $K \geq K_0(n, \mathbf{a})$. Since $D^{\mathrm{eff}}_{N+R} > 0$ and a non-zero integer has absolute value at least 1, (7.16) implies Liouville's integrality lower bound:

$$D^{\mathrm{eff}}_{N+R} \cdot |\Delta_{N+R}| \geq 1 \qquad \text{for every admissible } K \geq K_0(n, \mathbf{a}). \tag{7.17}$$

On the other hand, by Theorem 5.4(ii) and Theorem 6.5(ii) (6.23),

$$\limsup_{K\to\infty} \frac{1}{K^2} \log\Big(D^{\mathrm{eff}}_{N+R} \cdot |\Delta_{N+R}|\Big) \leq \Lambda^{(N+R)}_{\mathrm{net,full}}(n) \leq -0.000146 < 0, \tag{7.18}$$

which implies $\lim_{K\to\infty} D^{\mathrm{eff}}_{N+R} \cdot |\Delta_{N+R}| = 0$, and in particular $0 < D^{\mathrm{eff}}_{N+R} \cdot |\Delta_{N+R}| < 1$ for all sufficiently large admissible K. This contradicts (7.17), completing the induction on n and establishing Theorem 1.1 for all positive integers $n \geq 1$. □

8. Computational Verification and Reproducibility

To support independent verification and reproducibility of all algebraic identities, central factorial derivative formulas, matrix rank assertions, closed-form determinant factorizations, midpoint harmonic centering identities, Euler sine-square eigenvalue laws, and asymptotic scaling bounds in this paper, we provide the standalone Python 3 verification suite `reproducibility2_final.py`.

8.1. **Central factorial derivatives and harmonic telescoping ($m = 2, \ldots, 10000$).** Suite 1 of `reproducibility2_final.py` verifies Proposition 2.2 (both (2.3) and (2.4)), Proposition 2.3 (all three forms of (2.7) across $0 \leq r < m$), Proposition 2.4, Theorem 2.5 (including (2.20) and (2.23)), and Theorem 2.6. Table 1 displays the rational telescoping sum $T(m,1) = \frac{1}{4} + \frac{1}{2}(\frac{1}{m-1} + \frac{1}{m})$, the scaled remainder $4T(m,1) - 1 = \frac{2(2m-1)}{m(m-1)}$, and the decimal logarithms $\log_{10}|\mathcal{E}'_m(1)|$ and $\log_{10}|\frac{1}{2}\mathcal{E}''_m(1)|$ across $m = 2, \ldots, 10$ and $m = 100, 200, 500, 10000$.

m	$T(m,1)$ (exact / decimal)	$4T(m,1) - 1 = \frac{2(2m-1)}{m(m-1)}$	$\log_{10}\lvert\mathcal{E}'_m(1)\rvert$	$\log_{10}\lvert\frac{1}{2}\mathcal{E}''_m(1)\rvert$
2	1	3	−1.079181	−1.079181
3	2/3	5/3	−2.079181	−2.255273
4	13/24	7/6	−2.924279	−3.190547
5	19/40	9/10	−3.702431	−4.025737
6	13/30	11/15	−4.442793	−4.805971
7	17/42	13/21	−5.158797	−5.551597
8	43/112	15/28	−5.857767	−6.273516
9	53/144	17/36	−6.544147	−6.978234
10	16/45	19/45	−7.220841	−7.669934
100	0.2600505051	4.020202×10^{-2}	−62.952517	−63.537459
200	0.2550125628	2.005025×10^{-2}	−123.612520	−124.205958
500	0.2520020040	8.008016×10^{-3}	−304.828898	−305.427494
10000	0.2501000050	4.000200×10^{-4}	−6026.351289	−6026.953176

Table 1. Central factorial harmonic telescoping sums $T(m,1) = \frac{1}{4} + \frac{1}{2}(\frac{1}{m-1} + \frac{1}{m})$, remainder $4T(m,1) - 1$, and derivative magnitudes at $r = 1$.

8.2. **Finite-matrix verification of jet separation, tail determinant, and transition matrix convergence.** Suites 2 and 3 of `reproducibility2_final.py` verify Lemma 3.2, Theorem 3.3, Theorem 3.5, Lemma 4.2, Theorem 4.3, Theorem 4.4, Proposition 5.2, Theorem 5.3, Theorem 6.3, Proposition 6.4(i), and Theorem 7.1 in exact rational and 100-digit arithmetic across $(n, A_0K, AK) \in \{(2,1,4), (2,2,7), (2,3,10), (3,1,6), (3,2,11), (4,1,8)\}$, confirming $\mathrm{rank}(L_{\mathrm{top}}) = R$, $\mathrm{rank}(S_{2n+1,\mathrm{bot}}) = N$, $N_0 = 0$, and exact ratio 1 in (4.24). Table 2 displays all six rational entries $\Gamma_{m-N,d}$ $(0 \leq m - N < 3, 0 \leq d < 2)$ of the 3×2 transition matrix $\Gamma = (L_{\mathrm{top}}[:, J_{\mathrm{tail}}])^{-1} L_{\mathrm{top}}[:, J_{\mathrm{head}}]$ at $n = 2, A_0K = 1$ as AK increases from 3 to 20.

AK	$\Gamma_{0,0}$ $(m=2)$	$\Gamma_{0,1}$ $(m=2)$	$\Gamma_{1,0}$ $(m=3)$	$\Gamma_{1,1}$ $(m=3)$	$\Gamma_{2,0}$ $(m=4)$	$\Gamma_{2,1}$ $(m=4)$	lcm(denom(Γ))
3	−193.833333	+83.500000	−7940.000000	+2820.000000	−41848.333333	+13405.000000	6
4	−197.620000	+89.630000	−7999.300000	+2915.700000	−42067.200000	+13757.800000	100
5	−199.048000	+92.660000	−8018.240000	+2955.800000	−42132.720000	+13896.400000	250
6	−199.719184	+94.420816	−8025.910204	+2975.889796	−42157.695238	+13961.771429	7350
8	−200.299470	+96.342443	−8031.457965	+2994.162188	−42174.451190	+14016.830952	352800
10	−200.532967	+97.352863	−8033.231388	+3001.809043	−42179.275695	+14037.599226	53361000
12	−200.648549	+97.969738	−8033.954786	+3005.660190	−42181.067116	+14047.124279	360720360
16	−200.754073	+98.679622	−8034.485065	+3009.203543	−42182.240869	+14054.939484	16679709446400
20	−200.799099	+99.074162	−8034.658244	+3010.714648	−42182.570782	+14057.811213	3763359443844000

TABLE 2. Convergence of all six entries $\Gamma_{m-N,d}$ of the 3×2 transition matrix $\Gamma = (L_{\mathrm{top}}[:, J_{\mathrm{tail}}])^{-1} L_{\mathrm{top}}[:, J_{\mathrm{head}}]$ at $n = 2, A_0K = 1, (a_3, a_5) = (1, 2)$ as AK increases from 3 to 20.

8.3. **Full-range continuum exponents across $n = 1, \ldots, 10000$.** Suites 4 and 5 of `reproducibility2_final.py` evaluate the archimedean upper bounds $\Lambda^{\mathrm{UB}}_{\mathrm{arch}}(n)$ (6.21) and $\Lambda^{\mathrm{tight}}_{\mathrm{arch}}(n)$, the closed-form p-adic denominator bound $\Phi^{\mathrm{UB}}_{\mathrm{den}}(n) = -C_{\mathrm{fac}}(n) - \mathcal{I}^{\mathrm{LB}}_{\mathrm{occ}}(n)$ (6.22), and 60-digit outward-rounded interval enclosures (`mpmath.iv`). Table 3 presents the parameter scan at $n = 2$ across $A_0 \in \{0.008, 0.010, 0.012, 0.015\}$, Table 4 presents the asymptotic scaling across $n = 1, \ldots, 10$ and $n = 100, 200, 500, 10000$, and Table 5 records the 60-digit interval-certified closed-form bounds.

Kernel Family	$A_0 = C$	Rank R_c	Outer A	$\Lambda^{\mathrm{tight}}_{\mathrm{arch}}$	$\Phi^{\mathrm{UB}}_{\mathrm{den}}$	Closed-Form Net UB
$\mathcal{F}^{\mathrm{odd},[A_0,A]}_{5,2}$ $(L_{\mathrm{in}} = 5)$	0.0080	0.0160	0.0600	+0.001901	−0.004740	−0.002839
$\mathcal{F}^{\mathrm{odd},[A_0,A]}_{5,2}$ $(L_{\mathrm{in}} = 5)$	0.0100	0.0200	0.0750	+0.002970	−0.007094	−0.004124
$\mathcal{F}^{\mathrm{odd},[A_0,A]}_{5,2}$ $(L_{\mathrm{in}} = 5)$	0.0120	0.0240	0.0900	+0.004277	−0.009848	−0.005571
$\mathcal{F}^{\mathrm{odd},[A_0,A]}_{5,2}$ $(L_{\mathrm{in}} = 5)$	0.0150	0.0300	0.1125	+0.006683	−0.014685	−0.008001

TABLE 3. Closed-form archimedean, p-adic denominator, and net upper bounds for $n = 2$ ($\{1, \zeta(3), \zeta(5)\}$) across $A_0 \in \{0.008, 0.010, 0.012, 0.015\}$ with $A = 2.5(C + R_c) = 7.5A_0$.

n	L_{in}	$A_0(n) = C(n)$	$R_c(n)$	$C_{\mathrm{fac}}(n)$	$\Lambda^{\mathrm{UB}}_{\mathrm{arch}}(n)$ (6.21)	$\Lambda^{\mathrm{tight}}_{\mathrm{arch}}(n)$	$\Phi^{\mathrm{UB}}_{\mathrm{den}}(n)$ (6.22)	Net UB (Std)	Net UB (Tight)
1	3	3.000000×10^{-2}	0.000000	+0.007764	−0.001848	−0.001848	+0.000786	−0.001062	−0.001062
2	5	1.000000×10^{-2}	0.020000	+0.006311	+0.003356	+0.002970	−0.007094	−0.003738	−0.004124
3	7	6.000000×10^{-3}	0.024000	+0.005601	+0.004065	+0.003792	−0.011199	−0.007134	−0.007407
4	9	4.285714×10^{-3}	0.025714	+0.005224	+0.004246	+0.004039	−0.009145	−0.004899	−0.005106
5	11	3.333333×10^{-3}	0.026667	+0.004989	+0.004283	+0.004117	−0.007872	−0.003589	−0.003755
6	13	2.727273×10^{-3}	0.027273	+0.004827	+0.004267	+0.004129	−0.007010	−0.002743	−0.002882
7	15	2.307692×10^{-3}	0.027692	+0.004709	+0.004230	+0.004111	−0.006389	−0.002160	−0.002278
8	17	2.000000×10^{-3}	0.028000	+0.004619	+0.004183	+0.004079	−0.005921	−0.001738	−0.001842
9	19	1.764706×10^{-3}	0.028235	+0.004547	+0.004133	+0.004041	−0.005555	−0.001422	−0.001514
10	21	1.578947×10^{-3}	0.028421	+0.004489	+0.004082	+0.003999	−0.005261	−0.001179	−0.001262
100	201	1.507538×10^{-4}	0.029849	+0.003963	+0.002337	+0.002328	−0.002824	−0.000488	−0.000496
200	401	7.518797×10^{-5}	0.029925	+0.003925	+0.001729	+0.001725	−0.002681	−0.000952	−0.000956
500	1001	3.003003×10^{-5}	0.029970	+0.003901	+0.000915	+0.000913	−0.002593	−0.001679	−0.001680
10000	20001	1.500075×10^{-6}	0.029998	+0.003883	−0.001775	−0.001775	−0.002535	−0.004311	−0.004311

TABLE 4. Closed-form archimedean upper bounds $\Lambda^{\mathrm{UB}}_{\mathrm{arch}}(n)$ (6.21) and $\Lambda^{\mathrm{tight}}_{\mathrm{arch}}(n)$, effective p-adic denominator bound $\Phi^{\mathrm{UB}}_{\mathrm{den}}(n)$ (6.22), and net upper bounds across $n = 1, \ldots, 10$ and $n = 100, 200, 500, 10000$.

n	$\Lambda^{\mathrm{UB}}_{\mathrm{arch}}(n)$ Interval $[a,b]$	$\Lambda^{\mathrm{tight}}_{\mathrm{arch}}(n)$ Interval $[a,b]$	$\Phi^{\mathrm{UB}}_{\mathrm{den}}(n)$ Interval $[a,b]$	Net UB (Std) Upper Endpoint b	Net UB (Tight) Upper Endpoint b
1	$[-0.001849, -0.001847]$	$[-0.001849, -0.001847]$	$[+0.000785, +0.000787]$	-0.001062	-0.001062
2	$[+0.003355, +0.003357]$	$[+0.002969, +0.002971]$	$[-0.007095, -0.007093]$	-0.003738	-0.004124
3	$[+0.004064, +0.004066]$	$[+0.003791, +0.003793]$	$[-0.011200, -0.011198]$	-0.007134	-0.007407
4	$[+0.004245, +0.004247]$	$[+0.004038, +0.004040]$	$[-0.009146, -0.009144]$	-0.004899	-0.005106
5	$[+0.004282, +0.004284]$	$[+0.004116, +0.004118]$	$[-0.007873, -0.007871]$	-0.003589	-0.003755
10	$[+0.004081, +0.004083]$	$[+0.003998, +0.004000]$	$[-0.005262, -0.005260]$	-0.001179	-0.001262
100	$[+0.002336, +0.002338]$	$[+0.002327, +0.002329]$	$[-0.002825, -0.002823]$	-0.000488	-0.000496
10000	$[-0.001776, -0.001774]$	$[-0.001776, -0.001774]$	$[-0.002536, -0.002534]$	-0.004311	-0.004311

TABLE 5. Certified 60-digit outward-rounded interval enclosures $[a, b]$ (`mpmath.iv`) for the closed-form archimedean upper bounds $\Lambda^{\mathrm{UB}}_{\mathrm{arch}}(n)$ and $\Lambda^{\mathrm{tight}}_{\mathrm{arch}}(n)$, the p-adic denominator bound $\Phi^{\mathrm{UB}}_{\mathrm{den}}(n)$, and their net upper endpoints across $n \in \{1, 2, 3, 4, 5, 10, 100, 10000\}$.

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References

[1] R. Apéry, *Irrationalité de $\zeta(2)$ et $\zeta(3)$*, Astérisque **61** (1979), 11–13.
[2] K. Ball and T. Rivoal, *Irrationalité d'une infinité de valeurs de la fonction zêta aux entiers impairs*, Invent. Math. **146** (2001), no. 1, 193–207.
[3] F. Beukers, *A note on the irrationality of $\zeta(2)$ and $\zeta(3)$*, Bull. London Math. Soc. **11** (1979), no. 3, 268–272.
[4] F. Brown, *Mixed Tate motives over $\mathbb{Z}$*, Ann. of Math. (2) **175** (2012), no. 2, 949–976.
[5] F. Brown and W. Zudilin, *On cellular rational approximations to $\zeta(5)$*, arXiv preprint `arXiv:2210.03391`, 2022 (revised 2026).
[6] F. Calegari, V. Dimitrov, and Y. Tang, *The linear independence of $1, \zeta(2)$, and $L(2, \chi_{-3})$*, arXiv preprint `arXiv:2408.15403`, 2024.
[7] L. Euler, *De summis serierum reciprocarum*, Comment. Acad. Sci. Petrop. **7** (1740), 123–134.
[8] A. Fauzan, *$\zeta(5)$ is irrational*, Zenodo preprint, `doi:10.5281/zenodo.22826419`, September 2026.
[9] S. Fischler, *Linear independence of odd zeta values using Siegel's lemma*, J. Lond. Math. Soc. (2) **113** (2026), no. 4, Paper No. e70535, 42 pp.
[10] S. Fischler, J. Sprang, and W. Zudilin, *Many odd zeta values are irrational*, Compos. Math. **155** (2019), no. 5, 938–952.
[11] S. Fischler and W. Zudilin, *A refinement of Nesterenko's linear independence criterion with applications to zeta values*, Math. Ann. **347** (2010), no. 4, 739–763.
[12] I. Gessel and G. Viennot, *Binomial determinants, paths, and hook length formulae*, Adv. in Math. **58** (1985), no. 3, 300–321.
[13] S. Karlin, *Total Positivity, Vol. I*, Stanford Univ. Press, Stanford, CA, 1968.
[14] L. Lai, *Small improvements to the Ball–Rivoal theorem and its p-adic variant*, arXiv preprint `arXiv:2407.14236`, 2024.
[15] L. Lai, *The linear independence of $1, \zeta(3)$, and $\zeta(4)$*, arXiv preprint `arXiv:2610.06838v2`, October 2026.
[16] L. Lai and P. Yu, *A note on the number of irrational odd zeta values*, Compos. Math. **156** (2020), no. 8, 1699–1717.
[17] F. Lindemann, *Über die Zahl π*, Math. Ann. **20** (1882), no. 2, 213–225.
[18] B. Lindström, *On the vector representations of induced matroids*, Bull. London Math. Soc. **5** (1973), 85–90.
[19] Yu. V. Nesterenko, *On the linear independence of numbers*, Vestnik Moskov. Univ. Ser. I Mat. Mekh. **40** (1985), no. 1, 46–49.
[20] T. Rivoal, *La fonction zêta de Riemann prend une infinité de valeurs irrationnelles aux entiers impairs*, C. R. Acad. Sci. Paris Sér. I Math. **331** (2000), no. 4, 267–270.
[21] E. B. Saff and V. Totik, *Logarithmic Potentials with External Fields*, Grundlehren Math. Wiss. **316**, Springer-Verlag, Berlin, 1997.
[22] Q. Tang, *`zeta2-zeta3-linear-independence`: Lean 4 formalization of the linear independence of $1, \zeta(2)$, and $\zeta(3)$ over $\mathbb{Q}$*, GitHub repository `dtq1997/zeta2-zeta3-linear-independence`, September 2026.
[23] W. Zudilin, *One of the numbers $\zeta(5), \zeta(7), \zeta(9), \zeta(11)$ is irrational*, Uspekhi Mat. Nauk **56** (2001), no. 4, 149–150.
[24] W. Zudilin, *Arithmetic of linear forms involving odd zeta values*, J. Théor. Nombres Bordeaux **16** (2004), no. 1, 251–291.